

Complete Course on Vector Calculus

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1 Review of Vectors & Geometry

1.1 Goals

- Define what a vector is.
- Add vectors.
- Length of a vector.
- Unit vectors & direction.

Definition 1.1

In this course, a *vector* is an element of the vector space $\mathbb{R}^n$, usually with $n = 2$ or 3 . We write its components as $\vec{v} = \langle v_1, \dots, v_n \rangle$. Vectors can be added and multiplied by real numbers componentwise.

Geometrically, a vector represents a displacement. A point P specifies a location, whereas a vector $\vec{v}$ specifies a change in location: starting at P and moving by $\vec{v}$ gives the point $P + \vec{v}$. The same vector can be drawn starting at any point.

Points and vectors both use coordinate tuples, but play different roles. We use parentheses for points and angle brackets for vectors. The difference $Q - P$ is a vector. After choosing an origin O , the point P has a position vector $\overrightarrow{OP}$. Velocity and force are further examples of vectors.

The vector $\overrightarrow{AB}$ from $A = (a_1, a_2, a_3)$ to $B = (b_1, b_2, b_3)$ is

$$\overrightarrow{AB} = \langle b_1 - a_1, b_2 - a_2, b_3 - a_3 \rangle.$$

The *magnitude* (length) of a vector $\vec{u} = \langle u_1, u_2, u_3 \rangle$ is given by

$$\|\vec{u}\| = \sqrt{u_1^2 + u_2^2 + u_3^2}.$$

Example 1.1. Let $A = (1, 2, 3)$ and $B = (4, 6, 3)$.

1. Find the vector $\overrightarrow{AB}$.
2. Compute its magnitude.

1.2 Vector Addition & Scalar Multiplication

Add vectors componentwise:

$$\langle u_1, u_2, u_3 \rangle + \langle v_1, v_2, v_3 \rangle = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3 \rangle.$$

Geometrically, this is the *triangle rule*: place the tail of $\vec{v}$ at the head of $\vec{u}$. Their sum runs from the initial tail to the final head (see [Figure 1.1](#)).

Example 1.2. Find the sum of $\vec{u} = \langle 1, 2 \rangle$ and $\vec{v} = \langle 3, -5 \rangle$.

Solution.

$$\vec{u} + \vec{v} = \langle 1 + 3, 2 + (-5) \rangle = \langle 4, -3 \rangle.$$

■

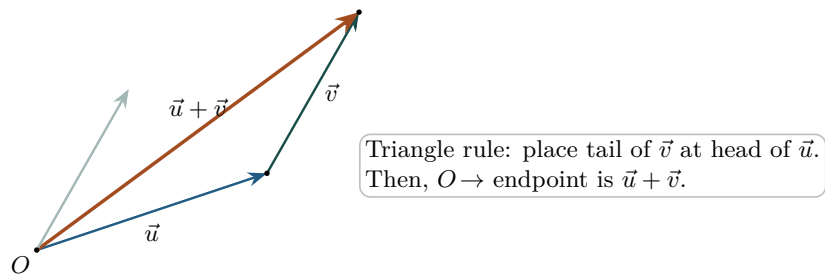


Figure 1.1: Triangle rule.

1.3 Scalar Multiplication

For $\lambda \in \mathbb{R}$, define $\lambda \vec{u}$ by multiplying each component by λ . Its length is $|\lambda| \|\vec{u}\|$. If $\vec{u} \neq \vec{0}$, multiplication by a positive scalar preserves its direction and multiplication by a negative scalar reverses it. If $\lambda = 0$, the result is the zero vector, which has no direction.

Example 1.3. When $\vec{u} = \langle u_1, u_2, u_3 \rangle$, then $\lambda \vec{u} = \langle \lambda u_1, \lambda u_2, \lambda u_3 \rangle$.

1.4 Standard Basis:

The vectors $\hat{i} = \langle 1, 0, 0 \rangle$, $\hat{j} = \langle 0, 1, 0 \rangle$, and $\hat{k} = \langle 0, 0, 1 \rangle$ are the standard basis vectors of the 3-dimensional vector space $\mathbb{R}^3$.

Example 1.4. Let $\vec{u} = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{v} = 2\hat{i} + 3\hat{j} - \hat{k}$. Find $2\vec{u} + 3\vec{v}$.

Solution.

$$\begin{aligned} 2\vec{u} + 3\vec{v} &= 2(\hat{i} - 2\hat{j} + \hat{k}) + 3(2\hat{i} + 3\hat{j} - \hat{k}) \\ &= (2\hat{i} - 4\hat{j} + 2\hat{k}) + (6\hat{i} + 9\hat{j} - 3\hat{k}) \\ &= 8\hat{i} + 5\hat{j} - \hat{k}. \end{aligned}$$

■

Definition 1.2

A vector $\vec{u}$ is a *unit vector* if it has length 1, that is,

$$\|\vec{u}\| = 1.$$

Note

Given a nonzero vector $\vec{w}$, the unit vector in the direction of $\vec{w}$ is

$$\text{unit vector in the direction of } \vec{w} = \frac{\vec{w}}{\|\vec{w}\|}.$$

Example 1.5.

1. Find a unit vector in the direction of $\vec{u} = \langle -5, 4 \rangle$.
2. Find a unit vector in the direction of $\vec{w} = 6\hat{i} - 3\hat{j} + 4\hat{k}$.

Solution. 1.

$$\|\vec{u}\| = \sqrt{(-5)^2 + 4^2} = \sqrt{25 + 16} = \sqrt{41},$$

$$\hat{u} = \frac{\vec{u}}{\|\vec{u}\|} = \left\langle \frac{-5}{\sqrt{41}}, \frac{4}{\sqrt{41}} \right\rangle.$$

2.

$$\|\vec{w}\| = \sqrt{6^2 + (-3)^2 + 4^2} = \sqrt{36 + 9 + 16} = \sqrt{61},$$

$$\hat{w} = \frac{\vec{w}}{\|\vec{w}\|} = \frac{6}{\sqrt{61}}\hat{i} - \frac{3}{\sqrt{61}}\hat{j} + \frac{4}{\sqrt{61}}\hat{k}.$$

■

Problem 1.6. A kayaker starts from the south bank of a straight river flowing due east at 4 mph. The kayak's speed relative to the water is 8 mph.

1. In what direction should the kayaker point the kayak to land straight across the river?
2. If the river is 0.75 miles wide, how long will the crossing take?

Solution. 1. To land straight north, cancel the river's 4 mph eastward drift with a 4 mph westward component of the kayak's 8 mph speed. Let $\vec{v}_{\text{rel}} = \langle -4, \sqrt{8^2 - 4^2} \rangle = \langle -4, 4\sqrt{3} \rangle$ (west, north). Thus the heading is 30° *west of north* (since $\tan \theta = \frac{4}{4\sqrt{3}} = \frac{1}{\sqrt{3}}$).

2. Ground speed across (north) is $4\sqrt{3}$ mph. For width 0.75 mi:

$$t = \frac{0.75}{4\sqrt{3}} = \frac{3}{16\sqrt{3}} \text{ hr} = \frac{\sqrt{3}}{16} \text{ hr} \approx 0.108 \text{ hr} \approx 6.5 \text{ min.}$$

■

2 The Dot Product

2.1 Goals

- Define the dot product between two vectors.
- Geometry of the dot product: magnitude and angles between vectors.
- Orthogonality.
- Projection of vectors.

Definition 2.1

The dot product of two vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{w} = \langle w_1, w_2, w_3 \rangle$ is

$$\mathbf{u} \cdot \mathbf{w} := u_1w_1 + u_2w_2 + u_3w_3.$$

The same definition applies in $\mathbb{R}^n$: multiply corresponding components and add the results. The dot product is a scalar, while a sum of vectors is a vector.

Example 2.1. Find the dot product of:

1. $\langle 2, 5 \rangle$ and $\langle -2, 4 \rangle$.
2. $4\hat{i} - \hat{j} + 3\hat{k}$ and $2\hat{i} + 3\hat{j} - \hat{k}$.

Solution. 1.

$$\begin{aligned}\langle 2, 5 \rangle \cdot \langle -2, 4 \rangle &= (2)(-2) + (5)(4) \\ &= -4 + 20 \\ &= 16.\end{aligned}$$

2.

$$\begin{aligned}(4\hat{i} - \hat{j} + 3\hat{k}) \cdot (2\hat{i} + 3\hat{j} - \hat{k}) &= (4)(2) + (-1)(3) + (3)(-1) \\ &= 8 - 3 - 3 \\ &= 2.\end{aligned}$$

■

Proposition 2.1

Use the definition of the dot product to show that it is commutative—i.e., $\mathbf{u} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{u}$.

2.2 Other properties:

1. $\mathbf{u} \cdot \mathbf{u} = \|\mathbf{u}\|^2$;
2. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$;
3. $\mathbf{0} \cdot \mathbf{u} = 0$;
4. $(\lambda \mathbf{u}) \cdot \mathbf{w} = \lambda(\mathbf{u} \cdot \mathbf{w})$.

Theorem 2.2

If $\mathbf{u}$ and $\mathbf{w}$ are nonzero and $\theta \in [0, \pi]$ is the angle between them, then

$$\mathbf{u} \cdot \mathbf{w} = \|\mathbf{u}\| \|\mathbf{w}\| \cos(\theta).$$

Remark 2.2. We can use Theorem 2.2 to find the angle between vectors.

Example 2.3. Find the angle between the vectors $\mathbf{a} = \langle 1, 1, 0 \rangle$ and $\mathbf{b} = \langle 0, 1, 1 \rangle$.

Solution. We use

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}.$$

$$\mathbf{a} \cdot \mathbf{b} = (1)(0) + (1)(1) + (0)(1) = 1,$$

$$\|\mathbf{a}\| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}, \quad \|\mathbf{b}\| = \sqrt{0^2 + 1^2 + 1^2} = \sqrt{2}.$$

$$\cos \theta = \frac{1}{\sqrt{2} \cdot \sqrt{2}} = \frac{1}{2}.$$

Thus,

$$\theta = \cos^{-1}\left(\frac{1}{2}\right) = 60^\circ.$$

■

Two vectors are *orthogonal*, written $\mathbf{u} \perp \mathbf{w}$, if $\mathbf{u} \cdot \mathbf{w} = 0$. For nonzero vectors, this means that the angle between them is $\pi/2$. The zero vector is orthogonal to every vector.

Remark 2.4. We can test whether two vectors $\mathbf{u}$ and $\mathbf{w}$ are orthogonal by computing their dot product.

Example 2.5. Decide whether each pair of vectors is orthogonal.

1. $\mathbf{u} = \langle 1, 2, 3 \rangle$ and $\mathbf{v} = \langle 4, -5, 2 \rangle$;
2. $\mathbf{u} = 2\hat{i} - 5\hat{j}$ and $\mathbf{v} = 5\hat{i} + 2\hat{j}$.

2.3 Vector and Scalar Projections

For $\mathbf{v} \neq \mathbf{0}$, the projection of $\mathbf{u}$ onto $\mathbf{v}$ is the component of $\mathbf{u}$ along the line spanned by $\mathbf{v}$ (see Figure 2.1). It may point in the same or the opposite direction as $\mathbf{v}$.

The *vector projection* of $\mathbf{u}$ onto $\mathbf{v}$ is

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \right) \mathbf{v} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \right) \cdot \frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

The *scalar projection* of $\mathbf{u}$ onto $\mathbf{v}$ is the (signed) magnitude of the vector projection—i.e.,

$$\text{comp}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|}.$$

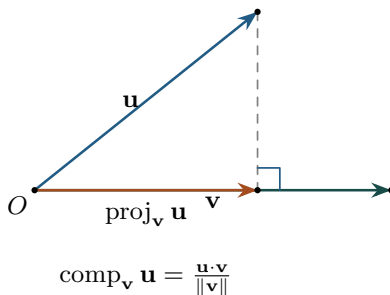


Figure 2.1: Vector projection and scalar projection.

Example 2.6. Let $\mathbf{u} = \langle 4, -1, 2 \rangle$ and $\mathbf{v} = \langle -2, 3, 5 \rangle$.

1. Find the scalar projection of $\mathbf{u}$ onto $\mathbf{v}$;
2. Find the vector projection of $\mathbf{u}$ onto $\mathbf{v}$.

Solution. 1. The scalar projection is

$$\text{comp}_{\mathbf{v}}(\mathbf{u}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|}.$$

Compute

$$\mathbf{u} \cdot \mathbf{v} = (4)(-2) + (-1)(3) + (2)(5) = -8 - 3 + 10 = -1,$$

$$\|\mathbf{v}\| = \sqrt{(-2)^2 + 3^2 + 5^2} = \sqrt{4 + 9 + 25} = \sqrt{38}.$$

Hence

$$\text{comp}_{\mathbf{v}}(\mathbf{u}) = \frac{-1}{\sqrt{38}}.$$

2. The vector projection is

$$\text{proj}_{\mathbf{v}}(\mathbf{u}) = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \right) \mathbf{v}.$$

Since $\|\mathbf{v}\|^2 = 38$, we get

$$\text{proj}_{\mathbf{v}}(\mathbf{u}) = \frac{-1}{38} \langle -2, 3, 5 \rangle = \left\langle \frac{2}{38}, \frac{-3}{38}, \frac{-5}{38} \right\rangle.$$

■

An application of vector projection in physics is calculating work. More specifically:

When a constant force $\vec{F}$ is applied to an object so that the object moves in a straight line from P to Q , the work W done by the force $\vec{F}$ acting at an angle θ from the line of motion is

$$W = \vec{F} \cdot \overrightarrow{PQ} = \|\vec{F}\| \|\overrightarrow{PQ}\| \cos \theta.$$

Example 2.7. A sled moves 10 m due east $\vec{D} = \langle 10, 0 \rangle$ m. A constant pull acts with force $\vec{F} = \langle 6, 8 \rangle$ N. The work done by the pulling force is:

$$\text{Work} = \vec{F} \cdot \vec{D} = \langle 6, 8 \rangle \cdot \langle 10, 0 \rangle = 60 \text{ J}.$$

3 The Cross Product

3.1 Goals

- Compute 2×2 and 3×3 determinants.
- Learn cross products; properties.
- Scalar triple products; volume.
- Torque and Physics

3.2 Determinants.

Recall how one computes:

1. For a 2×2 matrix:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

2. For a 3×3 matrix:

$$\det \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}.$$

Definition 3.1

Let $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$ be two vectors. The *cross product* of $\mathbf{a}$ and $\mathbf{b}$ is the vector

$$\mathbf{a} \times \mathbf{b} := \det \begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{pmatrix} = \langle a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1 \rangle.$$

Example 3.1. Let $\mathbf{u} = \langle 2, -1, 3 \rangle$ and $\mathbf{v} = \langle 4, 0, -2 \rangle$. Find $\mathbf{u} \times \mathbf{v}$.

Solution.

$$\begin{aligned} \mathbf{u} \times \mathbf{v} &= \det \begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & -1 & 3 \\ 4 & 0 & -2 \end{pmatrix} \\ &= \langle (-1)(-2) - 3(0), 3(4) - 2(-2), 2(0) - (-1)(4) \rangle \\ &= \langle 2, 16, 4 \rangle. \end{aligned}$$

■

Example 3.2. Show that $\mathbf{a} \times \mathbf{a} = \vec{0}$ for all vectors $\mathbf{a}$.

Proof. Let $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$. Then

$$\mathbf{a} \times \mathbf{a} = \langle a_2a_3 - a_3a_2, a_3a_1 - a_1a_3, a_1a_2 - a_2a_1 \rangle = \langle 0, 0, 0 \rangle = \vec{0}.$$

■

3.3 Properties of the Cross Product

Theorem 3.1

The cross product $\mathbf{a} \times \mathbf{b}$ is orthogonal to both $\mathbf{a}$ and $\mathbf{b}$. If it is nonzero, its direction is determined by the right-hand rule: curl the fingers of your right hand from $\mathbf{a}$ toward $\mathbf{b}$, and your thumb points along $\mathbf{a} \times \mathbf{b}$.

Proof. Let $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$. Set $\mathbf{a} \times \mathbf{b} = \langle c_1, c_2, c_3 \rangle$. Then

$$\begin{aligned} \mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) &= a_1c_1 + a_2c_2 + a_3c_3 \\ &= a_1(a_2b_3 - a_3b_2) + a_2(a_3b_1 - a_1b_3) + a_3(a_1b_2 - a_2b_1) \\ &= 0. \end{aligned}$$

Similarly,

$$\begin{aligned} \mathbf{b} \cdot (\mathbf{a} \times \mathbf{b}) &= b_1c_1 + b_2c_2 + b_3c_3 \\ &= b_1(a_2b_3 - a_3b_2) + b_2(a_3b_1 - a_1b_3) + b_3(a_1b_2 - a_2b_1) \\ &= 0. \end{aligned}$$

Hence $\mathbf{a} \times \mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

■

Beyond direction, the other key property of the cross product is its magnitude. What is $\|\mathbf{a} \times \mathbf{b}\|$ in terms of $\mathbf{a}$ and $\mathbf{b}$? The answer is given in the following theorem:

Theorem 3.2

If $\mathbf{a}$ and $\mathbf{b}$ are nonzero and $\theta \in [0, \pi]$ is the angle between them, then

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta.$$

Corollary 3.3

Two nonzero vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ are parallel if and only if $\mathbf{a} \times \mathbf{b} = \vec{0}$. More generally, the cross product vanishes exactly when the vectors are linearly dependent.

Proof.

$$\begin{aligned} \mathbf{a} \parallel \mathbf{b} &\iff \theta = 0 \text{ or } \pi \\ &\iff \sin \theta = 0 \\ &\iff \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta = 0 \iff \|\mathbf{a} \times \mathbf{b}\| = 0 \end{aligned}$$

■

Remark 3.3. The magnitude $\|\mathbf{a} \times \mathbf{b}\|$ equals the area of the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$.

Example 3.4. Find the area of the triangle with vertices $A = (2, -1, 0)$, $B = (9, 3, 0)$, and $C = (5, 11, 0)$.

Solution. The area of $\triangle ABC$ is

$$\frac{1}{2} \|\vec{AB} \times \vec{AC}\|.$$

We have $\vec{AB} = \langle 7, 4, 0 \rangle$ and $\vec{AC} = \langle 3, 12, 0 \rangle$. Thus

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 7 & 4 & 0 \\ 3 & 12 & 0 \end{vmatrix} = \langle 0, 0, 72 \rangle,$$

so

$$\text{Area} = \frac{1}{2} \|\vec{AB} \times \vec{AC}\| = \frac{1}{2} \cdot 72 = 36.$$

■

3.4 Other key properties:

1. $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$
2. $(\lambda \mathbf{a}) \times \mathbf{b} = \lambda(\mathbf{a} \times \mathbf{b}) = \mathbf{a} \times (\lambda \mathbf{b})$
3. $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$
4. $(\mathbf{a} + \mathbf{b}) \times \mathbf{c} = \mathbf{a} \times \mathbf{c} + \mathbf{b} \times \mathbf{c}$
5. $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$

3.5 Triple Products:

The *scalar triple product* of vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ is defined as

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}).$$

Its absolute value, $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$, equals the volume of the parallelepiped determined by $\mathbf{a}, \mathbf{b}, \mathbf{c}$.

Example 3.5. Find the volume of the parallelepiped determined by

$$\mathbf{a} = \langle 2, 1, 0 \rangle, \quad \mathbf{b} = \langle 1, 3, 1 \rangle, \quad \mathbf{c} = \langle 0, 2, 4 \rangle.$$

Solution. The volume is the absolute value of the scalar triple product:

$$V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|.$$

Compute

$$\mathbf{b} \times \mathbf{c} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 3 & 1 \\ 0 & 2 & 4 \end{vmatrix} = \langle 10, -4, 2 \rangle, \quad \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \langle 2, 1, 0 \rangle \cdot \langle 10, -4, 2 \rangle = 16.$$

Hence $V = 16$. ■

Example 3.6. A force $\mathbf{F} = \langle 0, 5, 0 \rangle$ newtons is applied at $P = (3, 0, 0)$ meters. Relative to a pivot at the origin, its position vector is $\mathbf{r} = \langle 3, 0, 0 \rangle$ meters. Find the torque $\boldsymbol{\tau}$ about the origin.

Solution. The cross product gives the torque

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}.$$

Compute

$$\mathbf{r} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 0 & 0 \\ 0 & 5 & 0 \end{vmatrix} = \langle 0, 0, 15 \rangle.$$

Thus the torque vector is $\boldsymbol{\tau} = \langle 0, 0, 15 \rangle \text{ N} \cdot \text{m}$, pointing in the positive z -direction. ■

4 Lines and Planes in $\mathbb{R}^3$

4.1 Goals

- Express lines in parametric and symmetric form.
- Equations of planes in $\mathbb{R}^3$.

A line in $\mathbb{R}^3$ is determined by a point and a nonzero direction vector (see [Figure 4.1](#)). In vector form,

$$\overrightarrow{P_0P} \parallel \vec{u} \iff \vec{r}(t) - \vec{r}_0 = t\vec{u}, \quad t \in \mathbb{R}.$$

Let $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$ and $\vec{u} = \langle a, b, c \rangle$. Then the parametric equations are

$$x(t) = x_0 + at, \quad y(t) = y_0 + bt, \quad z(t) = z_0 + ct.$$

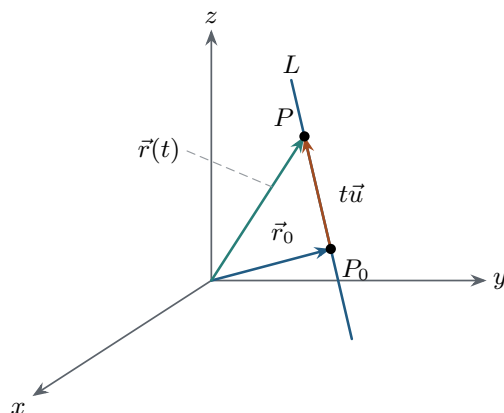


Figure 4.1: Line L in $\mathbb{R}^3$ through P_0 with direction $\vec{u}$, that is, $\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{u}$, where $\mathbf{r}_0 = \overrightarrow{OP_0}$.

Example 4.1. Find parametric equations for the line passing through $P(1, -2, 3)$ with direction vector $\mathbf{v} = \langle 4, 1, -5 \rangle$.

Solution. A line through point $P(x_0, y_0, z_0)$ with direction vector $\mathbf{v} = \langle a, b, c \rangle$ has parametric equations

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct, \quad t \in \mathbb{R}.$$

Substituting $P = (1, -2, 3)$ and $\mathbf{v} = \langle 4, 1, -5 \rangle$, we get

$$x = 1 + 4t, \quad y = -2 + t, \quad z = 3 - 5t.$$

Equivalently, in vector form,

$$\mathbf{r}(t) = \langle 1, -2, 3 \rangle + t\langle 4, 1, -5 \rangle.$$

■

Another way to describe the line $L \subset \mathbb{R}^3$ is to eliminate the parameter t . If $a, b, c \neq 0$, then from

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct$$

we obtain

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

Definition 4.1

The symmetric equations of a line are

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c},$$

provided $a, b, c \neq 0$.

If one of a, b, c is zero, keep that coordinate constant and omit its fraction. For example, if $a = 0$, then $x = x_0$ and

$$\frac{y - y_0}{b} = \frac{z - z_0}{c}.$$

Similarly,

$$\begin{aligned} b = 0 &\implies y = y_0, & \frac{x - x_0}{a} &= \frac{z - z_0}{c}, \\ c = 0 &\implies z = z_0, & \frac{x - x_0}{a} &= \frac{y - y_0}{b}. \end{aligned}$$

If two components of $\vec{u}$ are zero, the line is parallel to a coordinate axis; e.g., $b = c = 0$ gives $y = y_0$ and $z = z_0$ (with x free).

Example 4.2. Consider the line passing through the points $A(-3, 2, 5)$ and $B(1, -1, -2)$.

1. Find parametric and symmetric equations of the line.
2. Find the xy -trace (its intersection with the plane $z = 0$).

Solution. (a) Direction vector $\mathbf{v} = \overrightarrow{AB} = \langle 1 - (-3), -1 - 2, -2 - 5 \rangle = \langle 4, -3, -7 \rangle$. Parametric form:

$$x = -3 + 4t, \quad y = 2 - 3t, \quad z = 5 - 7t, \quad t \in \mathbb{R}.$$

Symmetric form:

$$\frac{x + 3}{4} = \frac{y - 2}{-3} = \frac{z - 5}{-7}.$$

(b) For the xy -trace, set $z = 0$ in the parametric equations:

$$5 - 7t = 0 \implies t = \frac{5}{7}.$$

Then

$$x = -3 + 4\left(\frac{5}{7}\right) = -\frac{1}{7}, \quad y = 2 - 3\left(\frac{5}{7}\right) = -\frac{1}{7}.$$

Thus the line meets the xy -plane at $\left(-\frac{1}{7}, -\frac{1}{7}, 0\right)$. ■

4.2 Line segments.

The segment from P_0 to P_1 is the set of convex combinations of the endpoints:

$$\vec{r}(t) = (1 - t)\vec{r}_0 + t\vec{r}_1, \quad 0 \leq t \leq 1,$$

where $\vec{r}_0$ and $\vec{r}_1$ are the position vectors of P_0 and P_1 (see Figure 4.2).

Example 4.3. Consider the lines

$$\ell_1 : \mathbf{r}(t) = \langle 2, -1, 0 \rangle + t\langle 1, 3, -2 \rangle, \quad \ell_2 : \mathbf{s}(s) = \langle 5, 5, -6 \rangle + s\langle -2, -3, 4 \rangle.$$

1. Are the lines parallel?
2. Do the lines intersect? If so, find the point of intersection.

Solution. (1) Direction vectors are $\mathbf{v}_1 = \langle 1, 3, -2 \rangle$ and $\mathbf{v}_2 = \langle -2, -3, 4 \rangle$. Since no scalar k satisfies $\mathbf{v}_2 = k\mathbf{v}_1$ (ratios $-2/1 = -2$, $-3/3 = -1$, $4/(-2) = -2$ disagree), the lines are *not parallel*.

(2) Set coordinates equal:

$$2 + t = 5 - 2s, \quad -1 + 3t = 5 - 3s, \quad -2t = -6 + 4s.$$

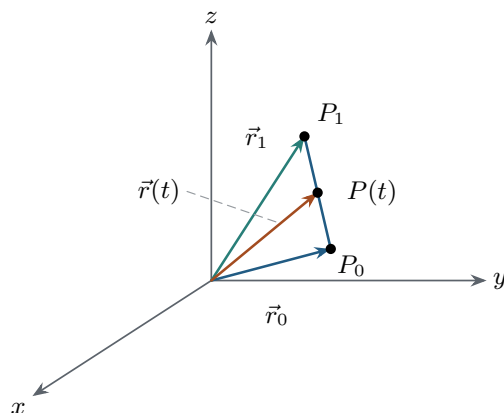


Figure 4.2: Line segment between the points P_0 and P_1

From the first two, $t + 2s = 3$ and $t + s = 2 \implies s = 1, t = 1$. These satisfy the third equation, so the lines intersect at

$$\mathbf{r}(1) = \langle 2, -1, 0 \rangle + \langle 1, 3, -2 \rangle = \boxed{\langle 3, 2, -2 \rangle}.$$

■

4.3 Planes

A plane in $\mathbb{R}^3$ is fully determined by a point $P_0(x_0, y_0, z_0)$ on the plane and a nonzero normal vector $\vec{n} = \langle a, b, c \rangle$ orthogonal to every displacement in the plane (see Figure 4.3).

A point $P(x, y, z)$ lies on the plane if and only if the vector $\overrightarrow{P_0P}$ is orthogonal to $\vec{n}$. This happens if and only if their dot product is zero:

$$\begin{aligned} \overrightarrow{P_0P} \cdot \vec{n} = 0 &\iff \langle x - x_0, y - y_0, z - z_0 \rangle \cdot \langle a, b, c \rangle = 0 \\ &\iff a(x - x_0) + b(y - y_0) + c(z - z_0) = 0. \end{aligned}$$

Definition 4.2

The *point-normal equation* of a plane passing through (x_0, y_0, z_0) with normal vector $\vec{n} = \langle a, b, c \rangle$ is

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

This is often expanded into the *general form* $ax + by + cz + d = 0$ and is called the *scalar equation* of the plane.

Example 4.4. Find an equation of the plane passing through $(-1, 3, 2)$ with normal vector $\vec{n} = \langle 2, -2, 4 \rangle$.

Solution. Equation:

$$2(x + 1) - 2(y - 3) + 4(z - 2) = 0 \implies 2x - 2y + 4z = 0.$$

■

Example 4.5. Find the scalar equation of the plane passing through the points $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$.

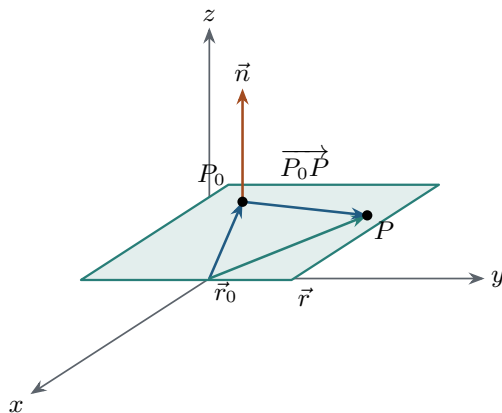


Figure 4.3: A plane determined by a point P_0 and a normal vector $\vec{n}$. Any vector $\overrightarrow{P_0P}$ in the plane is orthogonal to $\vec{n}$.

Solution. Vectors: $\langle -1, 1, 0 \rangle$, $\langle -1, 0, 1 \rangle$. Normal: $\langle 1, 1, 1 \rangle$. Equation:

$$x + y + z = 1.$$

Example 4.6. Show that the distance between a point $P(x_0, y_0, z_0)$ and a plane $ax + by + cz + d = 0$ is

$$\text{dist} = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Solution. Plane normal: $\vec{n} = \langle a, b, c \rangle$. Pick point Q on plane (satisfying $ax + by + cz + d = 0$). Then

$$\text{dist}(P, \text{plane}) = \frac{|(P - Q) \cdot \vec{n}|}{\|\vec{n}\|}.$$

Substituting gives the stated formula.

Example 4.7. Find a point on the line of intersection of the planes:

$$\text{plane } Q : \quad x + 3y - 2z = 1$$

$$\text{plane } R : \quad x - y + z = -3$$

Solution. We want (x, y, z) satisfying

$$\begin{cases} x + 3y - 2z = 1, \\ x - y + z = -3. \end{cases}$$

Set $z = 0$. Then

$$\begin{cases} x + 3y = 1, \\ x - y = -3. \end{cases}$$

Subtracting gives $4y = 4 \implies y = 1$, and then $x = -2$.

Hence one point on the line of intersection is $(-2, 1, 0)$.

5 Quadric Surfaces

5.1 Goals

- Define quadratic surfaces and sketch them using traces and intercepts.

Definition 5.1

A *cylinder* is a surface consisting of all lines parallel to a fixed direction that pass through every point of a given plane curve (the *generating curve*).

Example 5.1. Sketch $y = x^2$ in $\mathbb{R}^3$. (See Figure 5.1.)

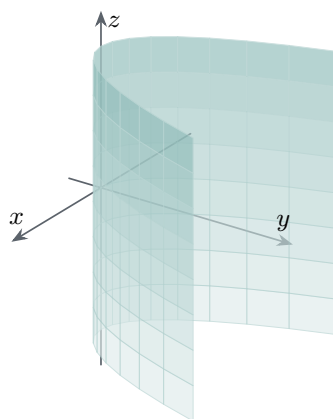


Figure 5.1: A sketch of the parabolic cylinder $y = x^2$ in $\mathbb{R}^3$.

Solution. This is the usual parabola $y = x^2$ in the xy -plane ($z = 0$). Since z is unrestricted, the curve extends parallel to the z -axis, producing a *parabolic cylinder*. ■

Example 5.2. Sketch $x^2 + \frac{z^2}{4} = 1$ in $\mathbb{R}^3$. (See Figure 5.2.)

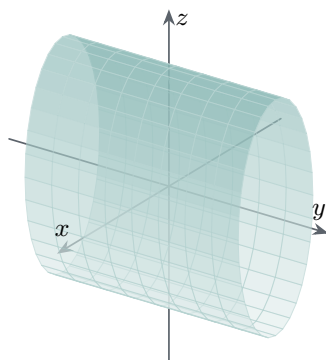


Figure 5.2: The elliptic cylinder $x^2 + \frac{z^2}{4} = 1$ in $\mathbb{R}^3$.

Solution. This is an ellipse in the xz -plane ($y = 0$). Since y is unrestricted, the ellipse extends parallel to the y -axis, producing an *elliptic cylinder*. ■

Definition 5.2

A *quadric surface* is a surface in $\mathbb{R}^3$ defined as the set of all points (x, y, z) satisfying a second-degree polynomial equation

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0,$$

where the coefficients $A, \dots, J$ are real constants and at least one of $A, \dots, F$ is nonzero. Such an equation can also describe a degenerate quadric, such as two intersecting planes.

Example 5.3. Sketch the surface

$$\frac{x^2}{4} + \frac{y^2}{9} - \frac{z^2}{16} = 1$$

using intercepts and traces. (See Figure 5.3)

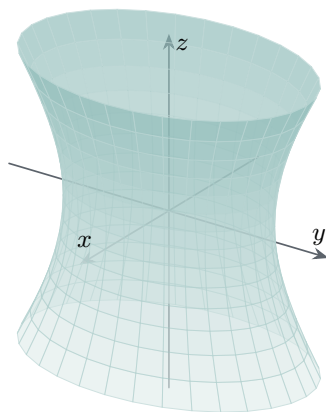


Figure 5.3: Wireframe sketch of $\frac{x^2}{4} + \frac{y^2}{9} - \frac{z^2}{16} = 1$ in $\mathbb{R}^3$ via a parametric mesh.

Solution. **Intercepts:** $x = \pm 2$, $y = \pm 3$, no z -intercepts.

Traces:

$$z = 0 : \frac{x^2}{4} + \frac{y^2}{9} = 1 \text{ (ellipse),}$$

$$x = 0 : \frac{y^2}{9} - \frac{z^2}{16} = 1 \text{ (hyperbola),}$$

$$y = 0 : \frac{x^2}{4} - \frac{z^2}{16} = 1 \text{ (hyperbola).}$$

Horizontal slices: For $z = z_0$, $\frac{x^2}{4} + \frac{y^2}{9} = 1 + \frac{z_0^2}{16}$ (ellipses expanding with $|z_0|$). ■

Example 5.4. Sketch the surface

$$-\frac{x^2}{9} - \frac{y^2}{4} + \frac{z^2}{16} = 1$$

using intercepts and traces. (See Figure 5.4.)

Solution. **Intercepts:** $z = \pm 4$, no x - or y -intercepts.

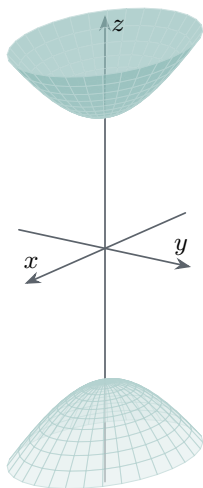


Figure 5.4: Wireframe sketch of $-\frac{x^2}{9} - \frac{y^2}{4} + \frac{z^2}{16} = 1$ in $\mathbb{R}^3$ (a hyperboloid of two sheets).

Traces:

$$z = 0 : -\frac{x^2}{9} - \frac{y^2}{4} = 1 \text{ (no real trace),}$$

$$x = 0 : -\frac{y^2}{4} + \frac{z^2}{16} = 1 \text{ (hyperbola),}$$

$$y = 0 : -\frac{x^2}{9} + \frac{z^2}{16} = 1 \text{ (hyperbola).}$$

Horizontal slices: For $z = z_0$ with $|z_0| > 4$, $\frac{x^2}{9} + \frac{y^2}{4} = \frac{z_0^2}{16} - 1$ (ellipses). For $|z_0| = 4$, the slice is a single point. For $|z_0| < 4$, it is empty. ■

Example 5.5. Sketch the surface defined by the equation

$$x^2 + \frac{y^2}{4} + \frac{z^2}{9} = 1$$

by identifying its intercepts and analyzing its traces in the coordinate planes. (See Figure 5.5)

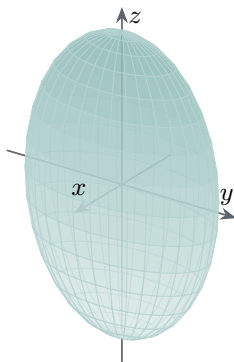


Figure 5.5: The ellipsoid $x^2 + \frac{y^2}{4} + \frac{z^2}{9} = 1$, showing intercepts at $x = \pm 1$, $y = \pm 2$, and $z = \pm 3$.

Solution. This is an **ellipsoid** centered at the origin.

- **Intercepts:** Setting two variables to zero gives the intercepts: x -intercepts at $(\pm 1, 0, 0)$, y -intercepts at $(0, \pm 2, 0)$, and z -intercepts at $(0, 0, \pm 3)$.
- **Traces:**
 - xy -trace ($z = 0$): The ellipse $x^2 + \frac{y^2}{4} = 1$ in the xy -plane.
 - xz -trace ($y = 0$): The ellipse $x^2 + \frac{z^2}{9} = 1$ in the xz -plane.
 - yz -trace ($x = 0$): The ellipse $\frac{y^2}{4} + \frac{z^2}{9} = 1$ in the yz -plane.

The surface is a smooth, closed “egg-shaped” surface stretching furthest along the z -axis. ■

Example 5.6. Identify and analyze the traces of the quadric surface

$$x^2 + y^2 - z^2 = 0$$

Determine the shape of the surface. (See Figure 5.6)

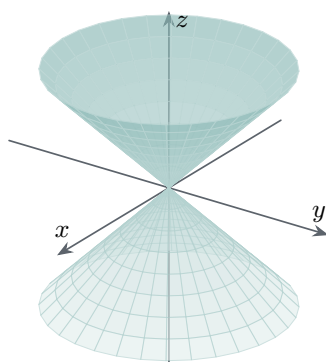


Figure 5.6: The circular cone $x^2 + y^2 = z^2$, illustrating the two nappes meeting at the origin.

Solution. This is a **circular cone** with its vertex at the origin and its axis along the z -axis.

- **Traces:**
 - xy -trace ($z = 0$): The single point $(0, 0, 0)$.
 - Horizontal traces ($z = k, k \neq 0$): The circles $x^2 + y^2 = k^2$ with radius $|k|$ in the plane $z = k$.
 - Vertical traces ($x = 0$ or $y = 0$): The pair of intersecting lines $z = \pm y$ or $z = \pm x$.

As $|k|$ increases, the circular traces expand, forming the two nappes of the cone. ■

Example 5.7. Identify the surface given by

$$x^2 + y^2 + z^2 = 4z - 3$$

by completing the square to rewrite the equation in standard form. (See Figure 5.7)

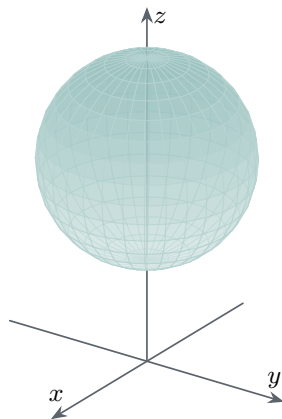


Figure 5.7: A sphere of radius 1 centered at $(0, 0, 2)$ on the z -axis.

Solution. First, rearrange the terms and complete the square for the z terms:

$$x^2 + y^2 + (z^2 - 4z) = -3$$

To complete the square, add $\left(\frac{-4}{2}\right)^2 = 4$ to both sides:

$$x^2 + y^2 + (z^2 - 4z + 4) = -3 + 4$$

$$x^2 + y^2 + (z - 2)^2 = 1$$

This is the standard form $(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$ of a **sphere**.

- **Center:** $(0, 0, 2)$
- **Radius:** $r = \sqrt{1} = 1$

■

6 Vector Valued Functions

6.1 Goals

- Define vector-valued functions.
- View them as curves in $\mathbb{R}^3$.
- Parametrize the curve of intersection of two surfaces.

Definition 6.1

A *vector-valued function* is a function whose outputs are vectors rather than scalars. More formally, it is a mapping

$$\mathbf{r}: D \subseteq \mathbb{R} \rightarrow \mathbb{R}^n,$$

where each input $t \in D$ (a real number) is mapped to an n -dimensional vector.

Example 6.1. Consider the function $\mathbf{r}(t) = \langle 1 - t, 3t^2, \ln t \rangle$. It's a function whose input is a real number and whose output is a vector. For example, $\mathbf{r}(2) = \langle -1, 12, \ln 2 \rangle$.

Consider a function $\mathbf{r}: D \subseteq \mathbb{R} \rightarrow \mathbb{R}^3$. A vector-valued function can be written in component form as

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle,$$

where $f(t), g(t), h(t)$ are real-valued functions of a single variable.

The natural domain of a component formula for $\mathbf{r}$ is the intersection of the domains of its components. A parametrization may restrict this domain to a specified interval.

Example 6.2. Find the domain of $\mathbf{r}(t) := \langle \sqrt{1 - t^2}, e^t, \ln(1 + t) \rangle$.

Solution. The components impose:

$$\begin{aligned} \sqrt{1 - t^2} &\implies 1 - t^2 \geq 0 \implies -1 \leq t \leq 1, \\ e^t &\implies t \in \mathbb{R}, \\ \ln(1 + t) &\implies 1 + t > 0 \implies t > -1. \end{aligned}$$

Intersecting: $[-1, 1] \cap \mathbb{R} \cap (-1, \infty) = (-1, 1]$.

Domain: $(-1, 1]$. ■

6.2 Limits and Continuity

For a vector-valued function $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$, the limit as $t \rightarrow a$ is defined componentwise:

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \left\langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \right\rangle.$$

Example 6.3. Find $\lim_{t \rightarrow 0} \mathbf{r}(t)$ where

$$\mathbf{r}(t) := \left\langle \frac{\sin t}{t}, 3t^2 + 1, te^{-t} \right\rangle.$$

Solution. Take limits componentwise:

$$\lim_{t \rightarrow 0} \frac{\sin t}{t} = 1, \quad \lim_{t \rightarrow 0} (3t^2 + 1) = 1, \quad \lim_{t \rightarrow 0} te^{-t} = \left(\lim_{t \rightarrow 0} t \right) \left(\lim_{t \rightarrow 0} e^{-t} \right) = 0 \cdot 1 = 0.$$

Therefore, $\lim_{t \rightarrow 0} \mathbf{r}(t) = \langle 1, 1, 0 \rangle$. ■

Definition 6.2

A vector-valued function $\mathbf{r}(t)$ is *continuous* at $t = a$ if:

1. The value $\mathbf{r}(a)$ is well defined, and
2. The limit $\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{r}(a)$.

6.3 Parametric Curves

On an interval I , a continuous function $\mathbf{r}: I \rightarrow \mathbb{R}^3$ parametrizes a curve: its image consists of the points with position vectors $\mathbf{r}(t)$. The curve lies in $\mathbb{R}^3$, whereas the graph $\{(t, \mathbf{r}(t)) : t \in I\}$ lies in $\mathbb{R}^4$. Different parametrizations can describe the same curve with different speeds or orientations.

Example 6.4. Describe the parametric curve $\mathbf{r}(t) = \langle 2 - t, 3 + 2t, t \rangle$.

Solution. The components are linear in t —cf. § 4. Eliminating the parameter:

$$x = 2 - t, \quad y = 3 + 2t, \quad z = t.$$

From $z = t$, we have $x = 2 - z$ and $y = 3 + 2z$. Thus the curve is a straight line in $\mathbb{R}^3$ passing through $(2, 3, 0)$ with direction vector $\langle -1, 2, 1 \rangle$. ■

Example 6.5. Sketch the curve given by $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$.

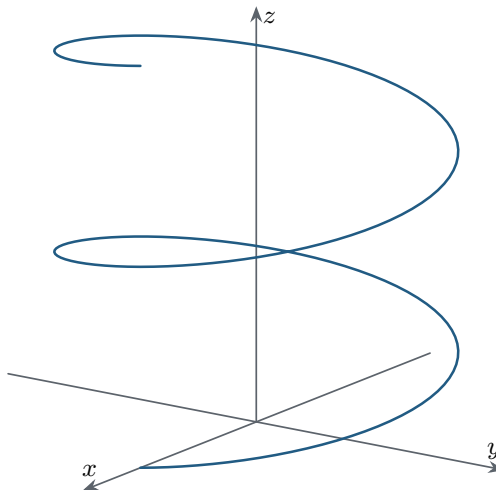


Figure 6.1: The helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ has radius 1 and winds around the z -axis.

Solution. The x - and y -components satisfy

$$x = \cos t, \quad y = \sin t \quad \implies \quad x^2 + y^2 = 1,$$

which describes the unit circle in the xy -plane. The z -component is $z = t$, so as t increases the point moves upward while circling around the z -axis. Hence the curve is a *helix* of radius 1 centered on the z -axis, oriented upward; see Figure 6.1. ■

Example 6.6. Find a parametric equation for the curve of intersection of the cylinder $x^2 + y^2 = 4$ and the plane $z = x + y$.

Solution. On the cylinder, set $x = 2 \cos t$, $y = 2 \sin t$ ($t \in [0, 2\pi)$). Then

$$z = x + y = 2 \cos t + 2 \sin t.$$

Thus a parametrization is

$$\mathbf{r}(t) = \langle 2 \cos t, 2 \sin t, 2 \cos t + 2 \sin t \rangle, \quad t \in [0, 2\pi).$$

This traces the intersection exactly once as t runs through $[0, 2\pi)$. ■

Example 6.7. Find a parametric equation for the curve of intersection of the sphere $x^2 + y^2 + z^2 = 9$ and the plane $z = 1$.

Solution. Intersecting gives a circle in the plane $z = 1$ with radius $\sqrt{9 - 1^2} = \sqrt{8}$. Parametrize that circle:

$$x = \sqrt{8} \cos t, \quad y = \sqrt{8} \sin t, \quad z = 1, \quad t \in [0, 2\pi).$$

Hence

$$\mathbf{r}(t) = \langle \sqrt{8} \cos t, \sqrt{8} \sin t, 1 \rangle, \quad t \in [0, 2\pi).$$

■

Example 6.8. Find a parametric equation for the curve of intersection of the elliptic cylinder $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and the plane $z = y - x$.

Solution. Parametrize the ellipse by $x = 3 \cos t$, $y = 2 \sin t$ ($t \in [0, 2\pi)$). Then the plane gives

$$z = y - x = 2 \sin t - 3 \cos t.$$

Therefore

$$\mathbf{r}(t) = \langle 3 \cos t, 2 \sin t, 2 \sin t - 3 \cos t \rangle, \quad t \in [0, 2\pi).$$

■

Example 6.9. Find a parametric equation for the curve of intersection of the paraboloid $z = x^2 + y^2$ and the plane $y = 1$.

Solution. Let $x = t$ and $y = 1$; then $z = t^2 + 1$. A convenient parametrization is

$$\mathbf{r}(t) = \langle t, 1, t^2 + 1 \rangle, \quad t \in \mathbb{R},$$

which traces the parabola opening in the $+z$ -direction within the plane $y = 1$.

■

7 Derivatives and Integrals of Vector Functions

7.1 Goals

- Define differentiation of vector-valued functions.
- Define integration of vector-valued functions.
- Define curvature as the rate of change of direction divided by speed.

Recall that if $y = f(t)$ is a scalar function, the derivative is

$$y' = f'(t) = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}.$$

Definition 7.1 (Derivative)

If $\mathbf{r}(t) = \langle x(t), y(t) \rangle$ (or $\langle x(t), y(t), z(t) \rangle$), the derivative is defined by

$$\mathbf{r}'(t) := \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h},$$

whenever this limit exists. Since subtraction and scalar division act componentwise,

$$\mathbf{r}'(t) = \langle x'(t), y'(t) \rangle \quad \text{and} \quad \mathbf{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle$$

in $\mathbb{R}^2$ and $\mathbb{R}^3$, respectively.

7.2 Geometric interpretation

$\mathbf{r}'(t)$ is the *velocity vector*, and its magnitude $\|\mathbf{r}'(t)\|$ is the *speed*. When $\mathbf{r}'(t) \neq \mathbf{0}$, it is tangent to the curve at $\mathbf{r}(t)$. (See Figure 7.1.)

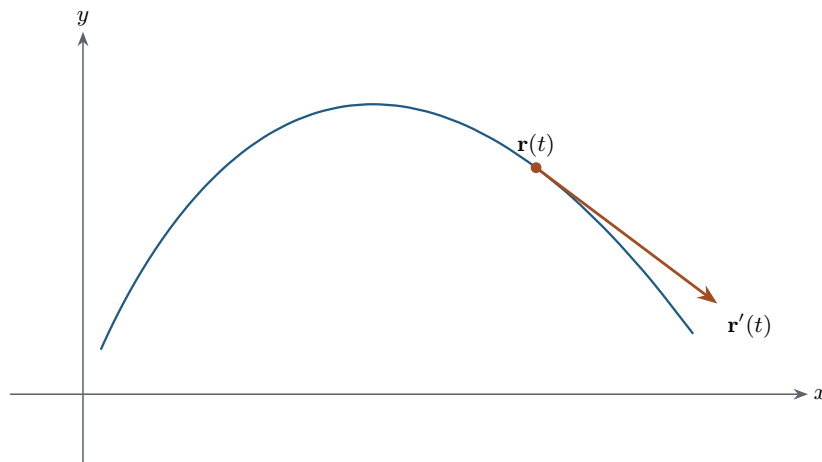


Figure 7.1: A visual representation of the relationship between the position vector $\mathbf{r}(t)$ and the tangent (velocity) vector $\mathbf{r}'(t)$.

Definition 7.2 (Unit tangent vector)

Let $\mathbf{r}$ be differentiable at t with $\mathbf{r}'(t) \neq \mathbf{0}$. The unit tangent vector is

$$\mathbf{T}(t) := \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}.$$

It points in the instantaneous direction of motion and satisfies $\|\mathbf{T}(t)\| = 1$.

Theorem 7.1 (Differentiation rules for vector-valued functions)

Suppose $\mathbf{u}, \mathbf{v} : I \rightarrow \mathbb{R}^n$ are differentiable vector functions on an interval I , $c \in \mathbb{R}$ is a constant, and $f : I \rightarrow \mathbb{R}$ is a differentiable scalar function. Then, for all $t \in I$,

1. $\frac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t).$
2. $\frac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t).$
3. $\frac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t).$
4. $\frac{d}{dt}[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t).$

$$5. (\text{In } \mathbb{R}^3) \frac{d}{dt} [\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t).$$

6. (Chain rule) If $g : I \rightarrow J$ and $\mathbf{r} : J \rightarrow \mathbb{R}^n$ are differentiable on intervals I, J , then

$$\frac{d}{dt} \mathbf{r}(g(t)) = g'(t) \mathbf{r}'(g(t)).$$

Reason: all identities follow from componentwise differentiation and the corresponding scalar rules.

7.3 Curvature: change in direction

Curvature measures how sharply a curve bends. Since $\mathbf{T}(t)$ records the direction of motion, its derivative $\mathbf{T}'(t)$ records how that direction changes. Thus $\|\mathbf{T}'(t)\|$ measures the rate of change of direction *per unit time*.

This rate depends on how quickly we traverse the curve. For example, if $\tilde{\mathbf{r}}(u) = \mathbf{r}(2u)$, then $\tilde{\mathbf{T}}(u) = \mathbf{T}(2u)$, so

$$\|\tilde{\mathbf{T}}'(u)\| = 2\|\mathbf{T}'(2u)\|.$$

We detect twice as much turning per unit time, even though the curve has the same shape. The speed also doubles. Dividing by speed cancels this effect and measures turning *per unit distance traveled*.

Definition 7.3 (Curvature)

Let $\mathbf{r} : I \rightarrow \mathbb{R}^n$ be twice continuously differentiable on an open interval I , with $\mathbf{r}'(t) \neq \mathbf{0}$ for every $t \in I$. The *curvature* at $\mathbf{r}(t)$ along this parametrized curve is

$$\kappa(t) := \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}, \quad \mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}.$$

For a small time increment h , the change in the unit tangent has magnitude approximately $\|\mathbf{T}'(t)\| |h|$, while the distance traveled is approximately $\|\mathbf{r}'(t)\| |h|$. Their quotient approaches $\kappa(t)$ as $h \rightarrow 0$.

Proposition 7.2 (Independence of parametrization)

Suppose $g : J \rightarrow I$ is a twice continuously differentiable bijection between open intervals, with $g'(u) \neq 0$. Then $\tilde{\mathbf{r}}(u) = \mathbf{r}(g(u))$ has curvature

$$\tilde{\kappa}(u) = \kappa(g(u)).$$

Thus curvature is unchanged at corresponding points of the curve.

Proof. The chain rule gives

$$\tilde{\mathbf{r}}'(u) = g'(u) \mathbf{r}'(g(u)), \quad \tilde{\mathbf{T}}(u) = \frac{g'(u)}{|g'(u)|} \mathbf{T}(g(u)).$$

Since g' is continuous and never zero, its sign is constant on J . Differentiating the second identity and taking norms gives

$$\begin{aligned}\|\tilde{\mathbf{T}}'(u)\| &= |g'(u)| \|\mathbf{T}'(g(u))\|, \\ \|\tilde{\mathbf{r}}'(u)\| &= |g'(u)| \|\mathbf{r}'(g(u))\|.\end{aligned}$$

The common factor $|g'(u)|$ cancels in the quotient. This also covers reversing the direction of travel, when $g' < 0$. ■

Example 7.1 (Curvature of a circle). Find the curvature of

$$\mathbf{r}(t) = \langle R \cos(\omega t), R \sin(\omega t) \rangle, \quad R > 0, \quad \omega > 0.$$

Solution. The velocity and speed are

$$\mathbf{r}'(t) = R\omega \langle -\sin(\omega t), \cos(\omega t) \rangle, \quad \|\mathbf{r}'(t)\| = R\omega.$$

Hence

$$\mathbf{T}(t) = \langle -\sin(\omega t), \cos(\omega t) \rangle, \quad \mathbf{T}'(t) = \omega \langle -\cos(\omega t), -\sin(\omega t) \rangle.$$

Therefore

$$\kappa(t) = \frac{\omega}{R\omega} = \frac{1}{R}.$$

The angular speed ω cancels. Smaller circles bend more sharply. ■

A straight line $\mathbf{r}(t) = \mathbf{p} + t\mathbf{v}$, with $\mathbf{v} \neq \mathbf{0}$, has constant unit tangent and therefore curvature 0.

7.4 Integration of vector functions

Definition 7.4 (Integral)

If $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ and X, Y, Z are chosen antiderivatives of x, y, z , respectively, then

$$\int \mathbf{r}(t) dt = \langle X(t), Y(t), Z(t) \rangle + \mathbf{C},$$

where $\mathbf{C} = \langle C_1, C_2, C_3 \rangle \in \mathbb{R}^3$ is a constant vector. For definite integrals,

$$\int_a^b \mathbf{r}(t) dt = \left\langle \int_a^b x(t) dt, \int_a^b y(t) dt, \int_a^b z(t) dt \right\rangle.$$

Theorem 7.3 (Fundamental Theorem)

If $\mathbf{r}$ is continuous on $[a, b]$, then

$$\frac{d}{dt} \left(\int_a^t \mathbf{r}(u) du \right) = \mathbf{r}(t), \quad a < t < b.$$

If, in addition, $\mathbf{r}$ is differentiable on $[a, b]$ with continuous derivative, then

$$\int_a^b \mathbf{r}'(t) dt = \mathbf{r}(b) - \mathbf{r}(a).$$

Example 7.2. Find the derivative of the expanding helix with parametric equation

$$\mathbf{r}(t) = \langle t \cos t, t \sin t, t \rangle.$$

Solution. Differentiate componentwise (product rule in x and y):

$$\mathbf{r}'(t) = \langle \cos t - t \sin t, \sin t + t \cos t, 1 \rangle.$$

The speed is

$$\|\mathbf{r}'(t)\| = \sqrt{(\cos t - t \sin t)^2 + (\sin t + t \cos t)^2 + 1} = \sqrt{t^2 + 2},$$

since $(\cos^2 + \sin^2) + t^2(\sin^2 + \cos^2) = 1 + t^2$ and cross terms cancel. ■

Example 7.3. Find parametric equations for the tangent line to the curve $\mathbf{r}(t) = \langle t \cos t, t \sin t, t \rangle$ at $t = \pi/2$.

Solution. Point on the curve:

$$\mathbf{r}\left(\frac{\pi}{2}\right) = \left\langle 0, \frac{\pi}{2}, \frac{\pi}{2} \right\rangle.$$

Tangent (velocity) vector:

$$\mathbf{r}'(t) = \langle \cos t - t \sin t, \sin t + t \cos t, 1 \rangle \implies \mathbf{r}'\left(\frac{\pi}{2}\right) = \left\langle -\frac{\pi}{2}, 1, 1 \right\rangle.$$

Thus a parametric form of the tangent line is

$$\ell(s) = \left\langle 0, \frac{\pi}{2}, \frac{\pi}{2} \right\rangle + s \left\langle -\frac{\pi}{2}, 1, 1 \right\rangle, \quad s \in \mathbb{R},$$

i.e.

$$x = -\frac{\pi}{2}s, \quad y = \frac{\pi}{2} + s, \quad z = \frac{\pi}{2} + s.$$

■

8 Arc Length and Line Integrals

8.1 Goals

- Derive and compute the arc length of a smooth parametric curve in $\mathbb{R}^n$.
- Formulate and interpret the differential arc length element $ds = \|\mathbf{r}'(t)\| dt$.
- Define and evaluate scalar line integrals along curves with respect to arc length.
- Apply scalar line integrals to physical problems, such as finding the total mass of a thin wire with variable density.

8.2 1. Arc Length of a Space Curve

Let $\mathbf{r} : [a, b] \rightarrow \mathbb{R}^n$ be a continuously differentiable parametrization of a curve C . Its length is the limit of sums of chord lengths as the partition becomes finer. Integrating the speed gives this limit.

Derivation of Differential Arc Length

To obtain an explicit formula for the differential arc length dL (conventionally denoted ds), we cut the curve into small pieces over time steps of size $h > 0$:

1. The straight-line chord vector connecting two nearby points on the curve is $\mathbf{r}(t+h) - \mathbf{r}(t)$.
2. The straight-line distance approximating the piece length is:

$$\Delta L \approx \|\mathbf{r}(t+h) - \mathbf{r}(t)\| = \left\| \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h} \right\| h.$$

3. As $h \rightarrow 0$, the difference quotient approaches $\mathbf{r}'(t)$. Thus the length over a short time interval is approximately $\|\mathbf{r}'(t)\|h$. Summing these contributions leads to the notation

$$dL = ds = \|\mathbf{r}'(t)\| dt.$$

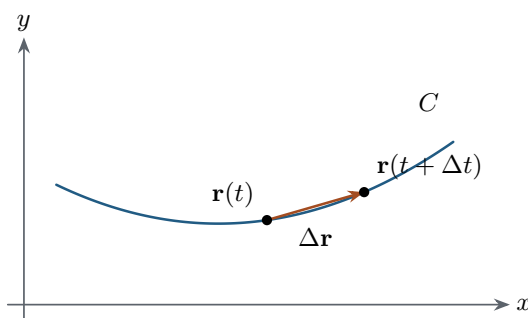


Figure 8.1: Secant chord approximation of the differential arc length element $dL = \|\mathbf{r}'(t)\| dt$ along the curve C .

Theorem 8.1 (Arc Length Formula)

If $\mathbf{r}$ is continuously differentiable on $[a, b]$, its length is

$$L = \int_a^b \|\mathbf{r}'(t)\| dt.$$

In component form for $\mathbb{R}^3$, where $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$:

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt.$$

If the parametrization retraces an arc, this integral counts each traversal. To compute the geometric length of C , trace it once. For a piecewise smooth curve, add the integrals over its smooth pieces.

Example 8.1. Find the arc length of the circular helix parameterized by

$$\mathbf{r}(t) = \langle a \cos t, a \sin t, bt \rangle, \quad t \in [0, 2\pi],$$

where $a > 0$ and $b > 0$ are constants.

Solution. First, compute the velocity vector $\mathbf{r}'(t)$:

$$\mathbf{r}'(t) = \langle -a \sin t, a \cos t, b \rangle.$$

Next, calculate the speed $\|\mathbf{r}'(t)\|$:

$$\|\mathbf{r}'(t)\| = \sqrt{(-a \sin t)^2 + (a \cos t)^2 + b^2} = \sqrt{a^2(\sin^2 t + \cos^2 t) + b^2} = \sqrt{a^2 + b^2}.$$

Because the speed is constant, the arc length integral evaluates to:

$$L = \int_0^{2\pi} \|\mathbf{r}'(t)\| dt = \int_0^{2\pi} \sqrt{a^2 + b^2} dt = 2\pi \sqrt{a^2 + b^2}.$$

■

Example 8.2. Find the length of the space curve $\mathbf{r}(t) = \langle 2t, t^2, \frac{1}{3}t^3 \rangle$ over the interval $0 \leq t \leq 1$.

Solution. Differentiating the position vector:

$$\mathbf{r}'(t) = \langle 2, 2t, t^2 \rangle.$$

The magnitude is:

$$\|\mathbf{r}'(t)\| = \sqrt{2^2 + (2t)^2 + (t^2)^2} = \sqrt{4 + 4t^2 + t^4} = \sqrt{(2 + t^2)^2} = 2 + t^2.$$

Integrating over $t \in [0, 1]$:

$$L = \int_0^1 (2 + t^2) dt = \left[2t + \frac{t^3}{3} \right]_0^1 = 2 + \frac{1}{3} = \frac{7}{3}.$$

■

8.3 2. Scalar Line Integrals and Mass of a Wire

Assume we have a thin wire shaped along a smooth space curve C parameterized by $\mathbf{r}(t)$ for $t \in [a, b]$. Suppose $\rho(x, y, z)$ gives the local linear mass density of the wire at any point (x, y, z) (in units of mass per unit length).

Derivation of Total Mass

To determine the total mass of the wire:

1. Cut the curve into small pieces of arc length ds .
2. Over an infinitesimal piece, assume the density is locally constant and equal to $\rho(\mathbf{r}(t))$.
3. The mass of an infinitesimal segment is:

$$d\text{Mass} = dm = \rho(\mathbf{r}(t)) ds = \rho(\mathbf{r}(t)) \|\mathbf{r}'(t)\| dt.$$

4. Accumulating over the whole parameter interval $[a, b]$ yields the total mass:

$$\text{Mass} = \int_a^b d\text{Mass} = \int_a^b \rho(\mathbf{r}(t)) \|\mathbf{r}'(t)\| dt.$$

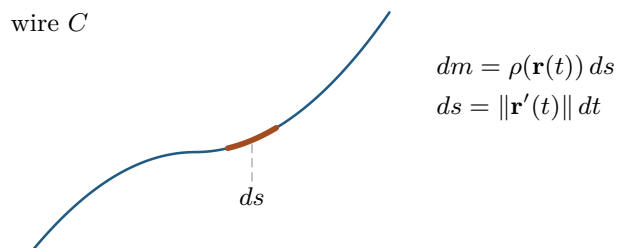


Figure 8.2: A thin wire C with varying linear density $\rho(x, y, z)$ and infinitesimal mass element $dm = \rho(\mathbf{r}(t)) ds$.

Definition 8.1 (Scalar Line Integral)

Let $f(x, y, z)$ be a continuous function defined on a domain containing a smooth curve C parameterized by $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ for $a \leq t \leq b$. The *line integral of f along C with respect to arc length* is defined as:

$$\int_C f(x, y, z) ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| dt.$$

In component form:

$$\int_C f(x, y, z) ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt.$$

Example 8.3. A thin wire is shaped along the helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ for $0 \leq t \leq 2\pi$. If its linear mass density at any point is proportional to its height, $\rho(x, y, z) = kz$ (where $k > 0$ is a constant), find the total mass of the wire.

Solution. First, find the velocity vector and speed:

$$\mathbf{r}'(t) = \langle -\sin t, \cos t, 1 \rangle,$$

$$\|\mathbf{r}'(t)\| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} = \sqrt{1 + 1} = \sqrt{2}.$$

Thus, the differential arc length element is $ds = \sqrt{2} dt$.

Evaluate the density along the curve:

$$\rho(\mathbf{r}(t)) = k z(t) = kt.$$

Now evaluate the scalar line integral:

$$M = \int_C \rho ds = \int_0^{2\pi} (kt) \sqrt{2} dt = k\sqrt{2} \left[\frac{t^2}{2} \right]_0^{2\pi} = k\sqrt{2} \left(\frac{4\pi^2}{2} \right) = 2\sqrt{2}k\pi^2.$$

■

Example 8.4. Evaluate $\int_C (x + 2y) ds$, where C is the upper half of the unit circle $x^2 + y^2 = 1$ traversed counterclockwise from $(1, 0)$ to $(-1, 0)$.

Solution. Parametrize the semicircle:

$$\mathbf{r}(t) = \langle \cos t, \sin t \rangle, \quad 0 \leq t \leq \pi.$$

The velocity and speed are:

$$\mathbf{r}'(t) = \langle -\sin t, \cos t \rangle \implies \|\mathbf{r}'(t)\| = \sqrt{\sin^2 t + \cos^2 t} = 1.$$

Hence, $ds = 1 dt$. Substitute into the line integral formula:

$$\begin{aligned} \int_C (x + 2y) ds &= \int_0^\pi (\cos t + 2 \sin t)(1) dt \\ &= \left[\sin t - 2 \cos t \right]_0^\pi \\ &= (\sin \pi - 2 \cos \pi) - (\sin 0 - 2 \cos 0) \\ &= (0 - 2(-1)) - (0 - 2(1)) \\ &= 2 + 2 = 4. \end{aligned}$$

■

9 Motion in Space: Velocity and Acceleration

9.1 Goals

- Define the tangent, normal, and binormal vectors.
- Understand how to compute the *velocity* and *acceleration* vectors.
- Interpret these vectors geometrically and connect them to physical motion.

Definition 9.1 (Tangent, Normal, and Binormal Vectors)

Let $\mathbf{r}(t)$ be a twice continuously differentiable curve in $\mathbb{R}^3$ with $\mathbf{r}'(t) \neq \mathbf{0}$. The normal and binormal below are defined at points where $\mathbf{T}'(t) \neq \mathbf{0}$.

- The *unit tangent vector* is

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|},$$

which points in the instantaneous direction of motion.

- The *unit normal vector* $\mathbf{N}(t)$ points in the direction of change of $\mathbf{T}(t)$ (toward the center of curvature). Formally,

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}.$$

- The *binormal vector* is defined by

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t).$$

It is orthogonal to both $\mathbf{T}$ and $\mathbf{N}$, and together the three vectors $\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ form a right-handed orthonormal basis called the *Frenet–Serret frame*.

Example 9.1. Find the unit tangent $\mathbf{T}$, unit normal $\mathbf{N}$, and binormal $\mathbf{B}$ of the curve

$$\mathbf{r}(t) = \langle 2t, t^2, \frac{1}{3}t^3 \rangle$$

at $t = 1$.

Solution. Tangent. $\mathbf{r}'(t) = \langle 2, 2t, t^2 \rangle$, and

$$v(t) = \|\mathbf{r}'(t)\| = \sqrt{4 + 4t^2 + t^4} = 2 + t^2.$$

Thus

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{v(t)} = \frac{\langle 2, 2t, t^2 \rangle}{2 + t^2}.$$

Normal. Differentiate $\mathbf{T}(t) = \mathbf{r}'(t)/v(t)$:

$$\mathbf{T}'(t) = \frac{\langle 0, 2, 2t \rangle v(t) - \langle 2, 2t, t^2 \rangle v'(t)}{v(t)^2}, \quad v'(t) = 2t.$$

A convenient numerator is

$$\mathbf{A}(t) = \langle -4t, 4 - 2t^2, 4t \rangle = 2\langle -2t, 2 - t^2, 2t \rangle,$$

and since $\|\mathbf{A}(t)\| = 2(2 + t^2) = 2v(t)$, we get

$$\mathbf{N}(t) = \frac{\mathbf{A}(t)}{\|\mathbf{A}(t)\|} = \frac{\langle -2t, 2 - t^2, 2t \rangle}{2 + t^2}.$$

Binormal. $\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t) = \frac{\langle t^2, -2t, 2 \rangle}{2 + t^2}.$

At $t = 1$:

$$\mathbf{T}(1) = \left\langle \frac{2}{3}, \frac{2}{3}, \frac{1}{3} \right\rangle, \quad \mathbf{N}(1) = \left\langle -\frac{2}{3}, \frac{1}{3}, \frac{2}{3} \right\rangle, \quad \mathbf{B}(1) = \left\langle \frac{1}{3}, -\frac{2}{3}, \frac{2}{3} \right\rangle.$$

Remark 9.2. On exams, you will be provided with the formulas for $\mathbf{N}$ and $\mathbf{B}$. However, you are expected to *know and use* the formula for the unit tangent vector $\mathbf{T}(t) = \mathbf{r}'(t)/\|\mathbf{r}'(t)\|$ without it being given.

Example 9.3 (Circular motion). Consider the parametrized curve

$$\mathbf{r}(t) = \langle \cos t, \sin t \rangle, \quad 0 \leq t \leq 2\pi.$$

Since $x(t) = \cos t$ and $y(t) = \sin t$, we have

$$x^2 + y^2 = \cos^2 t + \sin^2 t = 1,$$

so the particle moves along the unit circle centered at the origin.

The *velocity vector* is

$$\mathbf{v}(t) = \mathbf{r}'(t) = \langle -\sin t, \cos t \rangle.$$

Notice that

$$\mathbf{r}(t) \cdot \mathbf{v}(t) = \cos t(-\sin t) + \sin t(\cos t) = 0,$$

which shows that the velocity is always orthogonal to the position vector. Geometrically, the motion is tangent to the circle at each point.

The *speed* is the magnitude of the velocity:

$$\|\mathbf{v}(t)\| = \sqrt{(-\sin t)^2 + (\cos t)^2} = 1.$$

Thus, the particle moves with constant speed. However, since the direction of $\mathbf{v}(t)$ continually changes, the velocity vector itself is not constant.

The *acceleration vector* is

$$\mathbf{a}(t) = \mathbf{v}'(t) = \langle -\cos t, -\sin t \rangle = -\mathbf{r}(t).$$

Hence the acceleration points directly toward the center of the circle, opposite the position vector. This inward (centripetal) acceleration is the reason the particle remains on the circular path.

Example 9.4. Consider another circular motion given by

$$\mathbf{r}(t) = \langle \cos(2t), \sin(2t) \rangle, \quad 0 \leq t \leq \pi.$$

This represents the same circle as in 9.3, because $x^2 + y^2 = \cos^2(2t) + \sin^2(2t) = 1$.

The trajectory has the same image as before, but completes the circle in half the time. Its speed is therefore twice that of the previous example:

The *velocity vector* is

$$\mathbf{v}(t) = \mathbf{r}'(t) = \langle -2\sin(2t), 2\cos(2t) \rangle,$$

and the *speed* of the object is

$$\text{speed} = \|\mathbf{v}(t)\| = \sqrt{(-2\sin(2t))^2 + (2\cos(2t))^2} = 2.$$

Example 9.5 (Non-circular motion). Consider the parametric curve

$$\mathbf{r}(t) = \langle \cos t, 3\sin t \rangle \quad 0 \leq t \leq 2\pi.$$

This trajectory is not circular! Indeed,

$$x^2 + y^2 = \cos^2 t + 9\sin^2 t \neq \text{constant}.$$

However, we notice that

$$x^2 + \left(\frac{y}{3}\right)^2 = \cos^2 t + \sin^2 t = 1.$$

So, the object moves counterclockwise along an ellipse in the xy -plane. Its velocity is

$$\mathbf{v} = \mathbf{r}'(t) = \langle -\sin t, 3\cos t \rangle,$$

Here $\mathbf{r}(t) \cdot \mathbf{v}(t) = 8\sin t \cos t$, so position and velocity are not orthogonal in general. They are orthogonal when t is a multiple of $\pi/2$.

Example 9.6 (Long-Range Artillery). A shell is fired from a heavy artillery piece at the origin with an initial speed $v_0 = 600$ m/s at an elevation angle of 45° (due east; no air resistance). Let $g = 9.81$ m/s² and use coordinates so that $\mathbf{a}(t) = \langle 0, 0, -g \rangle$. The initial velocity is

$$\begin{aligned} \mathbf{v}(0) &= \langle v_0 \cos 45^\circ, 0, v_0 \sin 45^\circ \rangle = \langle 300\sqrt{2}, 0, 300\sqrt{2} \rangle \\ &\approx \langle 424.26, 0, 424.26 \rangle \text{ m/s.} \end{aligned}$$

Find:

1. How long the shell remains airborne.
2. The horizontal range (how far it travels along the ground).
3. The maximum height of the trajectory.

Solution. The acceleration is $\mathbf{a}(t) = \langle 0, 0, -9.81 \rangle$. Integrating with the initial velocity $\mathbf{v}(0) \approx \langle 424.26, 0, 424.26 \rangle$ and $\mathbf{r}(0) = \mathbf{0}$ gives

$$\mathbf{v}(t) = \langle 424.26, 0, 424.26 - 9.81t \rangle, \quad \mathbf{r}(t) = \langle 424.26t, 0, 424.26t - 4.905t^2 \rangle.$$

1. **Time airborne:** Set the vertical component $z(t) = 0$:

$$T = \frac{2 \cdot 424.26}{9.81} \approx 86.50 \text{ s.}$$

2. **Horizontal range:**

$$R = x(T) = 424.26 \cdot T \approx 3.67 \times 10^4 \text{ m} = 36.70 \text{ km.}$$

3. **Maximum height:** Occurs at $t = T/2 \approx 43.25$ s:

$$H = \frac{(424.26)^2}{2 \cdot 9.81} \approx 9174.5 \text{ m} \approx 9.17 \text{ km.}$$

■

Example 9.7 (Artillery trajectory). A shell is fired from the origin at speed $v_0 = 100$ m/s at elevation angle 30° (due east; no air resistance). Let $g = 10$ m/s² and use coordinates so that $\mathbf{a}(t) = \langle 0, 0, -10 \rangle$ m/s². The initial velocity is

$$\mathbf{v}(0) = \langle v_0 \cos 30^\circ, 0, v_0 \sin 30^\circ \rangle = \langle 50\sqrt{3}, 0, 50 \rangle \text{ m/s.}$$

Find:

1. How long the shell remains airborne.
2. The horizontal range (how far it travels along the ground).
3. The maximum height of the trajectory.

Solution. The acceleration is $\mathbf{a}(t) = \langle 0, 0, -10 \rangle$. Integrating with initial velocity $\mathbf{v}(0) = \langle 50\sqrt{3}, 0, 50 \rangle$ and $\mathbf{r}(0) = \mathbf{0}$ gives

$$\mathbf{v}(t) = \langle 50\sqrt{3}, 0, 50 - 10t \rangle, \quad \mathbf{r}(t) = \langle 50\sqrt{3}t, 0, 50t - 5t^2 \rangle.$$

1. **Time airborne:** Set vertical position $z(t) = 0$:

$$50t - 5t^2 = 0 \implies 5t(10 - t) = 0 \implies T = 10 \text{ s.}$$

2. **Horizontal range:**

$$R = x(T) = (50\sqrt{3})(10) = 500\sqrt{3} \text{ m} \approx 866 \text{ m.}$$

3. **Maximum height:** Peak height occurs when $v_z(t) = 50 - 10t = 0$, so $t = 5$ s:

$$H = z(5) = 50(5) - 5(5)^2 = 250 - 125 = 125 \text{ m.}$$

■

10 Functions of Several Variables

10.1 Goals

- Define functions of many variables.
- Understand the graph of a multivariable function.
- Use level curves and related techniques to sketch the graph.

Definition 10.1 (Function of Two Variables)

A *function of two variables* is a rule that assigns to each ordered pair (x, y) in a domain $D \subseteq \mathbb{R}^2$ a unique real number z . We usually write

$$z = f(x, y).$$

Here, D is called the *domain* of f , and the set of all values $z = f(x, y)$ is called the *range*.

The graph of $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ is the set

$$\{(x, y, f(x, y)) : (x, y) \in D\} \subseteq \mathbb{R}^3.$$

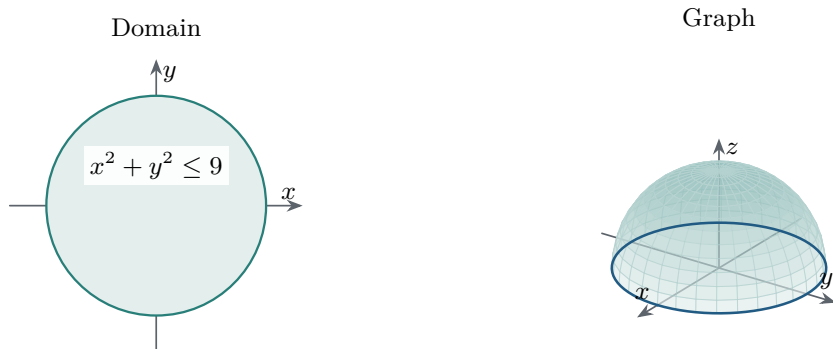


Figure 10.1: The domain of $f(x, y) = \sqrt{9 - x^2 - y^2}$ is the closed disk $x^2 + y^2 \leq 9$ in the xy -plane (left). Its graph is the upper hemisphere of the sphere $x^2 + y^2 + z^2 = 9$ (right).

Example 10.1. Consider the function $f(x, y) = \sqrt{9 - x^2 - y^2}$.

1. Find the domain of f .
2. Sketch the graph of f .

Solution. 1. The domain of f consists of all points $(x, y) \in \mathbb{R}^2$ satisfying

$$9 - x^2 - y^2 \geq 0 \implies x^2 + y^2 \leq 9.$$

So, the domain of f is the closed disk of radius 3 centered at the origin (see Figure 10.1).

2. Notice that

$$z = \sqrt{9 - x^2 - y^2} \iff x^2 + y^2 + z^2 = 9, \quad z \geq 0.$$

Thus, the graph of f is the upper hemisphere of a sphere of radius 3 centered at the origin. ■

Example 10.2. Find the domain of the following functions:

1. $f(x, y) = \frac{\ln(x - y + 1)}{2xy}$

2. $g(x, y) = \tan(x^2 + y^2)$

3. $h(x, y) = \sqrt{9 - x^2} + \sqrt{4 - y^2}$

Solution. 1. For $f(x, y) = \frac{\ln(x - y + 1)}{2xy}$ we need two conditions: the logarithm requires $x - y + 1 > 0$, i.e. $x > y - 1$, and the denominator requires $2xy \neq 0$, i.e. $x \neq 0$ and $y \neq 0$. Therefore, the domain is

$$D_f = \{(x, y) \in \mathbb{R}^2 \mid x > y - 1, x \neq 0, y \neq 0\}.$$

2. For $g(x, y) = \tan(x^2 + y^2)$ the tangent is undefined when its argument is an odd multiple of $\frac{\pi}{2}$, i.e. when $x^2 + y^2 = \frac{\pi}{2} + k\pi$ for $k \in \mathbb{Z}$. Therefore, the domain is

$$D_g = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \neq \frac{\pi}{2} + k\pi \text{ for every } k \in \mathbb{Z}\}.$$

3. For $h(x, y) = \sqrt{9 - x^2} + \sqrt{4 - y^2}$ both radicals must be nonnegative:

$$9 - x^2 \geq 0 \quad \text{and} \quad 4 - y^2 \geq 0.$$

Hence $-3 \leq x \leq 3$ and $-2 \leq y \leq 2$. Therefore, the domain is

$$D_h = \{(x, y) \in \mathbb{R}^2 \mid -3 \leq x \leq 3, -2 \leq y \leq 2\}.$$

■

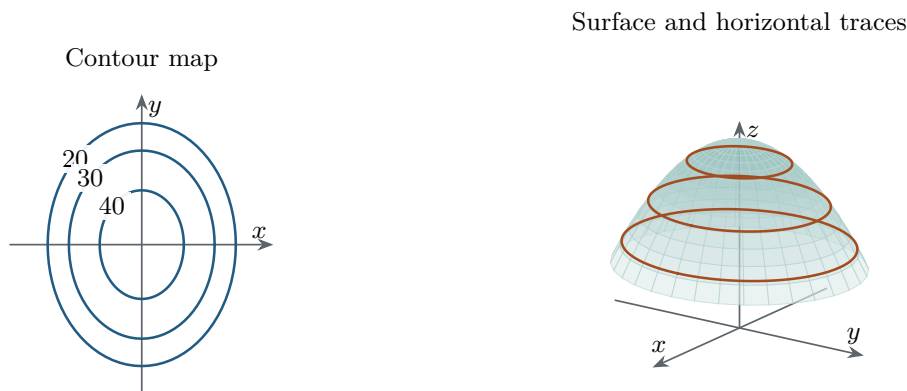


Figure 10.2: The graph of a paraboloid with selected level curves. Their projection onto the xy -plane (depicted on the left) is called the contour (i.e., topographic) map of the surface.

10.2 Level Curves

A *level set* of $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ consists of the points in D where $f(x, y) = c$ for a fixed value c . When this set is a curve, we call it a *level curve*. It is the projection onto the xy -plane of the horizontal slice $z = c$ of the graph:

$$\{(x, y) \in D : f(x, y) = c\}.$$

By plotting several such curves corresponding to different values of c , we gain a two-dimensional picture that reveals important geometric features of the surface such as symmetry, slope, and regions of increase or decrease. A collection of several level curves is called a *contour map* of f .

Figure 10.2 provides a geometric picture of level curves by showing how horizontal cuts of the surface correspond to curves in the xy -plane.

Example 10.3. Consider the surface defined by

$$z = f(x, y) = 16 - x^2 - y^2.$$

1. Find the equations of the level curves of f corresponding to $z = 12$ and $z = 8$.
2. Sketch these two level curves in the xy -plane.
3. Explain how these level curves help describe the shape of the graph of f in $\mathbb{R}^3$.

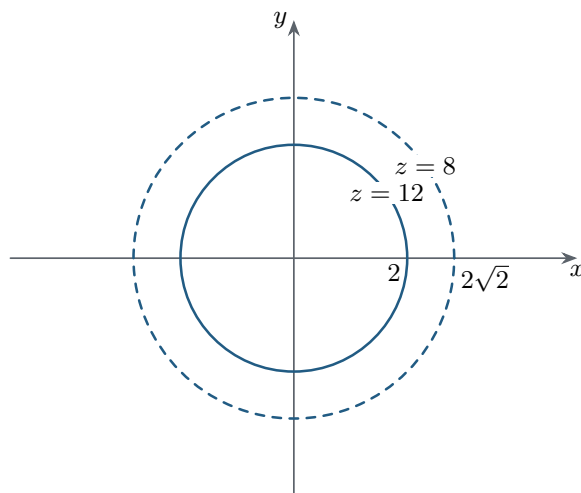


Figure 10.3: Level curves ($z = 12, 8$) of the paraboloid $z = 16 - x^2 - y^2$.

Solution. 1. For a level curve at height $z = c$ we set

$$f(x, y) = 16 - x^2 - y^2 = c \iff x^2 + y^2 = 16 - c.$$

- If $c = 12$, then $x^2 + y^2 = 4$, which is a circle of radius 2 centered at the origin.
- If $c = 8$, then $x^2 + y^2 = 8$, which is a circle of radius $\sqrt{8} = 2\sqrt{2}$ centered at the origin.

2. In the xy -plane, the two level curves are concentric circles with radii 2 and $2\sqrt{2}$.

3. Since $f(x, y) = 16 - x^2 - y^2$ is a downward-opening paraboloid, the level curves are nested circles. As z decreases, the circles get larger, reflecting how the paraboloid widens as we move down from its maximum point $(0, 0, 16)$. ■

Example 10.4. Sketch the level curves of the function $f(x, y) = 10 - 2x - 3y$ for the values $k = -6, 0, 6, 12$.

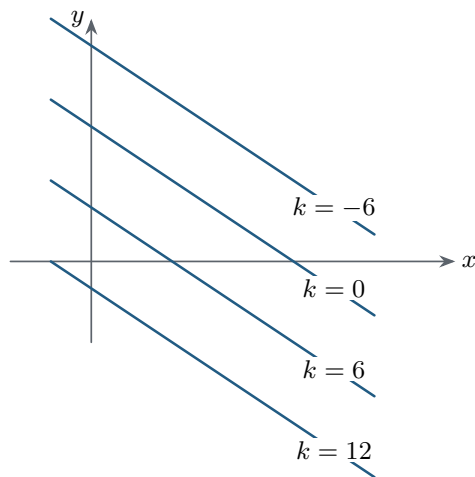


Figure 10.4: Level curves of $f(x, y) = 10 - 2x - 3y$ at $k = -6, 0, 6, 12$.

Solution. Level curves satisfy $10 - 2x - 3y = k$, i.e. $2x + 3y = 10 - k$ or

$$y = -\frac{2}{3}x + \frac{10-k}{3}.$$

So they are parallel lines with slope $-\frac{2}{3}$ and y -intercepts:

k	y -intercept $\frac{10-k}{3}$
-6	$\frac{16}{3}$
0	$\frac{10}{3}$
6	$\frac{4}{3}$
12	$-\frac{2}{3}$

That's the family of level curves. See Figure 10.4. ■

Example 10.5. Describe the level curves of the function $z = f(x, y) = \sin(xy)$ (where $-1 \leq z \leq 1$).

Solution. Fix $c \in [-1, 1]$. The level set $\{(x, y) : \sin(xy) = c\}$ is determined by

$$\sin(xy) = c \iff xy = \arcsin(c) + 2k\pi \quad \text{or} \quad xy = \pi - \arcsin(c) + 2k\pi, \quad k \in \mathbb{Z}.$$

Thus, for each $k \in \mathbb{Z}$ we get a rectangular hyperbola

$$xy = a_k \quad \text{or} \quad xy = b_k,$$

where

$$a_k = \arcsin(c) + 2k\pi, \quad b_k = \pi - \arcsin(c) + 2k\pi.$$

For a nonzero constant, $xy = \text{constant}$ is a rectangular hyperbola with the coordinate axes as asymptotes and two branches in opposite quadrants. The equation $xy = 0$ instead gives the two coordinate axes. The full level set is the union of the resulting curves.

Special cases.

- $c = 0$: $\sin(xy) = 0 \iff xy = k\pi, k \in \mathbb{Z}$. This includes the axes $xy = 0$ (i.e. $x = 0$ or $y = 0$) and the families $xy = \pm\pi, \pm2\pi, \dots$
- $c = 1$: $\sin(xy) = 1 \iff xy = \frac{\pi}{2} + 2k\pi, k \in \mathbb{Z}$.
- $c = -1$: $\sin(xy) = -1 \iff xy = -\frac{\pi}{2} + 2k\pi, k \in \mathbb{Z}$.

Symmetries. The level curves are invariant under $(x, y) \mapsto (y, x)$ and $(x, y) \mapsto (-x, -y)$, so each branch has a partner in the opposite quadrant. ■

11 Partial Derivatives

11.1 Goals

- Define partial differentiation. View it as rate of change and slope.
- Explain the rules of partial differentiation.
- Introduce higher-order partial derivatives.

Recall that in dimension 1, if $y = f(x)$, the derivative

$$y' = f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

measures the rate of change of y with respect to x .

When we extend this idea to functions of several variables, say $z = f(x, y)$, the situation becomes more interesting: z now depends on both x and y . So, the rate of change of z depends on the direction in which we move in the xy -plane. In other words, there are multiple possible directions of change to account for.

Definition 11.1

Let f be defined on a neighborhood of (x, y) . When the following limit exists, the *partial derivative* of f with respect to x (holding y constant) is

$$f_x(x, y) = \frac{\partial f}{\partial x}(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}.$$

Similarly, the partial derivative with respect to y is

$$f_y(x, y) = \frac{\partial f}{\partial y}(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h}.$$

In practice, we rarely use the limit definition. Instead, we rely on the usual differentiation rules to find the partial derivatives. Consider the following example.

Example 11.1. Find the partial derivatives of

$$f(x, y) = x^2y + e^{xy}.$$

Solution. To compute $\frac{\partial f}{\partial x}$, we treat y as a constant:

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (x^2y + e^{xy}) = 2xy + ye^{xy}.$$

To compute $\frac{\partial f}{\partial y}$, we treat x as a constant:

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (x^2y + e^{xy}) = x^2 + xe^{xy}.$$

■

Example 11.2. Find the partial derivatives of

$$f(x, y) = \sin(xy) + x \cos y.$$

Solution. For $\frac{\partial f}{\partial x}$, treat y as constant:

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (\sin(xy) + x \cos y) = y \cos(xy) + \cos y.$$

For $\frac{\partial f}{\partial y}$, treat x as constant:

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (\sin(xy) + x \cos y) = x \cos(xy) - x \sin y.$$

■

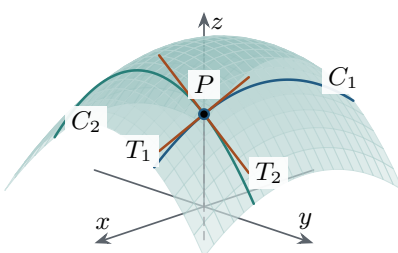


Figure 11.1: Geometric interpretation of partial derivatives. The surface $S : z = f(x, y)$ contains the point $P(a, b, c)$. The traces C_1 and C_2 correspond to fixing $y = b$ and $x = a$, respectively. The tangent lines T_1 and T_2 illustrate the slopes $f_x(a, b)$ and $f_y(a, b)$.

We can think of partial derivatives as the slopes of the surface when we move along the x - or y -direction. A geometric picture of this idea is shown in Figure 11.1.

Example 11.3. Compute the second-order partial derivatives f_{xx} , f_{yy} , f_{xy} , f_{yx} of

$$f(x, y) = \sin(xy) + x^2 \ln(1 + y), \quad (x, y) \in \mathbb{R} \times (-1, \infty),$$

and evaluate $f_{xy}(1, 0)$ and $f_{yx}(1, 0)$.

Solution. First derivatives:

$$f_x(x, y) = y \cos(xy) + 2x \ln(1 + y), \quad f_y(x, y) = x \cos(xy) + \frac{x^2}{1 + y}.$$

Second derivatives:

$$f_{xx}(x, y) = -y^2 \sin(xy) + 2 \ln(1 + y), \quad f_{yy}(x, y) = -x^2 \sin(xy) - \frac{x^2}{(1 + y)^2}.$$

Mixed derivatives (differentiate f_x w.r.t. y , and f_y w.r.t. x):

$$f_{xy}(x, y) = \cos(xy) - xy \sin(xy) + \frac{2x}{1 + y},$$

$$f_{yx}(x, y) = \cos(xy) - xy \sin(xy) + \frac{2x}{1 + y}.$$

At $(1, 0)$:

$$f_{xy}(1, 0) = \cos 0 - 0 \cdot \sin 0 + \frac{2 \cdot 1}{1 + 0} = 1 + 2 = 3, \quad f_{yx}(1, 0) = 3.$$

■

Remark 11.4. We see directly that $f_{xy}(x, y) = f_{yx}(x, y)$. This agrees with Clairaut's (or Schwarz's) theorem: since f has continuous second partials on its domain $\{y > -1\}$, the mixed partials commute.

Example 11.5. Find the 2-nd order derivatives of $f(x, y) = e^{x^2 y}$.

Solution. First derivatives:

$$f_x = 2xy e^{x^2 y}, \quad f_y = x^2 e^{x^2 y}.$$

Second derivatives:

$$f_{xx} = (2y + 4x^2 y^2) e^{x^2 y}, \quad f_{yy} = x^4 e^{x^2 y},$$

$$f_{xy} = (2x + 2x^3 y) e^{x^2 y}, \quad f_{yx} = (2x + 2x^3 y) e^{x^2 y}.$$

Hence $f_{xy} = f_{yx}$, as expected.

■

Finally, partial derivatives work the same way with functions of more than 2 variables. For example:

Example 11.6. Find the first-order partial derivatives of

$$f(x, y, z) = x^2 y + y z^3.$$

Solution.

$$f_x = 2xy, \quad f_y = x^2 + z^3, \quad f_z = 3yz^2.$$

■

12 The Chain Rule

12.1 Goals

- Describe the chain rule in multiple variables.
- Understand the total differential dF of a multivariable function.
- Derive and apply implicit differentiation formulas using differentials.
- Work through geometric and engineering applications.

We begin by recalling the chain rule in one variable. Suppose $y = f(x)$ and $x = g(t)$. Then y ultimately depends on t , and we may ask: how does the rate of change of y behave with respect to t ? The chain rule answers this question:

$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt} \quad \text{equivalently} \quad y'(t) = f'(g(t)) g'(t).$$

To remember this more easily, we often use a tree diagram showcasing the dependencies between the variables. See Figure 12.1.

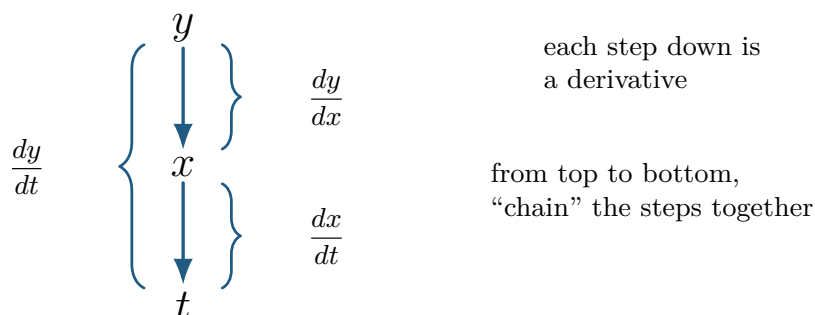


Figure 12.1: A tree diagram illustrating the chain rule. Each step down corresponds to a derivative, and from top to bottom the derivatives “chain” together to give $\frac{dy}{dt}$.

Ultimately, the main difference from the single-variable picture is that in several variables we have more dependencies. That is, the variables x, y (or more) may themselves depend on secondary variables $s, t, \dots$. We illustrate this with the following example.

Example 12.1. Consider $z = f(x, y) = x + y^3$ with $x(t) = e^t$ and $y(t) = \tan t$. Find $\frac{dz}{dt}$.

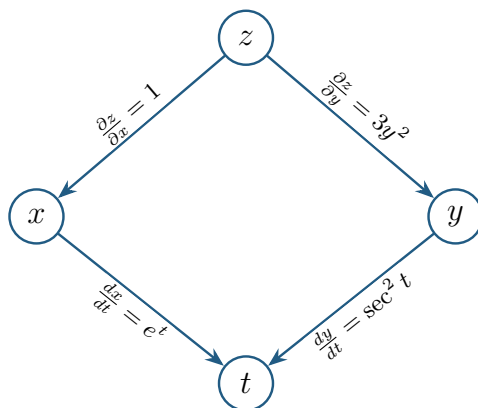
Solution. By the chain rule in two variables (see Figure 12.2),

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}.$$

Since $f_x = 1$, $f_y = 3y^2$, $\frac{dx}{dt} = e^t$, and $\frac{dy}{dt} = \sec^2 t$, we obtain

$$\frac{dz}{dt} = e^t + 3y^2 \sec^2 t = e^t + 3 \tan^2(t) \sec^2(t).$$

■



$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

Figure 12.2: A tree diagram showing the dependencies of z on x, y and ultimately on t .

Example 12.2. Let

$$z = x^2 y, \quad x = u + v, \quad y = uv.$$

Find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$.

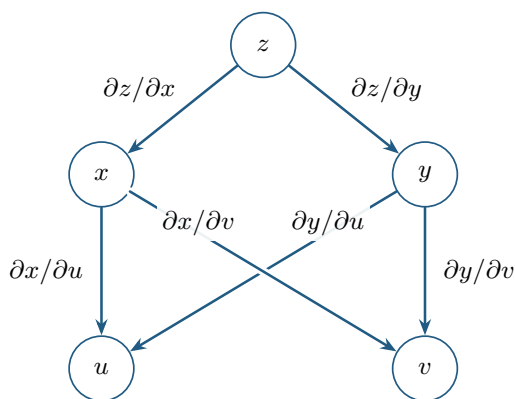


Figure 12.3: Dependency diagram for $z = z(x, y)$ with x, y functions of u, v .

Solution. By the chain rule,

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u}, \quad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v}.$$

Now

$$\frac{\partial z}{\partial x} = 2xy, \quad \frac{\partial z}{\partial y} = x^2, \quad \frac{\partial x}{\partial u} = 1, \quad \frac{\partial x}{\partial v} = 1, \quad \frac{\partial y}{\partial u} = v, \quad \frac{\partial y}{\partial v} = u.$$

So

$$\frac{\partial z}{\partial u} = (2xy)(1) + (x^2)(v), \quad \frac{\partial z}{\partial v} = (2xy)(1) + (x^2)(u).$$

Finally, substituting $x = u + v$, $y = uv$,

$$\frac{\partial z}{\partial u} = 2(u + v)(uv) + (u + v)^2 v, \quad \frac{\partial z}{\partial v} = 2(u + v)(uv) + (u + v)^2 u.$$

■

12.2 The Total Differential

The chain rule assumes that F is *differentiable* at the point in question. This means that its change is given by a linear map, up to an error small compared with the input displacement:

$$F(a + h, b + k) - F(a, b) = F_x(a, b)h + F_y(a, b)k + o(\sqrt{h^2 + k^2}).$$

Continuous first partial derivatives near (a, b) guarantee differentiability. The existence of partial derivatives at a point alone does not.

The chain rule also naturally reveals how small changes in inputs cause a change in the output. For a differentiable function of two variables $F(x, y)$, the **total differential** dF represents the principal linear part of the change ΔF :

$$dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy = F_x dx + F_y dy.$$

Similarly, for a function of three variables $F(x, y, z)$, the total differential is

$$dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy + \frac{\partial F}{\partial z} dz = F_x dx + F_y dy + F_z dz.$$

Along a differentiable path $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, apply the differential to the velocity vector to obtain the chain rule:

$$\frac{dF}{dt} = F_x \frac{dx}{dt} + F_y \frac{dy}{dt} + F_z \frac{dz}{dt}.$$

12.3 Implicit Differentiation via Differentials

Differentials provide a fast, elegant way to derive the formulas for implicit differentiation.

Case 1: One Independent Variable. Suppose an equation $F(x, y) = C$ defines y implicitly as a differentiable function of x , i.e., $y = y(x)$. Since the level curve $F(x, y) = C$ is constant, its total differential must vanish:

$$dF = 0 \implies F_x dx + F_y dy = 0.$$

Dividing formally by dx (or applying $\frac{d}{dx}$ along the curve):

$$F_x + F_y \frac{dy}{dx} = 0.$$

Assuming $F_y \neq 0$, solving directly for $\frac{dy}{dx}$ gives:

$$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

Example 12.3. Find $\frac{dy}{dx}$ if $x^2 + y^2 = 1$.

Solution. Set $F(x, y) = x^2 + y^2 - 1$. The curve is the level set $F = 0$. The partial derivatives are:

$$F_x = 2x, \quad F_y = 2y.$$

Using the differential relation $dF = 2x dx + 2y dy = 0$, we find:

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{2x}{2y} = -\frac{x}{y}, \quad y \neq 0.$$

■

Case 2: Multiple Independent Variables. Now suppose an equation $F(x, y, z) = C$ defines z implicitly as a function of two independent variables x and y , so $z = z(x, y)$.

Along this level surface, the total differential remains zero:

$$dF = F_x dx + F_y dy + F_z dz = 0.$$

To isolate the partial derivative with respect to x , we hold y constant (so $dy = 0$):

$$F_x dx + F_z dz = 0 \implies F_x + F_z \frac{\partial z}{\partial x} = 0 \implies \frac{\partial z}{\partial x} = -\frac{F_x}{F_z}.$$

Similarly, holding x constant (so $dx = 0$):

$$F_y dy + F_z dz = 0 \implies F_y + F_z \frac{\partial z}{\partial y} = 0 \implies \frac{\partial z}{\partial y} = -\frac{F_y}{F_z},$$

provided $F_z \neq 0$.

Example 12.4. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if

$$F(x, y, z) = x^2 + y^2 + z^2 - 1 = 0,$$

where $z = z(x, y)$.

Solution. Compute the partial derivatives of F :

$$F_x = 2x, \quad F_y = 2y, \quad F_z = 2z.$$

At points where $z \neq 0$, applying our formulas gives:

$$\begin{aligned} \frac{\partial z}{\partial x} &= -\frac{F_x}{F_z} = -\frac{2x}{2z} = -\frac{x}{z}, \\ \frac{\partial z}{\partial y} &= -\frac{F_y}{F_z} = -\frac{2y}{2z} = -\frac{y}{z}. \end{aligned}$$

■

12.4 Engineering Application: Sensitivity Analysis

In engineering, the chain rule and differentials are used for **sensitivity analysis**—determining how fluctuations in multiple physical components contribute to the total change of a system.

Example 12.5. The volume of a cylindrical fuel tank is given by $V = \pi r^2 h$. Due to thermal expansion, both the radius r and the height h are increasing over time t . At a specific moment, the measurements are:

$$r = 2 \text{ m}, \quad h = 5 \text{ m}, \quad \frac{dr}{dt} = 0.01 \text{ m/min}, \quad \frac{dh}{dt} = 0.03 \text{ m/min}.$$

Find the rate of change of the volume $\frac{dV}{dt}$ at this instant.

Solution. The total differential of volume is $dV = \frac{\partial V}{\partial r} dr + \frac{\partial V}{\partial h} dh$. Dividing by dt gives the multivariable chain rule:

$$\frac{dV}{dt} = \frac{\partial V}{\partial r} \frac{dr}{dt} + \frac{\partial V}{\partial h} \frac{dh}{dt}.$$

First, compute the partial derivatives (the sensitivities):

$$\frac{\partial V}{\partial r} = 2\pi r h, \quad \frac{\partial V}{\partial h} = \pi r^2.$$

Substituting the given values:

$$\frac{dV}{dt} = (2\pi \cdot 2 \cdot 5)(0.01) + (\pi \cdot 2^2)(0.03)$$

$$\frac{dV}{dt} = 20\pi(0.01) + 4\pi(0.03) = 0.2\pi + 0.12\pi = 0.32\pi \text{ m}^3/\text{min}.$$

Interpretation: The radius contributes 0.2π and the height contributes 0.12π to the rate of volume change. In relative terms, $dV/V = 2 dr/r + dh/h$: a 1% change in radius produces approximately a 2% change in volume when height is fixed. ■

13 Directional Derivatives and the Gradient Vector

13.1 Lesson Goals

- Derive the **directional derivative** via parametric curves and the multivariable Chain Rule.
- Define the **gradient vector** ∇f and connect it directly to directional rates of change.
- Interpret the gradient geometrically as encoding the **direction and magnitude of maximal increase**.
- Understand the orthogonality between the gradient vector and **level curves**.

13.2 1. Directional Derivatives via the Chain Rule

Assume f is differentiable on an open set containing the base point. The standard partial derivatives f_x and f_y measure the rate of change of $z = f(x, y)$ exclusively parallel to the coordinate axes. To determine the rate of change when moving in an arbitrary direction, we specify a **unit vector** $\mathbf{u} = \langle a, b \rangle$ in the domain (with $a^2 + b^2 = 1$).

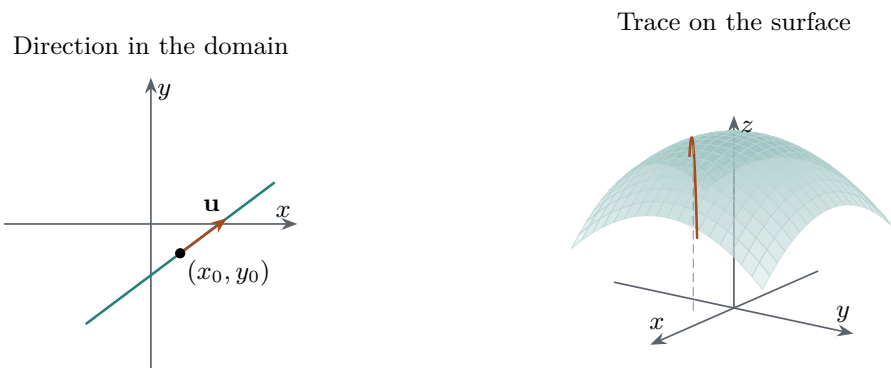


Figure 13.1: Moving along the line $\langle x_0 + at, y_0 + bt \rangle$ in the domain traces a curve on surface S , whose initial slope is $\left. \frac{dz}{dt} \right|_{t=0} = D_{\mathbf{u}}f(x_0, y_0)$.

Derivation via Parametrization

Consider a straight line through (x_0, y_0) in the direction of unit vector $\mathbf{u} = \langle a, b \rangle$:

$$x(t) = x_0 + at, \quad y(t) = y_0 + bt.$$

Because $\|\mathbf{u}\| = 1$, the parameter t represents the signed arc length distance traveled along the line. Restricting $z = f(x, y)$ along this path produces a single-variable function of t :

$$z(t) = f(x_0 + at, y_0 + bt).$$

By the multivariable Chain Rule, the rate of change of z with respect to distance t is:

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

Since $\frac{dx}{dt} = a$ and $\frac{dy}{dt} = b$, evaluating at $t = 0$ (the base point (x_0, y_0)) gives:

$$\left. \frac{dz}{dt} \right|_{t=0} = f_x(x_0, y_0) a + f_y(x_0, y_0) b.$$

Rewriting this sum as an algebraic dot product naturally separates the surface properties from the chosen direction:

$$\left. \frac{dz}{dt} \right|_{t=0} = \langle f_x(x_0, y_0), f_y(x_0, y_0) \rangle \cdot \langle a, b \rangle.$$

Definition 13.1 (Gradient Vector and Directional Derivative)

Let $f(x, y)$ be a differentiable function. The **gradient vector** of f , denoted ∇f , is:

$$\nabla f(x, y) = \left\langle \frac{\partial f}{\partial x}(x, y), \frac{\partial f}{\partial y}(x, y) \right\rangle = f_x \mathbf{i} + f_y \mathbf{j}.$$

The **directional derivative** in a unit direction $\mathbf{u} = \langle a, b \rangle$ is

$$D_{\mathbf{u}}f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}.$$

Differentiability gives $D_{\mathbf{u}}f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \mathbf{u}$. The limit can also exist for functions that are not differentiable, but the gradient formula then need not hold.

Vectors as differentiation operators

At a point $p = (x_0, y_0)$, any vector $\mathbf{v} = \langle a, b \rangle$ defines an operator on differentiable functions:

$$\mathbf{v}_p[f] = \left. \frac{d}{dt} \right|_{t=0} f(p + t\mathbf{v}) = af_x(p) + bf_y(p) = df_p(\mathbf{v}).$$

Thus we can identify a vector at p with $a \partial_x|_p + b \partial_y|_p$. This operator takes a function as input and returns a scalar. Unit vectors measure change per unit distance, while an arbitrary vector also specifies the speed of the path. The coordinate vectors give $\partial_x f = f_x$ and $\partial_y f = f_y$.

Example 13.1. Let $f(x, y) = x^2 \cos y$. Find the directional derivative of f in the direction of $\mathbf{v} = \langle -1, 2 \rangle$ at the point $(2, \pi/2)$.

Solution. **Step 1: Normalize the direction vector into a unit vector:**

$$\|\mathbf{v}\| = \sqrt{(-1)^2 + 2^2} = \sqrt{5} \implies \mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \left\langle -\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle.$$

Step 2: Compute the gradient vector $\nabla f(x, y)$:

$$\nabla f(x, y) = \langle f_x, f_y \rangle = \langle 2x \cos y, -x^2 \sin y \rangle.$$

Step 3: Evaluate the gradient at $(2, \pi/2)$:

$$\nabla f\left(2, \frac{\pi}{2}\right) = \left\langle 2(2) \cos\left(\frac{\pi}{2}\right), -(2)^2 \sin\left(\frac{\pi}{2}\right) \right\rangle = \langle 0, -4 \rangle.$$

Step 4: Take the dot product:

$$D_{\mathbf{u}}f\left(2, \frac{\pi}{2}\right) = \nabla f\left(2, \frac{\pi}{2}\right) \cdot \mathbf{u} = \langle 0, -4 \rangle \cdot \left\langle -\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle = 0 \left(-\frac{1}{\sqrt{5}}\right) + (-4) \left(\frac{2}{\sqrt{5}}\right) = -\frac{8}{\sqrt{5}}.$$

$$\boxed{D_{\mathbf{u}}f\left(2, \frac{\pi}{2}\right) = -\frac{8}{\sqrt{5}}}$$

■

13.3 2. Maximizing the Directional Derivative

When $\nabla f \neq \mathbf{0}$, use the geometric formula for the dot product: $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$, where $\theta \in [0, \pi]$ is the angle between the vectors. Since $\mathbf{u}$ is a unit vector ($\|\mathbf{u}\| = 1$):

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = \|\nabla f\| \|\mathbf{u}\| \cos \theta = \|\nabla f\| \cos \theta.$$

Theorem 13.1 (Optimal Rates of Change)

Let f be differentiable at (x_0, y_0) and suppose $\nabla f(x_0, y_0) \neq \mathbf{0}$:

- **Maximum Rate of Increase:** Occurs when $\cos \theta = 1$ ($\theta = 0$), meaning $\mathbf{u}$ points in the **same direction** as ∇f :

$$\max_{\|\mathbf{u}\|=1} D_{\mathbf{u}}f = \|\nabla f\|.$$

- **Maximum Rate of Decrease:** Occurs when $\cos \theta = -1$ ($\theta = \pi$), meaning $\mathbf{u}$ points in the **opposite direction** $-\nabla f$:

$$\min_{\|\mathbf{u}\|=1} D_{\mathbf{u}}f = -\|\nabla f\|.$$

- **Zero Rate of Change:** Occurs when $\cos \theta = 0$ ($\theta = \pi/2$), meaning $\mathbf{u}$ is **orthogonal** to ∇f :

$$D_{\mathbf{u}}f = 0.$$

If $\nabla f(x_0, y_0) = \mathbf{0}$, every directional derivative there is zero. There is no distinguished direction of increase or decrease to first order.

Extension to Three Variables

For $w = f(x, y, z)$, the gradient vector in $\mathbb{R}^3$ is:

$$\nabla f(x, y, z) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle.$$

The directional derivative in the direction of unit vector $\mathbf{u} = \langle a, b, c \rangle$ remains $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$.

13.4 3. Gradient Vector and Level Curves

Consider a level curve $f(x, y) = c$. Along this curve, the output value of f remains constant. If the curve is parameterized by $\mathbf{r}(t) = \langle x(t), y(t) \rangle$, applying the Chain Rule along the curve yields:

$$\frac{d}{dt}[f(x(t), y(t))] = \frac{d}{dt}[c] = 0,$$

$$\frac{\partial f}{\partial x}x'(t) + \frac{\partial f}{\partial y}y'(t) = \nabla f(x, y) \cdot \mathbf{r}'(t) = 0.$$

Geometric Principle: Gradient Orthogonality

At a point where $\nabla f \neq \mathbf{0}$, the level set of a continuously differentiable function is locally a smooth curve, and ∇f is normal to its tangent line. It points toward increasing values of f . At a point where $\nabla f = \mathbf{0}$, this calculation does not determine a normal direction.

Example 13.2. Let $f(x, y) = x^2 + y^2$. At the point $(1, 2)$:

1. Find the direction and value of the maximum rate of increase.
2. Find the direction and value of the maximum rate of decrease.
3. Find directions in which the instantaneous rate of change is zero.

Solution. **Step 1: Compute and evaluate the gradient:**

$$\nabla f(x, y) = \langle 2x, 2y \rangle \implies \nabla f(1, 2) = \langle 2, 4 \rangle.$$

$$\|\nabla f(1, 2)\| = \sqrt{2^2 + 4^2} = \sqrt{20} = 2\sqrt{5}.$$

Solutions:

- **Maximum increase:** Points in the direction of $\nabla f(1, 2) = \langle 2, 4 \rangle$ (unit direction $\mathbf{u} = \frac{1}{\sqrt{5}}\langle 1, 2 \rangle$). The maximum rate of increase is:

$$\|\nabla f(1, 2)\| = 2\sqrt{5}$$

- **Maximum decrease:** Points in the opposite direction $-\nabla f(1, 2) = \langle -2, -4 \rangle$ (unit direction $\mathbf{u} = -\frac{1}{\sqrt{5}}\langle 1, 2 \rangle$). The rate of decrease is:

$$-\|\nabla f(1, 2)\| = -2\sqrt{5}$$

- **Zero change:** Any direction orthogonal to ∇f . Solving $\langle 2, 4 \rangle \cdot \langle u_1, u_2 \rangle = 2u_1 + 4u_2 = 0$ gives unit directions:

$$\mathbf{u} = \pm \frac{1}{\sqrt{5}}\langle -2, 1 \rangle.$$

Along these directions (tangent to the level circle $x^2 + y^2 = 5$), $D_{\mathbf{u}}f = 0$.

■

14 Tangent Plane and Linear Approximation

14.1 Goals

- Define the tangent plane to a surface at a point.
- Use it to compute the linear approximation of a function at a point.

We begin with the one-dimensional case. If $y = f(x)$ is a curve in $\mathbb{R}^2$, then the tangent line to f at $x = a$ is

$$\ell : y = f(a) + f'(a)(x - a).$$

Near $x = a$, this line provides a linear approximation to f :

$$f(x) \approx \ell(x).$$

(See Figure 14.1.)

If f is continuously differentiable near (x_0, y_0) , the tangent vectors to curves on its graph span a plane. Translating this vector plane to $(x_0, y_0, f(x_0, y_0))$ gives the *tangent plane* (see Figure 14.2). Its vectors describe directions, while its points give the linear approximation to the surface.

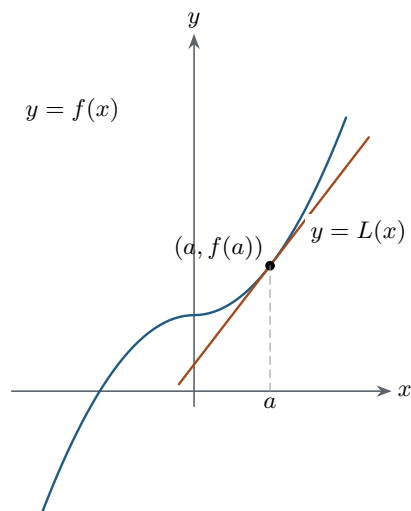


Figure 14.1: The curve $y = f(x)$ and its tangent line ℓ at $x = a$. Near $x = a$, the tangent line provides a linear approximation to f .

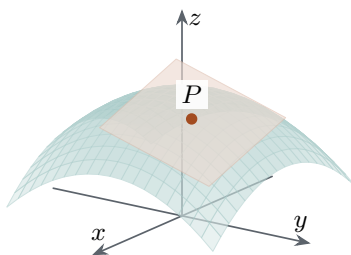


Figure 14.2: The tangent plane to the paraboloid $z = x^2 + y^2$ at the point (x_0, y_0, z_0) . The dashed teal curves are coordinate traces through the point, and the thin black lines show the infinitely many tangent lines contained in the tangent plane.

14.2 Equation of the Tangent Plane

We want the equation of the tangent plane to a surface at a point. The key idea is that the tangent plane consists of all tangent lines to curves that lie on the surface and pass through the point.

Assume f is continuously differentiable near (x_0, y_0) . Let

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$$

be a differentiable curve on the surface, with tangent vector $\mathbf{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle$. Since $z = f(x, y)$ describes the surface, any such curve satisfies

$$z(t) = f(x(t), y(t)).$$

Define

$$F(x, y, z) = f(x, y) - z.$$

Then the surface is the level set $F(x, y, z) = 0$.

By the chain rule (implicit differentiation),

$$\frac{\partial F}{\partial x} x'(t) + \frac{\partial F}{\partial y} y'(t) + \frac{\partial F}{\partial z} z'(t) = 0.$$

In vector form,

$$\underbrace{\langle F_x, F_y, F_z \rangle}_{\nabla F(x, y, z)} \cdot \underbrace{\langle x'(t), y'(t), z'(t) \rangle}_{\text{tangent vector}} = 0.$$

Thus, ∇F is orthogonal to every tangent vector, so $\nabla F(x_0, y_0, z_0)$ is a normal vector to the tangent plane at (x_0, y_0, z_0) .

Therefore, the tangent plane has equation

$$\boxed{\nabla F(x_0, y_0, z_0) \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0,}$$

where $F(x, y, z) = f(x, y) - z$.

Explicitly, since

$$\nabla F(x, y, z) = \langle f_x(x, y), f_y(x, y), -1 \rangle,$$

the tangent plane at (x_0, y_0, z_0) , with $z_0 = f(x_0, y_0)$, is

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - z_0) = 0,$$

or equivalently,

$$\boxed{z = z_0 + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).}$$

This is the standard formula for the tangent plane to the surface $z = f(x, y)$ at the point (x_0, y_0, z_0) .

Example 14.1. Find the equation of the tangent plane to the paraboloid

$$z = x^2 + y^2$$

at the point $(1, 2, 5)$.

Solution. We have $f(x, y) = x^2 + y^2$, so

$$f_x(x, y) = 2x, \quad f_y(x, y) = 2y.$$

At $(x_0, y_0) = (1, 2)$, this gives

$$f_x(1, 2) = 2, \quad f_y(1, 2) = 4, \quad z_0 = f(1, 2) = 5.$$

Thus the tangent plane is

$$z = z_0 + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) = 5 + 2(x - 1) + 4(y - 2).$$

Simplifying,

$$z = 2x + 4y - 5.$$

■

14.3 Linear Approximation and the Tangent Plane

In single-variable calculus, the tangent line to $y = f(x)$ at $x = a$ gives the best linear approximation to the function near that point:

$$L(x) = f(a) + f'(a)(x - a).$$

In two variables, the analogue of the tangent line is the *tangent plane*.

Let $z = f(x, y)$ be differentiable at (a, b) . The tangent plane to the surface at the point $(a, b, f(a, b))$ is given by:

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

When (x, y) is close to (a, b) , the values on the surface are well-approximated by the values on this plane. We define the *linearization* $L(x, y)$ of f at (a, b) by

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b),$$

so that for $(x, y) \approx (a, b)$, we have the *linear approximation*:

$$f(x, y) \approx L(x, y).$$

Example 14.2 (Approximating a distance). Find the linearization of $f(x, y) = \sqrt{x^2 + y^2}$ at $(3, 4)$, and use it to approximate $\sqrt{(3.02)^2 + (3.97)^2}$.

Solution. First, evaluate f and its first partial derivatives at the base point $(a, b) = (3, 4)$:

$$f(3, 4) = \sqrt{3^2 + 4^2} = 5,$$

$$f_x(x, y) = \frac{x}{\sqrt{x^2 + y^2}} \implies f_x(3, 4) = \frac{3}{5} = 0.6,$$

$$f_y(x, y) = \frac{y}{\sqrt{x^2 + y^2}} \implies f_y(3, 4) = \frac{4}{5} = 0.8.$$

The linearization of f at $(3, 4)$ is:

$$L(x, y) = 5 + \frac{3}{5}(x - 3) + \frac{4}{5}(y - 4).$$

To approximate $\sqrt{(3.02)^2 + (3.97)^2}$, we set $(x, y) = (3.02, 3.97)$:

$$\begin{aligned} f(3.02, 3.97) &\approx L(3.02, 3.97) \\ &= 5 + 0.6(3.02 - 3) + 0.8(3.97 - 4) \\ &= 5 + 0.6(0.02) + 0.8(-0.03) \\ &= 5 + 0.012 - 0.024 = 4.988. \end{aligned}$$

Note: The exact value to five decimal places is ≈ 4.98812 , demonstrating that the plane captures the change accurately while handling positive and negative increments simultaneously. ■

Example 14.3 (An exponential function). Find the linear approximation of $f(x, y) = xe^{xy}$ near the point $(2, 0)$, and use it to estimate $f(1.98, 0.05)$.

Solution. Evaluate the function and its partial derivatives at $(a, b) = (2, 0)$:

$$\begin{aligned} f(2, 0) &= 2e^0 = 2, \\ f_x(x, y) &= e^{xy} + x(ye^{xy}) = e^{xy}(1 + xy) \implies f_x(2, 0) = 1(1 + 0) = 1, \\ f_y(x, y) &= x(xe^{xy}) = x^2e^{xy} \implies f_y(2, 0) = 4e^0 = 4. \end{aligned}$$

The linear approximation is:

$$f(x, y) \approx 2 + 1(x - 2) + 4(y - 0).$$

Evaluating at $(x, y) = (1.98, 0.05)$:

$$\begin{aligned} f(1.98, 0.05) &\approx 2 + (1.98 - 2) + 4(0.05 - 0) \\ &= 2 - 0.02 + 0.20 = 2.18. \end{aligned}$$

■

15 Maximum and Minimum Values

15.1 Goals

- Find and classify critical points of functions of two variables.
- Interpret the Hessian and its eigenvalues geometrically.
- Derive and use the multivariable second derivative test.
- Find absolute extrema on closed and bounded domains.

15.2 Critical Points and the Hessian

Recall that the gradient $\nabla f(a, b)$ points in the direction of steepest increase of f . If f is differentiable and has a local maximum or minimum at an interior point (a, b) , then

$$\nabla f(a, b) = \vec{0}.$$

Thus, to locate possible extrema, we first solve

$$f_x(x, y) = 0, \quad f_y(x, y) = 0.$$

For a differentiable function, a point satisfying these equations is a **critical point**. In the broader convention, points where a first derivative does not exist are also critical candidates and must be checked separately. A critical point can be a local extremum, a saddle point, or a degenerate point that needs further analysis.

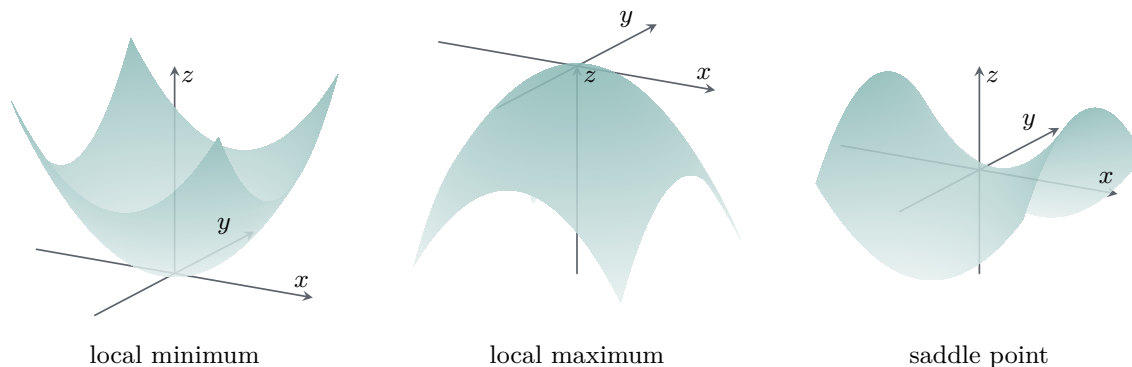


Figure 15.1: Typical critical points: a local minimum, a local maximum, and a saddle point.

Assume now that f has continuous second partial derivatives near the critical point. To classify it, use the matrix of second derivatives, called the **Hessian**:

$$H_f(a, b) = \begin{pmatrix} f_{xx}(a, b) & f_{xy}(a, b) \\ f_{yx}(a, b) & f_{yy}(a, b) \end{pmatrix}.$$

If f has continuous second partial derivatives near (a, b) , then $f_{xy} = f_{yx}$, so the Hessian is symmetric and therefore has real eigenvalues.

At a critical point, the second-order Taylor approximation is

$$f(a + h, b + k) - f(a, b) = \frac{1}{2} \begin{pmatrix} h & k \end{pmatrix} H_f(a, b) \begin{pmatrix} h \\ k \end{pmatrix} + o(h^2 + k^2).$$

Because the Hessian is symmetric, an orthogonal change of coordinates diagonalizes it. If its eigenvalues are λ_1 and λ_2 , then in suitable coordinates (u, v) the quadratic term becomes

$$\frac{1}{2}(\lambda_1 u^2 + \lambda_2 v^2).$$

The signs of the eigenvalues therefore determine the local shape of the graph:

- If $\lambda_1, \lambda_2 > 0$, the graph curves upward in every direction, so f has a local minimum.
- If $\lambda_1, \lambda_2 < 0$, the graph curves downward in every direction, so f has a local maximum.
- If the eigenvalues have opposite signs, the graph curves upward in some directions and downward in others, so the point is a saddle point.
- If an eigenvalue is zero, the quadratic approximation does not settle the question, so the test is inconclusive.

For a 2×2 Hessian, the product of the eigenvalues is its determinant:

$$D(a, b) = \det H_f(a, b) = f_{xx}(a, b)f_{yy}(a, b) - (f_{xy}(a, b))^2 = \lambda_1 \lambda_2.$$

This gives the familiar **multivariable second derivative test**:

- If $D > 0$ and $f_{xx} > 0$, then both eigenvalues are positive and f has a local minimum.
- If $D > 0$ and $f_{xx} < 0$, then both eigenvalues are negative and f has a local maximum.
- If $D < 0$, then the eigenvalues have opposite signs and the point is a saddle point.
- If $D = 0$, then the test is inconclusive.

In short, the discriminant test is an efficient two-variable way of checking the signs of the Hessian eigenvalues.

Example 15.1. Find and classify the critical points of

$$f(x, y) = x^2 + xy + y^2 - 2x + 3y.$$

Solution. The first-order partial derivatives are

$$f_x = 2x + y - 2, \quad f_y = x + 2y + 3.$$

Thus the critical points satisfy

$$\begin{cases} 2x + y = 2, \\ x + 2y = -3. \end{cases}$$

Solving gives the unique critical point

$$(x, y) = \left(\frac{7}{3}, -\frac{8}{3}\right).$$

The Hessian is

$$H_f = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Its eigenvalues are 1 and 3, both positive. Hence the critical point is a local minimum. In fact, since the Hessian is positive definite everywhere, f is strictly convex, so this is the unique global minimum. Its value is

$$f\left(\frac{7}{3}, -\frac{8}{3}\right) = -\frac{19}{3}.$$

Equivalently,

$$D = (2)(2) - 1^2 = 3 > 0, \quad f_{xx} = 2 > 0,$$

which gives the same conclusion by the discriminant test. ■

Problem 15.2. Find and classify all critical points of

$$f(x, y) = xy e^{-x^2-y^2}.$$

Solution. The first-order partial derivatives are

$$f_x = y(1 - 2x^2)e^{-x^2-y^2}, \quad f_y = x(1 - 2y^2)e^{-x^2-y^2}.$$

Since the exponential factor is never zero, the critical-point equations reduce to

$$y(1 - 2x^2) = 0, \quad x(1 - 2y^2) = 0.$$

Therefore the critical points are

$$(0, 0)$$

and

$$\left(\frac{\varepsilon_1}{\sqrt{2}}, \frac{\varepsilon_2}{\sqrt{2}}\right), \quad \varepsilon_1, \varepsilon_2 \in \{-1, 1\}.$$

The second derivatives are

$$\begin{aligned} f_{xx} &= 2xy(2x^2 - 3)e^{-x^2-y^2}, \\ f_{yy} &= 2xy(2y^2 - 3)e^{-x^2-y^2}, \\ f_{xy} &= (1 - 2x^2)(1 - 2y^2)e^{-x^2-y^2}. \end{aligned}$$

At $(0, 0)$,

$$H_f(0, 0) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

whose eigenvalues are 1 and -1 . Thus $(0, 0)$ is a saddle point.

At the other four critical points, $x^2 = y^2 = 1/2$, so $f_{xy} = 0$ and

$$f_{xx} = f_{yy} = -4xy e^{-1}.$$

Thus the Hessian is a scalar multiple of the identity, and both eigenvalues equal $-4xy/e$.

- If x and y have the same sign, then $xy = 1/2$ and both eigenvalues are $-2/e < 0$. Hence

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \quad \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$$

are local maxima.

- If x and y have opposite signs, then $xy = -1/2$ and both eigenvalues are $2/e > 0$. Hence

$$\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \quad \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

are local minima.

In fact, these are also the absolute maxima and minima, with values $1/(2e)$ and $-1/(2e)$, respectively. ■

15.3 Optimization by Reduction to Two Variables

A geometric optimization problem can often be reduced to finding extrema of a function of two variables. When minimizing a distance, it is usually simpler to minimize its square, since the square-root function is increasing on $[0, \infty)$.

Example 15.3. Consider the plane

$$2x + 6y + 3z = 6$$

in $\mathbb{R}^3$; see Figure 15.2. Find the point on the plane closest to the origin using the second derivative test.

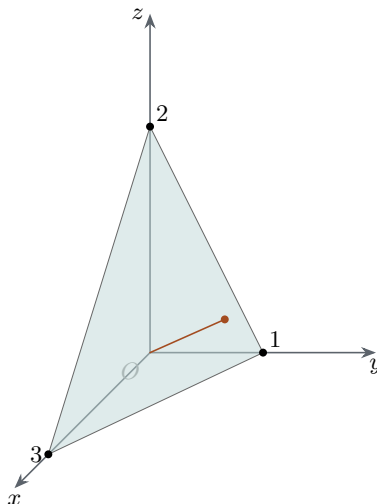


Figure 15.2: The plane $2x + 6y + 3z = 6$ and the perpendicular segment from the origin to the closest point.

Solution. Solve the plane equation for z :

$$z = 2 - \frac{2}{3}x - 2y.$$

The squared distance from the origin is therefore

$$F(x, y) = x^2 + y^2 + \left(2 - \frac{2}{3}x - 2y\right)^2.$$

Its first derivatives are

$$F_x = \frac{26}{9}x + \frac{8}{3}y - \frac{8}{3}, \quad F_y = \frac{8}{3}x + 10y - 8.$$

Solving $F_x = F_y = 0$ gives

$$(x, y) = \left(\frac{12}{49}, \frac{36}{49}\right), \quad z = \frac{18}{49}.$$

The Hessian is constant:

$$H_F = \begin{pmatrix} \frac{26}{9} & \frac{8}{3} \\ \frac{8}{3} & 10 \end{pmatrix}.$$

Its eigenvalues are

$$2 \quad \text{and} \quad \frac{98}{9},$$

so H_F is positive definite. Consequently, F is strictly convex and the critical point is the unique global minimum. Thus the closest point is

$$\boxed{\left(\frac{12}{49}, \frac{36}{49}, \frac{18}{49}\right)}.$$

The minimum squared distance is

$$\frac{36}{49},$$

so the minimum distance is

$$\boxed{\frac{6}{7}}.$$

■

15.4 Absolute Extrema on Closed and Bounded Domains

If f is continuous on a nonempty closed and bounded set $D \subset \mathbb{R}^2$, then the Extreme Value Theorem guarantees that f attains an absolute maximum and an absolute minimum on D .

For a region with a piecewise smooth boundary, check the interior and boundary, then compare the values:

1. **Interior:** Where f is differentiable, find critical points in the interior of D by solving

$$\nabla f = \vec{0}.$$

2. **Boundary:** Restrict f to each smooth boundary piece. Parametrization or substitution reduces the problem to one-variable optimization. Include endpoints, corners, and any feasible points where derivatives do not exist.
3. **Comparison:** Evaluate f at every interior and boundary candidate. The largest value is the absolute maximum, and the smallest is the absolute minimum.

The second derivative test can classify interior critical points, but absolute extrema are determined only after all candidate values have been compared.

Example 15.4. Find the absolute extrema of

$$f(x, y) = x^2 + y^2 - 2x - 4y + 6$$

on the triangular region with vertices $A(0, 0)$, $B(4, 0)$, and $C(0, 4)$.

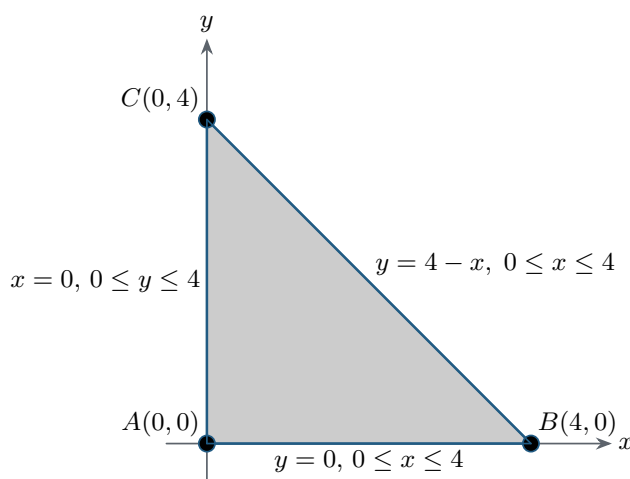


Figure 15.3: The triangular domain and its three boundary segments $\overline{AB}$, $\overline{AC}$, and $\overline{BC}$.

Solution. First consider the interior. Since

$$\nabla f = \langle 2x - 2, 2y - 4 \rangle,$$

the only critical point is

$$(x, y) = (1, 2).$$

It lies inside the triangle, and

$$f(1, 2) = 1.$$

Now check the three boundary segments shown in [Figure 15.3](#).

Segment AB : Here $y = 0$ and $0 \leq x \leq 4$. Then

$$f(x, 0) = x^2 - 2x + 6.$$

The derivative vanishes at $x = 1$, so the candidates and values are

$$f(0, 0) = 6, \quad f(1, 0) = 5, \quad f(4, 0) = 14.$$

Segment AC : Here $x = 0$ and $0 \leq y \leq 4$. Then

$$f(0, y) = y^2 - 4y + 6.$$

The derivative vanishes at $y = 2$, giving

$$f(0, 0) = 6, \quad f(0, 2) = 2, \quad f(0, 4) = 6.$$

Segment BC : Here $x + y = 4$, so $y = 4 - x$ with $0 \leq x \leq 4$. Thus

$$f(x, 4 - x) = 2x^2 - 6x + 6.$$

The derivative vanishes at $x = 3/2$, giving

$$f(0, 4) = 6, \quad f\left(\frac{3}{2}, \frac{5}{2}\right) = \frac{3}{2}, \quad f(4, 0) = 14.$$

Comparing all candidate values, we obtain

Absolute minimum: $f(1, 2) = 1$

and

Absolute maximum: $f(4, 0) = 14$.

■

Problem 15.5. Find the absolute extrema of

$$f(x, y) = 2x^2 + 3y^2$$

on the closed disk

$$x^2 + y^2 \leq 25.$$

Solution. The function is continuous on a closed and bounded domain, so absolute extrema exist.

In the interior,

$$\nabla f = \langle 4x, 6y \rangle = \vec{0}$$

gives the critical point $(0, 0)$, where

$$f(0, 0) = 0.$$

On the boundary circle, use the parameterization

$$x = 5 \cos t, \quad y = 5 \sin t.$$

Then

$$\begin{aligned} f(5 \cos t, 5 \sin t) &= 50 \cos^2 t + 75 \sin^2 t \\ &= 50 + 25 \sin^2 t. \end{aligned}$$

Therefore the boundary values range from 50 to 75:

- The boundary minimum 50 occurs when $\sin t = 0$, at $(\pm 5, 0)$.
- The boundary maximum 75 occurs when $\sin^2 t = 1$, at $(0, \pm 5)$.

Comparing these values with the interior value gives

Absolute minimum: 0 at $(0, 0)$

and

Absolute maximum: 75 at $(0, 5)$ and $(0, -5)$
--

■

16 Lagrange Multipliers

16.1 Goals

- Explain the geometric idea behind the method of Lagrange multipliers.
- Find constrained extrema of functions of two or more variables.
- Use the Lagrange multiplier to estimate how an optimal value changes when the constraint changes.
- Distinguish local constrained extrema from global constrained extrema.
- Apply the method to geometric, engineering, economic, and mathematical models.

16.2 Constrained Optimization

In constrained optimization, we seek the maximum or minimum of an objective function

$$f(x, y)$$

subject to a constraint

$$g(x, y) = k.$$

Unlike ordinary optimization, the variables are not free to move in every direction: they must remain on the constraint curve.

Problem 16.1 (Motivating problem). Minimize

$$f(x, y) = x^2 + y^2$$

subject to

$$xy = 1.$$

Geometrically, this asks for the points on the hyperbola $xy = 1$ that are closest to the origin.

16.3 Geometric Idea: Tangent Level Curves

A level curve of f is a curve of the form

$$f(x, y) = c.$$

For $f(x, y) = x^2 + y^2$, the level curves are circles centered at the origin. The constraint $g(x, y) = k$ is another curve in the plane.

At a regular constrained extremum, f has zero derivative in every direction tangent to the constraint. Thus ∇f is orthogonal to that tangent line, just as ∇g is. If $\nabla f \neq \mathbf{0}$, the objective's level curve is tangent to the constraint. In either case,

$$\nabla f(x, y) = \lambda \nabla g(x, y)$$

for some scalar λ , called a *Lagrange multiplier*.

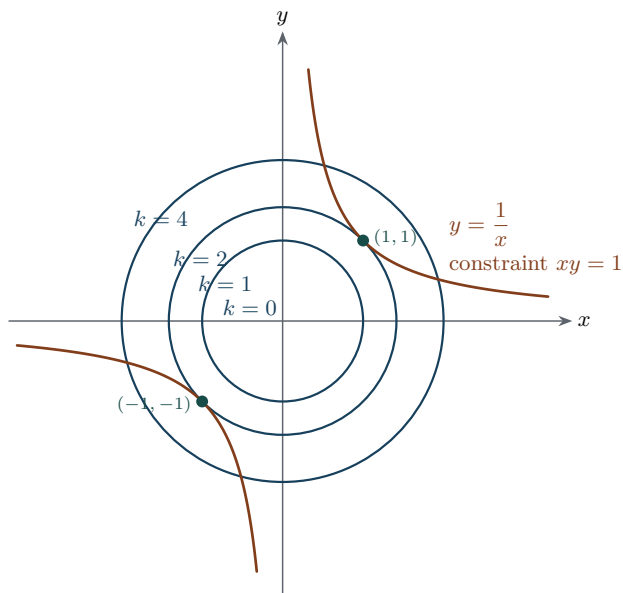


Figure 16.1: Level curves of $f(x, y) = x^2 + y^2$ and the constraint $xy = 1$. The minimum occurs where a level curve is tangent to the constraint.

16.4 The Method of Lagrange Multipliers

Assume f and g are continuously differentiable near the constraint. To optimize $f(x, y)$ subject to $g(x, y) = k$:

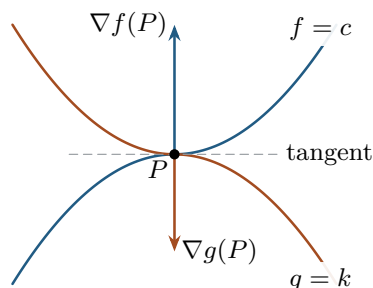


Figure 16.2: At a regular constrained extremum, ∇f and ∇g are parallel.

1. Compute ∇f and ∇g .
2. Solve the system

$$\begin{cases} \nabla f(x, y) = \lambda \nabla g(x, y), \\ g(x, y) = k. \end{cases}$$

3. Evaluate f at every candidate point.
4. Use the geometry of the constraint, compactness, or another classification method to decide which candidates are local or global extrema.

For a function of three variables subject to one constraint, the method is identical:

$$\nabla f(x, y, z) = \lambda \nabla g(x, y, z), \quad g(x, y, z) = k.$$

Important qualification. The condition $\nabla f = \lambda \nabla g$ applies at regular points where $\nabla g \neq \mathbf{0}$. Any feasible points where $\nabla g = \mathbf{0}$ must be checked separately. Also, solving the Lagrange equations produces candidates; it does not by itself prove whether they are maxima or minima.

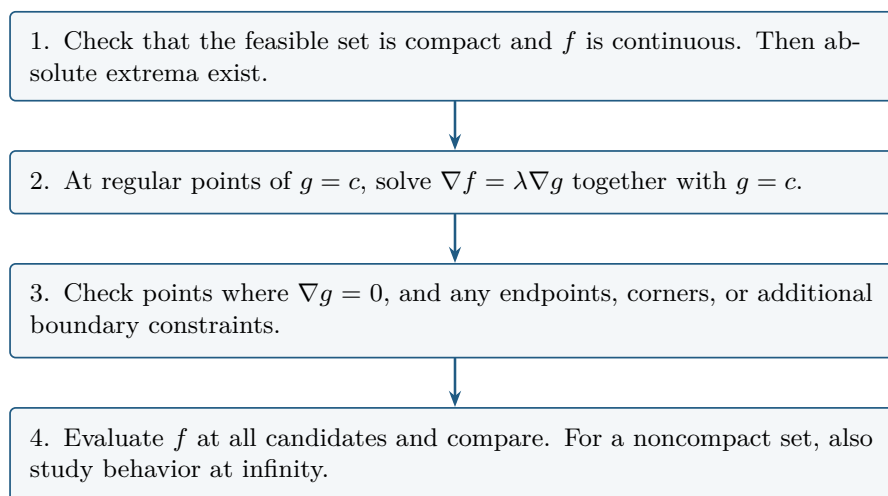


Figure 16.3: A workflow for solving constrained optimization problems with Lagrange multipliers.

16.5 Example 1: Closest Points on a Hyperbola

Return to the problem

$$f(x, y) = x^2 + y^2, \quad g(x, y) = xy = 1.$$

We have

$$\nabla f = \langle 2x, 2y \rangle, \quad \nabla g = \langle y, x \rangle.$$

The Lagrange system is

$$\begin{cases} 2x = \lambda y, \\ 2y = \lambda x, \\ xy = 1. \end{cases}$$

Since the constraint implies $x \neq 0$ and $y \neq 0$, the first two equations give

$$\frac{2x}{y} = \frac{2y}{x} \implies x^2 = y^2.$$

Thus $y = \pm x$. The constraint $xy = 1$ rules out $y = -x$, so $y = x$ and

$$x^2 = 1.$$

The candidate points are

$$(1, 1) \quad \text{and} \quad (-1, -1).$$

At both points,

$$f = x^2 + y^2 = 2.$$

Furthermore, on the constraint $xy = 1$,

$$x^2 + y^2 \geq 2|xy| = 2.$$

Therefore the absolute minimum value is

$$\boxed{f_{\min} = 2},$$

attained at $(1, 1)$ and $(-1, -1)$. There is no absolute maximum because $x^2 + y^2$ is unbounded on the hyperbola.

16.6 Example 2: Maximum Area with Fixed Perimeter

Problem 16.2. A rectangular poster has fixed perimeter 120 cm. Let its side lengths be x and y centimeters.

1. Find the dimensions that maximize the area $A(x, y) = xy$.
2. Find the maximum area.

Solution. The constraint is

$$2x + 2y = 120 \iff x + y = 60.$$

Let $g(x, y) = x + y$. Since

$$\nabla A = \langle y, x \rangle, \quad \nabla g = \langle 1, 1 \rangle,$$

the Lagrange equations are

$$\begin{cases} y = \lambda, \\ x = \lambda, \\ x + y = 60. \end{cases}$$

Hence $x = y$, and the constraint gives

$$x = y = 30.$$

Therefore

$$A_{\max} = A(30, 30) = 900 \text{ cm}^2.$$

To verify the classification, eliminate y using $y = 60 - x$:

$$A(x, 60 - x) = 60x - x^2.$$

This is a concave parabola, so its critical point at $x = 30$ is the global maximum. Thus, among rectangles with fixed perimeter, the square has the largest area. ■

16.7 The Multiplier as a Sensitivity Measure

Suppose a differentiable branch of optimizers $(x(k), y(k))$ satisfies $g(x(k), y(k)) = k$ and $\nabla f = \lambda(k)\nabla g$. For its objective value $F(k) = f(x(k), y(k))$, the chain rule gives

$$F'(k) = \nabla f \cdot \langle x'(k), y'(k) \rangle = \lambda(k) \frac{d}{dk} g(x(k), y(k)) = \lambda(k).$$

With the convention $\mathcal{L} = f - \lambda(g - k)$, the multiplier measures sensitivity to the constraint value. For a small change Δk , the optimal value changes by approximately $\lambda(k)\Delta k$.

Example 16.3 (Production cost and sensitivity). In a continuous production model, $x, y \geq 0$ measure the amounts of two items produced. The cost in dollars is

$$C(x, y) = 5x^2 + 2xy + 3y^2 + 800.$$

1. If the production quota is $x + y = 39$, find the minimum production cost.
2. Estimate the change in minimum cost if the quota is raised to 40 or lowered to 38.

Solution. Use the Lagrangian

$$\mathcal{L}(x, y, \lambda) = 5x^2 + 2xy + 3y^2 + 800 - \lambda(x + y - 39).$$

The first-order conditions are

$$10x + 2y = \lambda, \quad 2x + 6y = \lambda, \quad x + y = 39.$$

Subtracting the first two equations gives

$$8x - 4y = 0 \implies y = 2x.$$

The constraint then gives

$$x = 13, \quad y = 26.$$

Therefore

$$C(13, 26) = 5(13)^2 + 2(13)(26) + 3(26)^2 + 800 = \boxed{4349}.$$

The quadratic part of C is positive definite, so this point is the unique global constrained minimum.
At the optimum,

$$\lambda = 10(13) + 2(26) = 182.$$

Thus a one-unit increase in the quota changes the minimum cost by approximately \$182:

- Raising the quota from 39 to 40 increases the minimum cost by approximately \$182.
- Lowering the quota from 39 to 38 decreases the minimum cost by approximately \$182.

These are local linear estimates, not exact cost differences. ■

16.8 Example 3: A Three-Variable Quadratic Model

Problem 16.4. Minimize

$$Q(x, y, z) = x^2 + 2y^2 + 3z^2$$

subject to

$$x + y + z = 11.$$

Find both the minimizer and the minimum value.

Solution. Let

$$g(x, y, z) = x + y + z.$$

Since

$$\nabla Q = \langle 2x, 4y, 6z \rangle, \quad \nabla g = \langle 1, 1, 1 \rangle,$$

the Lagrange equations are

$$2x = \lambda, \quad 4y = \lambda, \quad 6z = \lambda, \quad x + y + z = 11.$$

Thus

$$x = \frac{\lambda}{2}, \quad y = \frac{\lambda}{4}, \quad z = \frac{\lambda}{6}.$$

Substituting into the constraint,

$$\frac{\lambda}{2} + \frac{\lambda}{4} + \frac{\lambda}{6} = 11 \implies \frac{11\lambda}{12} = 11 \implies \lambda = 12.$$

Therefore

$$\boxed{(x_0, y_0, z_0) = (6, 3, 2)}.$$

The minimum value is

$$Q_{\min} = 6^2 + 2(3^2) + 3(2^2) = 36 + 18 + 12 = \boxed{66}.$$

Because the Hessian of Q is

$$\text{diag}(2, 4, 6),$$

which is positive definite, Q is strictly convex. Hence this is the unique global minimum on the constraint plane. ■

16.9 Example 4: Minimum-Variance Portfolio

Problem 16.5. Three uncorrelated assets have variances 1, 2, and 3. Choose portfolio weights w_1, w_2, w_3 to minimize

$$V(w) = w_1^2 + 2w_2^2 + 3w_3^2$$

subject to the budget constraint

$$w_1 + w_2 + w_3 = 1.$$

Find the minimizing weights and the minimum variance.

Solution. The Lagrangian is

$$\mathcal{L} = w_1^2 + 2w_2^2 + 3w_3^2 - \lambda(w_1 + w_2 + w_3 - 1).$$

The first-order conditions are

$$2w_1 = \lambda, \quad 4w_2 = \lambda, \quad 6w_3 = \lambda, \quad w_1 + w_2 + w_3 = 1.$$

Therefore

$$w_1 = \frac{\lambda}{2}, \quad w_2 = \frac{\lambda}{4}, \quad w_3 = \frac{\lambda}{6}.$$

Using the constraint,

$$\frac{\lambda}{2} + \frac{\lambda}{4} + \frac{\lambda}{6} = 1 \implies \lambda = \frac{12}{11}.$$

Thus

$$w^* = \left(\frac{6}{11}, \frac{3}{11}, \frac{2}{11} \right).$$

The minimum variance is

$$V_{\min} = \left(\frac{6}{11} \right)^2 + 2 \left(\frac{3}{11} \right)^2 + 3 \left(\frac{2}{11} \right)^2 = \frac{66}{121} = \boxed{\frac{6}{11}}.$$

■

16.10 Advanced Example: Classification on a Noncompact Constraint

Problem 16.6. Let

$$f(x, y, z) = xyz.$$

Find and classify all constrained critical points subject to

$$x + 9y^2 + z^2 = 4.$$

Determine whether global extrema exist.

Solution. Let

$$g(x, y, z) = x + 9y^2 + z^2.$$

The Lagrange equations are

$$\begin{cases} yz = \lambda, \\ xz = 18\lambda y, \\ xy = 2\lambda z, \\ x + 9y^2 + z^2 = 4. \end{cases}$$

From the first equation, $\lambda = yz$. Substitution into the next two equations gives

$$xz = 18y^2z, \quad xy = 2yz^2.$$

Case 1: $y \neq 0$ and $z \neq 0$. Dividing by z and y , respectively, gives

$$x = 18y^2, \quad x = 2z^2.$$

Hence

$$z^2 = 9y^2, \quad z = \pm 3y.$$

The constraint becomes

$$18y^2 + 9y^2 + 9y^2 = 4,$$

so

$$y^2 = \frac{1}{9}, \quad z^2 = 1, \quad x = 2.$$

This produces four points:

$$\left(2, \frac{1}{3}, 1\right), \quad \left(2, \frac{1}{3}, -1\right), \quad \left(2, -\frac{1}{3}, 1\right), \quad \left(2, -\frac{1}{3}, -1\right).$$

Case 2: $z = 0$. Then $\lambda = 0$, and the third Lagrange equation gives $xy = 0$. Together with $x + 9y^2 = 4$, this yields

$$(4, 0, 0), \quad \left(0, \frac{2}{3}, 0\right), \quad \left(0, -\frac{2}{3}, 0\right).$$

Case 3: $y = 0$. Then $\lambda = 0$, and the second Lagrange equation gives $xz = 0$. Together with $x + z^2 = 4$, this yields

$$(4, 0, 0), \quad (0, 0, 2), \quad (0, 0, -2).$$

To classify the points, eliminate x using the constraint:

$$x = 4 - 9y^2 - z^2.$$

The restriction of f to the constraint surface is therefore

$$h(y, z) = (4 - 9y^2 - z^2)yz.$$

Its second derivatives are

$$h_{yy} = -54yz, \quad h_{zz} = -6yz, \quad h_{yz} = 4 - 27y^2 - 3z^2.$$

At $(y, z) = (\frac{1}{3}, 1)$ and $(-\frac{1}{3}, -1)$, the Hessian is

$$\begin{pmatrix} -18 & -2 \\ -2 & -2 \end{pmatrix},$$

which is negative definite. Therefore

$$\left(2, \frac{1}{3}, 1\right) \quad \text{and} \quad \left(2, -\frac{1}{3}, -1\right)$$

are local constrained maxima with

$$f = \frac{2}{3}.$$

At $(y, z) = (\frac{1}{3}, -1)$ and $(-\frac{1}{3}, 1)$, the Hessian is

$$\begin{pmatrix} 18 & -2 \\ -2 & 2 \end{pmatrix},$$

which is positive definite. Therefore

$$\left(2, \frac{1}{3}, -1\right) \quad \text{and} \quad \left(2, -\frac{1}{3}, 1\right)$$

are local constrained minima with

$$f = -\frac{2}{3}.$$

At the remaining critical points, the Hessian of h is indefinite, so

$$(4, 0, 0), \quad \left(0, \pm\frac{2}{3}, 0\right), \quad (0, 0, \pm 2)$$

are saddle points.

Finally, the constraint surface is not compact. For example, setting $y = t$ and $z = t$ gives

$$x = 4 - 10t^2, \quad f = (4 - 10t^2)t^2 \longrightarrow -\infty.$$

Setting $y = t$ and $z = -t$ gives

$$f = -(4 - 10t^2)t^2 \longrightarrow +\infty.$$

Therefore f is unbounded above and below on the constraint, so no global maximum or minimum exists.

Conclusion.

Local maxima:	$\left(2, \frac{1}{3}, 1\right), \left(2, -\frac{1}{3}, -1\right),$	$f = \frac{2}{3},$
Local minima:	$\left(2, \frac{1}{3}, -1\right), \left(2, -\frac{1}{3}, 1\right),$	$f = -\frac{2}{3},$
Saddle points:	$(4, 0, 0), \left(0, \pm\frac{2}{3}, 0\right), (0, 0, \pm 2),$	$f = 0,$
Global extrema:	none.	

■

16.11 Challenge: Hölder's Inequality

Problem 16.7 (Challenge—Hölder's inequality). Let $p > 1$ and $q > 1$ satisfy

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Let $y_1, \dots, y_n \geq 0$ be fixed constants. Find the maximum value of

$$f(x_1, \dots, x_n) = \sum_{i=1}^n x_i y_i$$

subject to

$$x_1^p + \cdots + x_n^p = A, \quad x_i \geq 0,$$

where $A > 0$ is constant.

Solution. The feasible set is nonempty and compact, so a maximum exists. Assume first that not all y_i are zero. At a maximizer, $x_i = 0$ whenever $y_i = 0$, since otherwise we could transfer that unused part of the constraint to a positive coefficient and increase f . Conversely, if $y_i > 0$ then $x_i > 0$: increasing a zero component by ε gains $y_i \varepsilon$, while the compensating decrease of a positive component costs only $O(\varepsilon^p)$. Since $p > 1$, this increases f for small ε .

We may therefore apply Lagrange multipliers on the positive components. Write

$$S = \sum_{i=1}^n y_i^q > 0.$$

Use the Lagrangian

$$\mathcal{L} = \sum_{i=1}^n x_i y_i - \lambda \left(\sum_{i=1}^n x_i^p - A \right).$$

For every index with $y_i > 0$, the first-order condition is

$$y_i = \lambda p x_i^{p-1}.$$

Thus the maximizing vector must have the form

$$x_i = c y_i^{1/(p-1)} = c y_i^{q-1}$$

for some constant $c > 0$. If $y_i = 0$, an optimizer may be chosen with $x_i = 0$.

Using the constraint and the identity $p(q-1) = q$,

$$A = \sum_{i=1}^n x_i^p = c^p \sum_{i=1}^n y_i^{p(q-1)} = c^p \sum_{i=1}^n y_i^q = c^p S.$$

Therefore

$$c = \left(\frac{A}{S} \right)^{1/p}.$$

The maximum objective value is

$$\begin{aligned} f_{\max} &= \sum_{i=1}^n x_i y_i = c \sum_{i=1}^n y_i^q \\ &= \left(\frac{A}{S} \right)^{1/p} S = A^{1/p} S^{1-1/p} = A^{1/p} S^{1/q}. \end{aligned}$$

Hence

$$f_{\max} = A^{1/p} \left(\sum_{i=1}^n y_i^q \right)^{1/q}.$$

Equality is attained when

$$x_i = \left(\frac{A}{\sum_{j=1}^n y_j^q} \right)^{1/p} y_i^{q-1}.$$

If all $y_i = 0$, then $f \equiv 0$, so the maximum value is 0 and every feasible vector is a maximizer.

This optimization result is precisely the equality case of Hölder's inequality. ■

17 Double Integrals over Rectangles

17.1 Goals

- Define $\iint_R f \, dA$ on a rectangle via Riemann sums; continuity $\Rightarrow$ integrable; interpret as volume/mass.
- Compute via Fubini as iterated integrals, swap order correctly.
- Use linearity, additivity, and comparison bounds; compute the average value $\frac{1}{|R|} \iint_R f \, dA$.

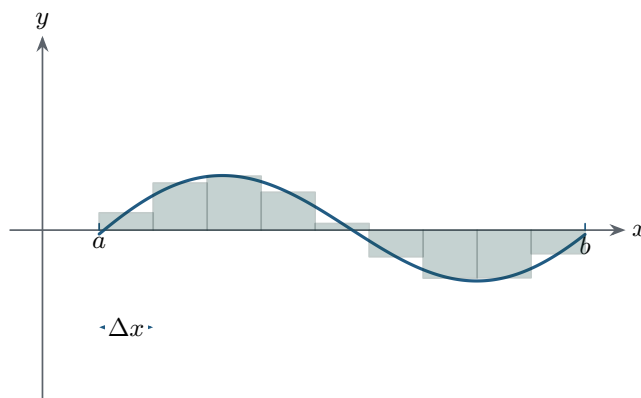


Figure 17.1: Approximating $\int_a^b f(x) \, dx$ by Riemann rectangles (midpoint samples). Each slab has width Δx and height $f(x_i^*)$; the signed-area limit is the integral.

Recall: if $y = f(x)$ is continuous on $[a, b]$, then

$$\int_a^b f(x) \, dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x, \quad \Delta x = \frac{b-a}{n},$$

is the (signed) area between the graph of f and the x -axis. Each term is the area of a thin rectangle of width Δx and height $f(x_i^*)$. See Figure 17.1 for a geometric depiction of the Riemann sum.

If $z = f(x, y)$ is continuous on the rectangle $R = [a, b] \times [c, d]$, we define the double integral similarly by partitioning R into an $m \times n$ grid—see Figure 17.2. Let

$$\Delta x = \frac{b-a}{m}, \quad \Delta y = \frac{d-c}{n}, \quad \Delta A = \Delta x \Delta y,$$

and choose a sample point (x_{ij}^*, y_{ij}^*) in each subrectangle. Then

$$\iint_R f(x, y) \, dA := \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*) \Delta A.$$

For $f \geq 0$, this integral equals the volume under the surface $z = f(x, y)$ above R ; in general it is a signed volume. On rectangles, when f is continuous,

$$\iint_R f \, dA = \int_a^b \left(\int_c^d f(x, y) \, dy \right) dx = \int_c^d \left(\int_a^b f(x, y) \, dx \right) dy.$$

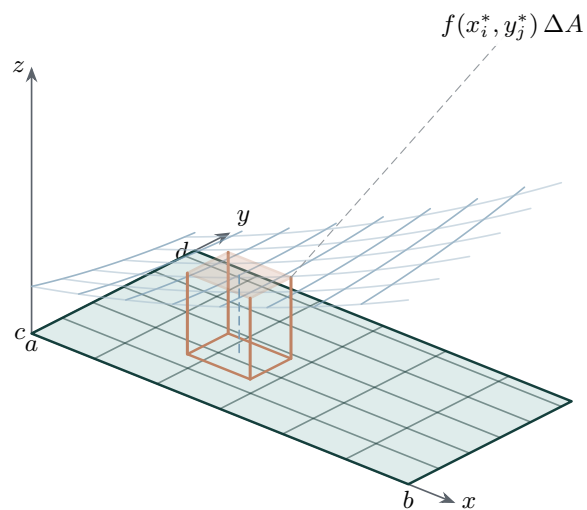


Figure 17.2: Volume under $z = f(x, y)$ over $R = [a, b] \times [c, d]$. One subrectangle contributes a prism of height $f(x_i^*, y_j^*)$ and base $\Delta A = \Delta x \Delta y$, i.e., a term $f(x_i^*, y_j^*) \Delta A$ in the double Riemann sum.

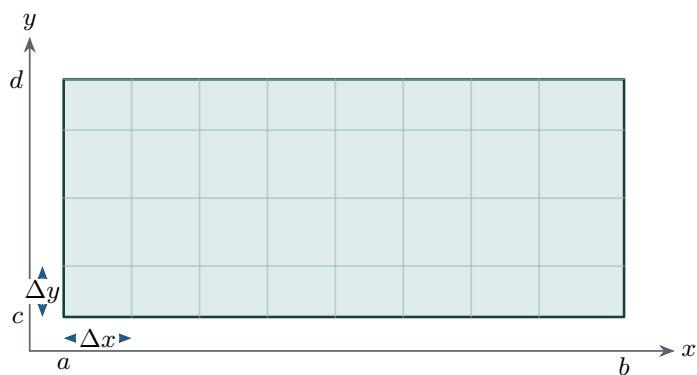


Figure 17.3: The base rectangle $R = [a, b] \times [c, d]$ with an $m \times n$ partition; Δx marked along x . Each cell has area $\Delta A = \Delta x \Delta y$.

Example 17.1. Evaluate the double integral $\int_0^2 \int_0^3 (6 - 2x - 3y) dx dy$.

Solution. Compute the inner integral first:

$$\int_0^3 (6 - 2x - 3y) dx = \left[6x - x^2 - 3yx \right]_0^3 = 18 - 9 - 9y = 9 - 9y.$$

Now integrate in y :

$$\int_0^2 (9 - 9y) dy = \left[9y - \frac{9}{2}y^2 \right]_0^2 = 18 - 18 = \boxed{0}.$$

Physical interpretation. If $z = 6 - 2x - 3y$ is a height (or a signed density), then over R the contributions above and below the xy -plane cancel exactly. The net signed volume (or total signed mass/charge) is zero. ■

Example 17.2. Evaluate $\int_0^1 \int_0^2 (x^2 + xy) e^y dx dy$.

Solution. Inner integral (treat y as constant):

$$\int_0^2 (x^2 + xy) dx = \left[\frac{x^3}{3} + \frac{yx^2}{2} \right]_0^2 = \frac{8}{3} + 2y.$$

Now integrate in y :

$$\int_0^1 \left(\frac{8}{3} + 2y \right) e^y dy = \left[\frac{8}{3}e^y + 2(y-1)e^y \right]_0^1 = \frac{8}{3}e - \frac{2}{3} = \boxed{\frac{8e-2}{3}}.$$

Example 17.3. Evaluate $\int_0^1 \int_0^2 y^5 x^2 e^{x^3 y^3} dx dy$.

Solution. For $y > 0$, let $u = y^3 x^3$, so $du = 3y^3 x^2 dx$. At $y = 0$, both the inner integral and its resulting formula are zero:

$$\int_0^2 y^5 x^2 e^{x^3 y^3} dx = y^5 \int_0^2 x^2 e^u dx = \frac{y^5}{3y^3} \int_0^{8y^3} e^u du = \frac{y^2}{3} (e^{8y^3} - 1).$$

Now integrate in y :

$$\int_0^1 \frac{y^2}{3} (e^{8y^3} - 1) dy = \frac{1}{3} \int_0^1 y^2 e^{8y^3} dy - \frac{1}{3} \int_0^1 y^2 dy.$$

For the first term, set $t = y^3$, so $dt = 3y^2 dy$:

$$\frac{1}{3} \int_0^1 y^2 e^{8y^3} dy = \frac{1}{9} \int_0^1 e^{8t} dt = \frac{e^8 - 1}{72}.$$

The second term is $\frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$. Therefore

$$\boxed{\int_0^1 \int_0^2 y^5 x^2 e^{x^3 y^3} dx dy = \frac{e^8 - 9}{72}}.$$

Remark 17.4. For a continuous function f on a bounded region R with $0 < |R| < \infty$ and piecewise smooth boundary, the *average value* of f is

$$f_{\text{avg}} = \frac{1}{|R|} \iint_R f(x, y) dA.$$

Geometrically, this is the height of a flat “average” surface whose volume over R equals the volume under $z = f(x, y)$. In physical settings, it corresponds to average density, temperature, or pressure over a region.

Example 17.5. Compute the *average value* of $f(x, y) = x^2 + y$ over the rectangle $R = [0, 2] \times [0, 1]$.

Solution. The area of R is $|R| = (2 - 0)(1 - 0) = 2$. By definition,

$$f_{\text{avg}} = \frac{1}{|R|} \iint_R f(x, y) dA = \frac{1}{2} \int_0^2 \int_0^1 (x^2 + y) dy dx.$$

Compute the inner integral:

$$\int_0^1 (x^2 + y) dy = x^2 y + \frac{1}{2} y^2 \Big|_0^1 = x^2 + \frac{1}{2}.$$

Now integrate in x :

$$\int_0^2 \left(x^2 + \frac{1}{2}\right) dx = \left[\frac{x^3}{3} + \frac{x}{2}\right]_0^2 = \frac{8}{3} + 1 = \frac{11}{3}.$$

Hence

$$f_{\text{avg}} = \frac{1}{2} \cdot \frac{11}{3} = \boxed{\frac{11}{6}}.$$

Interpretation: if the surface $z = f(x, y)$ represents height or density, then f_{avg} is the constant value that would produce the same total volume (or total mass) over the region R . ■

18 Double Integrals over General Regions

18.1 Goals

- Describe regions using vertical (Type I) or horizontal (Type II) slices.
- Choose an order with simple bounds and an easy inner integral.
- Change the order by sketching the region and rewriting the bounds.

18.2 From Rectangles to Curved Regions

Last time: for a rectangle $R = [a, b] \times [c, d]$ and continuous f ,

$$\iint_R f dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy.$$

Think of an iterated integral as *adding along a slice, then sweeping the slices across the region*.

In a rectangle, every vertical slice has the same endpoints c and d . For a curved region, those endpoints can depend on x . Horizontal slices work the same way, with endpoints that can depend on y .

18.3 Type I and Type II Regions

These names describe two ways to slice a region so that each slice is a single interval. In the formulas below, the boundary functions are continuous, the lower bound does not exceed the upper bound, and f is continuous on D .

Type I (vertical slices). Fix x : where does the slice enter and leave D ? If it runs from $y = g_1(x)$ to $y = g_2(x)$ as x ranges from a to b , then

$$D = \{(x, y) : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}.$$

For continuous f ,

$$\iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx.$$

The inner integral goes *bottom to top*; the outer integral sweeps *left to right*.

Type II (horizontal slices). Fix y : where does the slice enter and leave D ? If it runs from $x = h_1(y)$ to $x = h_2(y)$ as y ranges from c to d , then

$$D = \{(x, y) : c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}.$$

The integral is

$$\iint_D f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy.$$

The inner integral goes *left to right*; the outer integral sweeps *bottom to top*.

A region can be both types. Choose the direction that makes the bounds and integration easier. If a slice has gaps, split the region into pieces. If a boundary changes formula, split the integral where it changes.

Changing order: sketch the region, draw a slice in the new direction, and read its endpoints. For continuous f , both orders give the same integral when they describe the same region. *Rewrite the bounds; do not just exchange them.*

Example 18.1. Evaluate $\iint_D xy^2 dA$, where D lies between $y = x^2$ and $y = x$ for $0 \leq x \leq 1$.

Solution. Start by sketching the region; see Figure 18.1.

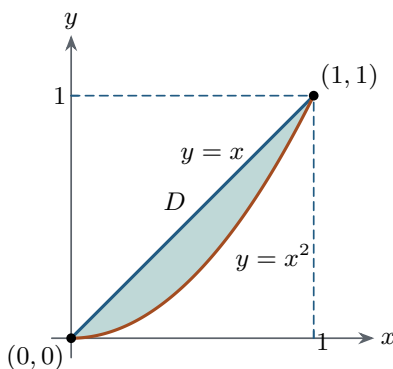


Figure 18.1: The region between $y = x^2$ and $y = x$.

Vertical slices: x ranges from 0 to 1, and each slice runs from $y = x^2$ to $y = x$. Thus D is Type I, and

$$\begin{aligned}\iint_D xy^2 dA &= \int_0^1 \int_{x^2}^x xy^2 dy dx \\ &= \int_0^1 \frac{x}{3} (x^3 - x^6) dx \\ &= \frac{1}{3} \left(\frac{1}{5} - \frac{1}{8} \right) = \boxed{\frac{1}{40}}.\end{aligned}$$

Horizontal slices: y ranges from 0 to 1. The slice starts at the line $x = y$ and ends at the parabola $x = \sqrt{y}$. Thus D is also Type II, and

$$\begin{aligned}\iint_D xy^2 dA &= \int_0^1 \int_y^{\sqrt{y}} xy^2 dx dy \\ &= \int_0^1 \frac{y^2}{2} (y - y^2) dy \\ &= \frac{1}{2} \left(\frac{1}{4} - \frac{1}{5} \right) = \boxed{\frac{1}{40}}.\end{aligned}$$

We swept the same region in two directions, so the answers agree. ■

Example 18.2. Evaluate by changing the order of integration:

$$\int_0^{27} \int_{\sqrt[3]{x}}^3 4e^{y^4} dy dx.$$

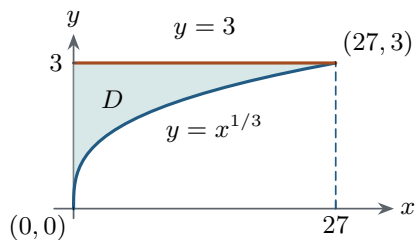


Figure 18.2: The region $0 \leq x \leq 27$, $\sqrt[3]{x} \leq y \leq 3$.

Solution. The inner integral has no elementary antiderivative in y . Integrating in x first is easy because $4e^{y^4}$ is constant along a horizontal slice.

The original bounds describe

$$D = \{(x, y) : 0 \leq x \leq 27, \sqrt[3]{x} \leq y \leq 3\}.$$

For horizontal slices, y ranges from 0 to 3, and x runs from 0 to y^3 . Therefore,

$$\begin{aligned}\int_0^{27} \int_{\sqrt[3]{x}}^3 4e^{y^4} dy dx &= \int_0^3 \int_0^{y^3} 4e^{y^4} dx dy \\ &= \int_0^3 4y^3 e^{y^4} dy \\ &= \left[e^{y^4} \right]_0^3 = \boxed{e^{81} - 1}.\end{aligned}$$

The horizontal slice has length y^3 , which supplies the factor needed for the substitution $u = y^4$. ■

Example 18.3. Consider the region R (see Figure 18.3) enclosed by

$$y = \sqrt{4 - x^2} \quad \text{and} \quad y = \frac{1}{2}(x + 2).$$

Set up $\iint_R f(x, y) dA$ in both orders:

(a) $dy dx$.

(b) $dx dy$. Where does the right boundary of a horizontal slice change?

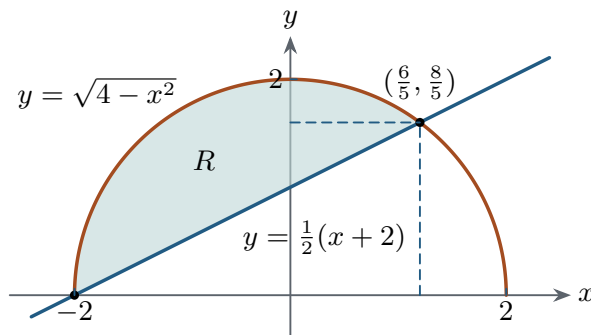


Figure 18.3: The region enclosed by the semicircle and the line.

Solution. The curves intersect at $(-2, 0)$ and $(6/5, 8/5)$.

(a) **Vertical slices.** Each slice runs from the line to the semicircle, with $-2 \leq x \leq 6/5$:

$$\iint_R f(x, y) dA = \int_{-2}^{6/5} \int_{(x+2)/2}^{\sqrt{4-x^2}} f(x, y) dy dx.$$

(b) **Horizontal slices.** The region reaches from $y = 0$ to $y = 2$. The left endpoint is always $x = -\sqrt{4 - y^2}$. Below $y = 8/5$, the right endpoint is on the line $x = 2y - 2$; above it, the right endpoint is on the semicircle $x = \sqrt{4 - y^2}$. Split where that boundary changes:

$$\begin{aligned} \iint_R f(x, y) dA &= \int_0^{8/5} \int_{-\sqrt{4-y^2}}^{2y-2} f(x, y) dx dy \\ &\quad + \int_{8/5}^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} f(x, y) dx dy. \end{aligned}$$

Here, vertical slices give the simpler setup: one integral instead of two. ■

19 Double Integrals in Polar Coordinates

19.1 Goals

- Describe points and regions using polar coordinates.
- Derive the polar area element from the area of an annular sector.

- Set up and evaluate double integrals over circular and radial regions.

Circles and rays are awkward to describe with rectangular coordinates. Polar coordinates describe a point by its distance from the origin and its direction, so these regions often have simpler bounds.

19.2 Polar coordinates

Definition 19.1

The **polar coordinate map** is

$$\Phi: [0, \infty) \times \mathbb{R} \longrightarrow \mathbb{R}^2, \quad \Phi(r, \theta) = (r \cos \theta, r \sin \theta).$$

A pair (r, θ) is a set of **polar coordinates** for $P = (x, y)$ if $P = \Phi(r, \theta)$. Here r is the distance from the origin and θ is the angle, measured in radians counterclockwise from the positive x -axis.

Thus

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r = \sqrt{x^2 + y^2}.$$

For $P \neq (0, 0)$, $r > 0$ is unique and θ is determined up to an integer multiple of 2π . Choosing $0 \leq \theta < 2\pi$ gives a unique pair for each nonzero point. The quadrant determines the angle, so $\tan \theta = y/x$ alone does not determine θ .

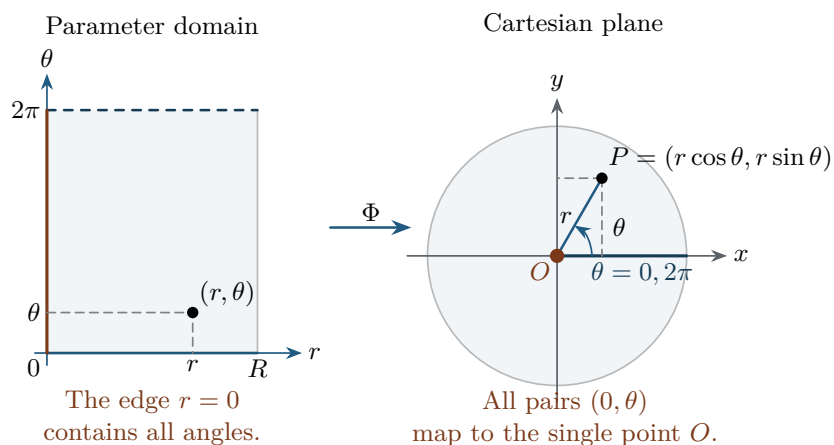


Figure 19.1: The polar map sends a parameter pair (r, θ) to a point at distance r and angle θ . The whole edge $r = 0$ maps to the origin. The edges $\theta = 0$ and $\theta = 2\pi$ map to the same ray.

What happens at the origin? Every pair $(0, \theta)$ represents $(0, 0)$, so the angle is undefined there. In the parameter domain, the whole edge $r = 0$ collapses to one point. Away from this edge, choosing a half-open angle interval of length 2π avoids repeated points. More precisely, restricting to $r > 0$ and $0 < \theta < 2\pi$ gives a smooth one-to-one change of variables onto the plane with the nonnegative x -axis removed. The omitted ray has area zero, so it does not affect an integral of a bounded continuous function. We may include boundary values in our integration limits. An unbounded integrand still requires a separate convergence check.

Reading polar bounds. For $r > 0$, $r = c$ describes a circle and $\theta = \alpha$ describes a ray. A region described by

$$\alpha \leq \theta \leq \beta, \quad g_1(\theta) \leq r \leq g_2(\theta)$$

is swept out by moving along each ray from the inner boundary to the outer boundary. Here $0 \leq g_1 \leq g_2$, and we choose the angles so that the interior is covered once. For a full turn, the endpoint ray may be repeated.

19.3 Deriving the area element

A rectangle in the (r, θ) -plane becomes an **annular sector** in the (x, y) -plane. Consider

$$r \leq \rho \leq r + \Delta r, \quad \theta \leq \varphi \leq \theta + \Delta \theta,$$

where $\Delta r > 0$ and $0 < \Delta \theta \leq 2\pi$.

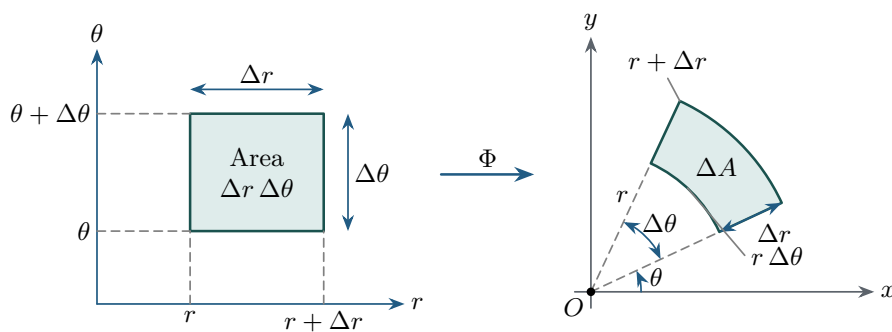


Figure 19.2: A parameter rectangle of area $\Delta r \Delta \theta$ becomes an annular sector. Its radial width is Δr and its inner arc has length $r \Delta \theta$.

The area of a sector of radius s and angle $\Delta \theta$ is $\frac{1}{2}s^2\Delta \theta$. Subtract the inner sector from the outer one:

$$\begin{aligned} \Delta A &= \frac{1}{2}((r + \Delta r)^2 - r^2)\Delta \theta \\ &= \left(r + \frac{\Delta r}{2}\right) \Delta r \Delta \theta. \end{aligned}$$

Thus the exact area is the radial width times the arc length at the midpoint radius. Dividing by the parameter area gives

$$\frac{\Delta A}{\Delta r \Delta \theta} = r + \frac{\Delta r}{2} \longrightarrow r.$$

The term $\frac{1}{2}(\Delta r)^2\Delta \theta$ is of higher order, so the *differential* area element is

$$\boxed{dA = r dr d\theta}.$$

This is a limiting statement, not an exact formula for a finite sector. It also agrees with the Jacobian calculation:

$$\det D\Phi = \det \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix} = r.$$

Theorem 19.1 (Integration in polar coordinates)

Let D be a bounded region swept out once in its interior by

$$\alpha \leq \theta \leq \beta, \quad 0 \leq g_1(\theta) \leq r \leq g_2(\theta),$$

where the boundary functions are piecewise continuous. If f is continuous on the closure of D , then

$$\iint_D f(x, y) dA = \int_{\alpha}^{\beta} \int_{g_1(\theta)}^{g_2(\theta)} f(r \cos \theta, r \sin \theta) r dr d\theta.$$

Regions requiring several radial intervals are split into pieces.

Example 19.1. Evaluate

$$\iint_D (x^2 + y^2) dA,$$

where $a > 0$ and D is the disk $x^2 + y^2 \leq a^2$ (See Figure 19.3).

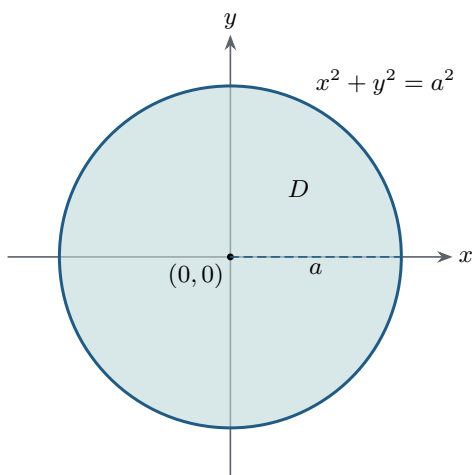


Figure 19.3: The disk $x^2 + y^2 \leq a^2$.

Solution. In polar coordinates, $x^2 + y^2 = r^2$, and the region D is described by $0 \leq r \leq a$, $0 \leq \theta \leq 2\pi$. Thus

$$\iint_D (x^2 + y^2) dA = \int_0^{2\pi} \int_0^a r^2 \cdot r dr d\theta = \int_0^{2\pi} \int_0^a r^3 dr d\theta.$$

Compute:

$$= \int_0^{2\pi} \left[\frac{r^4}{4} \right]_0^a d\theta = \int_0^{2\pi} \frac{a^4}{4} d\theta = \frac{a^4}{4} \cdot 2\pi = \frac{\pi a^4}{2}.$$

■

Example 19.2. D is the region between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 2$. Find

$$\iint_D e^{x^2+y^2} dA.$$

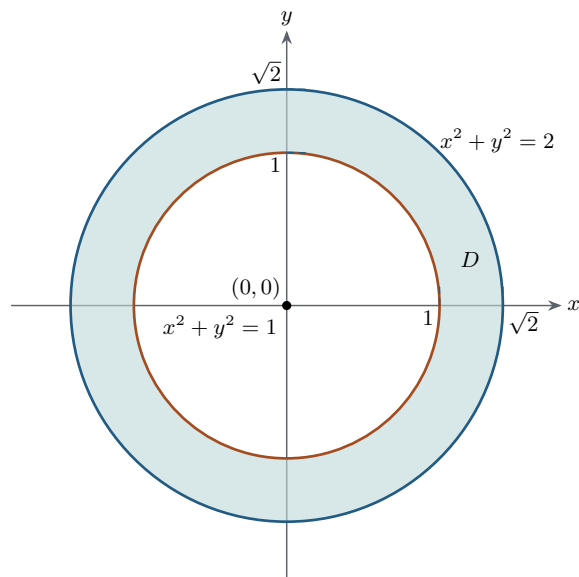


Figure 19.4: The annulus $1 \leq x^2 + y^2 \leq 2$.

Solution. Switch to polar coordinates: $x = r \cos \theta$, $y = r \sin \theta$, so $x^2 + y^2 = r^2$ and $dA = r dr d\theta$. The region between $x^2 + y^2 = 1$ and $x^2 + y^2 = 2$ is

$$1 \leq r \leq \sqrt{2}, \quad 0 \leq \theta \leq 2\pi.$$

Hence

$$\begin{aligned} \iint_D e^{x^2+y^2} dA &= \int_0^{2\pi} \int_1^{\sqrt{2}} e^{r^2} r dr d\theta \\ &= \int_0^{2\pi} \left[\frac{1}{2} e^{r^2} \right]_{r=1}^{\sqrt{2}} d\theta \\ &= \int_0^{2\pi} \frac{1}{2} (e^2 - e) d\theta \\ &= \pi (e^2 - e). \end{aligned}$$

$$\boxed{\pi(e^2 - e)}$$

■

Example 19.3. Evaluate the double integral

$$\iint_D \frac{1}{\sqrt{x^2 + y^2}} dA$$

where D is enclosed by the cardioid $r = 1 + \cos \theta$ (see Figure 19.5). Interpret the integral as an improper integral at the origin.

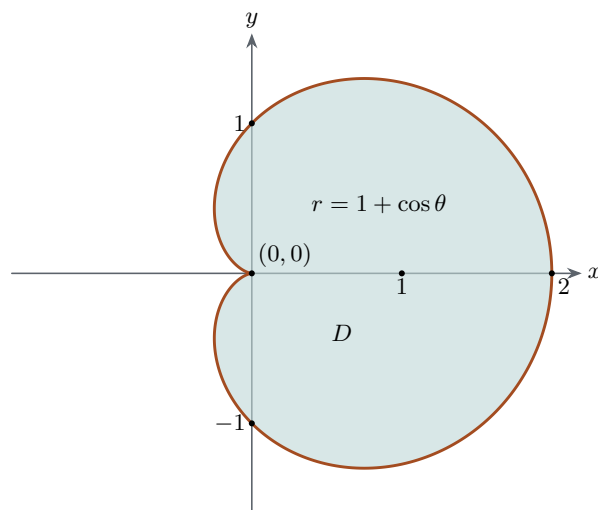


Figure 19.5: The cardioid $r = 1 + \cos \theta$ and its enclosed region.

Solution. Switch to polar: $x = r \cos \theta$, $y = r \sin \theta$, $dA = r dr d\theta$. Then

$$\begin{aligned} \iint_D \frac{1}{\sqrt{x^2 + y^2}} dA &= \int_0^{2\pi} \int_0^{1+\cos \theta} \frac{1}{r} r dr d\theta \\ &= \int_0^{2\pi} \int_0^{1+\cos \theta} 1 dr d\theta \\ &= \int_0^{2\pi} (1 + \cos \theta) d\theta \\ &= 2\pi. \end{aligned}$$

To justify the singularity at $r = 0$, integrate first over $D \cap \{r \geq \varepsilon\}$ and let $\varepsilon \downarrow 0$. The removed contribution tends to zero, since $(1/r)dA = dr d\theta$ for $r > 0$. ■

Example 19.4. Consider the solid bounded above by $z = \sqrt{9 - x^2 - y^2}$ and below by $z = \sqrt{x^2 + y^2}$. Use a double integral with polar coordinates to calculate its volume.

Solution. The region is between the upper hemisphere $z = \sqrt{9 - x^2 - y^2}$ and the cone $z = \sqrt{x^2 + y^2}$. In polar (r, θ) , they intersect when

$$\sqrt{9 - r^2} = r \implies r^2 = \frac{9}{2} \implies r = \frac{3}{\sqrt{2}}.$$

The volume is

$$V = \iint_D (z_{\text{top}} - z_{\text{bot}}) dA = \int_0^{2\pi} \int_0^{3/\sqrt{2}} (\sqrt{9 - r^2} - r) r dr d\theta.$$

Compute the radial integral using $u = 9 - r^2$, $du = -2r dr$:

$$\int_0^{3/\sqrt{2}} (r\sqrt{9 - r^2} - r^2) dr = \left[-\frac{1}{3}(9 - r^2)^{3/2} - \frac{r^3}{3} \right]_0^{3/\sqrt{2}} = 9 - \frac{9\sqrt{2}}{2}.$$

Thus

$$V = 2\pi \left(9 - \frac{9\sqrt{2}}{2} \right) = 9\pi (2 - \sqrt{2}).$$

■

20 Triple Integrals

20.1 Goals

- Set up and evaluate triple integrals over regions in $\mathbb{R}^3$ using Cartesian coordinates.

Intuitively, we have the following description:

1. $\iint_R f(x, y) dA$ is the accumulation of $f(x, y)$ over the region R . See Figure 20.1.

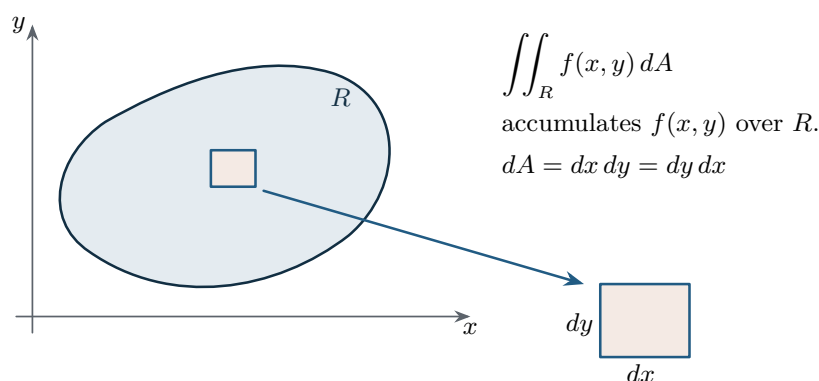


Figure 20.1: Area element dA .

2. $\iiint_D f(x, y, z) dV$ is the accumulation of $f(x, y, z)$ over the volume D . See Figure 20.2 for a geometric representation.

Definition 20.1 (Triple integral)

Let $E \subset \mathbb{R}^3$ be a bounded, closed solid with piecewise smooth boundary, and let f be continuous on E . Partition a containing box into small rectangular cells. For every cell meeting E , choose a sample point $p_{ijk}^* \in E$ and let ΔV_{ijk} be the volume of its intersection with E . The triple integral is the limit

$$\iiint_E f dV = \lim_{\text{mesh} \rightarrow 0} \sum_{i,j,k} f(p_{ijk}^*) \Delta V_{ijk}.$$

For $f = 1$ this gives volume, and for a nonnegative density it gives mass.

Under these hypotheses, Fubini's theorem allows any of the six integration orders. The bounds

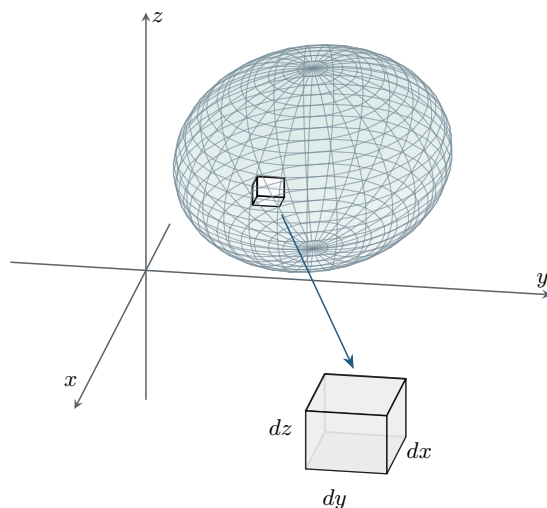


Figure 20.2: Volume element dV .

must describe the same solid, sometimes in several pieces.

$$\begin{aligned}
 dV &= dx \, dy \, dz \\
 dV &= dy \, dx \, dz \\
 dV &= dx \, dz \, dy \\
 dV &= dy \, dz \, dx \\
 dV &= dz \, dx \, dy \\
 dV &= dz \, dy \, dx
 \end{aligned}$$

Pick the order that makes bounds clean.

Example 20.1. Find the volume under $2x + 3y + z = 6$, above the xy -plane, in the first octant (i.e., $x, y, z \geq 0$).

Solution. The plane gives $z = 6 - 2x - 3y$. In the xy -plane the shadow is the triangle

$$x \geq 0, \quad y \geq 0, \quad 2x + 3y \leq 6,$$

with intercepts $(3, 0)$ and $(0, 2)$.

Choose the order $dz \, dy \, dx$:

$$V = \int_{x=0}^3 \int_{y=0}^{(6-2x)/3} \int_{z=0}^{6-2x-3y} 1 \, dz \, dy \, dx.$$

First,

$$\int_0^{6-2x-3y} 1 \, dz = 6 - 2x - 3y,$$

then

$$\int_0^{(6-2x)/3} (6 - 2x - 3y) \, dy = \frac{(6 - 2x)^2}{6},$$

so

$$V = \int_0^3 \frac{(6 - 2x)^2}{6} \, dx = \frac{1}{6} \int_0^3 (36 - 24x + 4x^2) \, dx = 6.$$

$$\boxed{V = 6}$$

Check: The region is a tetrahedron with intercepts $(3, 0, 0)$, $(0, 2, 0)$, and $(0, 0, 6)$. Its volume is

$$\frac{abc}{6} = \frac{3 \cdot 2 \cdot 6}{6} = 6.$$

■

Example 20.2. Find the volume under the paraboloid $z = 4 - x^2 - y^2$, above the xy -plane, in the first octant ($x, y, z \geq 0$).

Solution. We have z running from 0 up to $4 - x^2 - y^2$. In the xy -plane, the shadow is the quarter disk $x \geq 0$, $y \geq 0$, $x^2 + y^2 \leq 4$, i.e.

$$0 \leq x \leq 2, \quad 0 \leq y \leq \sqrt{4 - x^2}.$$

Choose the order $dz \, dy \, dx$:

$$V = \int_{x=0}^2 \int_{y=0}^{\sqrt{4-x^2}} \int_{z=0}^{4-x^2-y^2} 1 \, dz \, dy \, dx.$$

Inner integral:

$$\int_0^{4-x^2-y^2} 1 \, dz = 4 - x^2 - y^2.$$

Next in y :

$$\int_0^{\sqrt{4-x^2}} (4 - x^2 - y^2) \, dy = (4 - x^2) \sqrt{4 - x^2} - \frac{(4 - x^2)^{3/2}}{3} = \frac{2}{3} (4 - x^2)^{3/2}.$$

So

$$V = \int_0^2 \frac{2}{3} (4 - x^2)^{3/2} \, dx.$$

Use $x = 2 \sin \theta$ ($0 \leq \theta \leq \pi/2$):

$$4 - x^2 = 4 \cos^2 \theta, \quad (4 - x^2)^{3/2} = 8 \cos^3 \theta, \quad dx = 2 \cos \theta \, d\theta.$$

Thus

$$V = \int_0^{\pi/2} \frac{2}{3} \cdot 8 \cos^3 \theta \cdot 2 \cos \theta \, d\theta = \frac{32}{3} \int_0^{\pi/2} \cos^4 \theta \, d\theta.$$

Recall $\int_0^{\pi/2} \cos^4 \theta \, d\theta = \frac{3\pi}{16}$. Hence

$$V = \frac{32}{3} \cdot \frac{3\pi}{16} = 2\pi.$$

$$\boxed{V = 2\pi}$$

Remark: Cylindrical coordinates would simplify this problem, but the Cartesian setup works well too. ■

21 Triple Integrals in Cylindrical Coordinates

21.1 Goals

- Recognize when cylindrical coordinates simplify a 3D region/integrand.
- Set up triple integrals in cylindrical coordinates.
- Convert surfaces/regions between Cartesian and cylindrical forms to determine bounds.
- Evaluate applications (volume, mass, center of mass, moments) using this coordinate system.

21.2 Cylindrical coordinates

A cylinder is described by a distance from an axis, an angle around that axis, and a height. We use polar coordinates in the xy -plane and keep the Cartesian coordinate z .

Definition 21.1

The **cylindrical coordinate map** is

$$\Psi: [0, \infty) \times \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R}^3, \quad \Psi(r, \theta, z) = (r \cos \theta, r \sin \theta, z).$$

A triple (r, θ, z) is a set of **cylindrical coordinates** for P if $P = \Psi(r, \theta, z)$. Here r is the distance to the z -axis, θ is the polar angle of the projection onto the xy -plane, and z is the height.

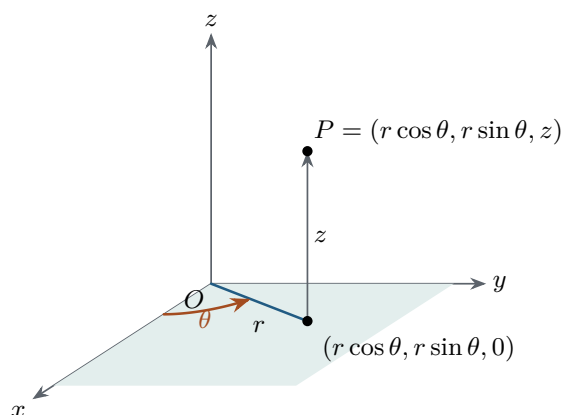


Figure 21.1: Project P onto the xy -plane to read r and θ . The vertical displacement from the projection to P is z .

Transformations and uniqueness.

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r = \sqrt{x^2 + y^2}.$$

Off the z -axis, $r > 0$ and choosing $0 \leq \theta < 2\pi$ gives a unique triple. Choose θ from the quadrant of (x, y) , rather than from $\tan \theta = y/x$ alone. On the z -axis, every $(0, \theta, z)$ represents the same point $(0, 0, z)$, so θ is undefined there. The entire parameter face $r = 0$ collapses onto that axis.

For $r > 0$ and $0 < \theta < 2\pi$, Ψ is a smooth one-to-one change of variables onto space with the half-plane $y = 0$, $x \geq 0$ removed. This seam and the axis have volume zero, so boundary values may be included in limits for continuous integrands. Singular integrands require a separate convergence check.

Reading coordinate surfaces. For $a > 0$, $r = a$ is a circular cylinder. The surface $z = c$ is a horizontal plane. For fixed $\theta = \alpha$, the points with $r \geq 0$ form a vertical half-plane bounded by the z -axis.

Example 21.1. Convert $(3, 3, 2)$ to cylindrical coordinates.

Solution. The point lies in the first quadrant of the xy -plane, so

$$r = \sqrt{3^2 + 3^2} = 3\sqrt{2}, \quad \theta = \frac{\pi}{4}, \quad z = 2.$$

Thus $(r, \theta, z) = (3\sqrt{2}, \pi/4, 2)$. ■

21.3 Deriving the volume element

A small parameter box becomes an annular sector of height Δz . Using the polar area calculation, its *exact* volume is

$$\Delta V = \left(r + \frac{\Delta r}{2}\right) \Delta r \Delta \theta \Delta z.$$

Therefore $\Delta V/(\Delta r \Delta \theta \Delta z) \rightarrow r$ as the box shrinks. Equivalently,

$$\det D\Psi = \det \begin{pmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = r, \quad \boxed{dV = r \, dr \, d\theta \, dz}.$$

21.4 Setting up triple integrals

Let E be a bounded solid described, once in its interior, by

$$\alpha \leq \theta \leq \beta, \quad 0 \leq r_1(\theta) \leq r \leq r_2(\theta), \quad z_1(r, \theta) \leq z \leq z_2(r, \theta),$$

with piecewise continuous boundary functions. For f continuous on the closure of E ,

$$\iiint_E f(x, y, z) \, dV = \int_{\alpha}^{\beta} \int_{r_1(\theta)}^{r_2(\theta)} \int_{z_1(r, \theta)}^{z_2(r, \theta)} f(r \cos \theta, r \sin \theta, z) \, r \, dz \, dr \, d\theta.$$

Rewrite the integrand and *every boundary* in the new variables. Split the solid if one set of bounds does not describe it.

Example 21.2. Find the volume inside the cylinder $x^2 + y^2 \leq 4$ that lies *below* the plane $z = 3$ and above the xy -plane.

Solution. In cylindrical coordinates (r, θ, z) with

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z,$$

the cylinder is $r \leq 2$, the xy -plane is $z = 0$, and the plane is $z = 3$. Thus the region is described by

$$0 \leq r \leq 2, \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq z \leq 3.$$

The Jacobian is r . Therefore,

$$V = \int_{\theta=0}^{2\pi} \int_{r=0}^2 \int_{z=0}^3 r \, dz \, dr \, d\theta.$$

Integrate from the inside out:

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^2 3r \, dr \, d\theta \\ &= \int_0^{2\pi} \left[\frac{3}{2} r^2 \right]_0^2 d\theta = \int_0^{2\pi} 6 \, d\theta = 12\pi. \end{aligned}$$

Therefore

$$\boxed{V = 12\pi}.$$

Check (geometry): The base is a disk of radius 2, with area $\pi(2)^2 = 4\pi$. The region is a right circular cylinder of height 3. So $V = \text{area of base} \times \text{height} = 4\pi \cdot 3 = 12\pi$, matching the integral. ■

Example 21.3. Find the volume of the solid $z^2 \leq x^2 + y^2 \leq 4$: it lies outside the two axial regions of the double cone $z^2 = x^2 + y^2$ and inside the cylinder $x^2 + y^2 = 4$.

Solution. We use cylindrical coordinates (r, θ, z) , where $x^2 + y^2 = r^2$. The cylinder is given by $r = 2$, and the cone is $z^2 = r^2$, or $z = \pm r$. The region is bounded laterally by the cylinder ($r = 2$) and vertically by the cone ($z = \pm r$).

The volume element in cylindrical coordinates is $r \, dz \, dr \, d\theta$. The limits are:

$$0 \leq \theta \leq 2\pi, \quad 0 \leq r \leq 2, \quad -r \leq z \leq r.$$

Thus, the volume is

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^2 \int_{-r}^r r \, dz \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^2 [rz]_{z=-r}^{z=r} dr \, d\theta \\ &= \int_0^{2\pi} \int_0^2 2r^2 dr \, d\theta \\ &= \int_0^{2\pi} \left[\frac{2r^3}{3} \right]_0^2 d\theta \\ &= \int_0^{2\pi} \frac{16}{3} d\theta \\ &= \frac{16}{3}(2\pi) = \frac{32\pi}{3}. \end{aligned}$$

Therefore, the volume of the specified solid is

$$\boxed{\frac{32\pi}{3}}.$$

■

22 Triple Integrals in Spherical Coordinates

22.1 Goals

- Recognize when spherical coordinates simplify a 3D region/integrand.
- Set up triple integrals in spherical coordinates.
- Convert surfaces/regions between Cartesian and spherical forms to determine bounds.
- Evaluate applications (volume, mass, center of mass, moments) using this coordinate system.

22.2 Spherical coordinates

For a sphere or a cone with vertex at the origin, it is natural to choose a direction first and then move along the corresponding ray. Two angles specify the direction, and a distance locates the point.

Definition 22.1

The **spherical coordinate map** is

$$\Sigma: [0, \infty) \times \mathbb{R} \times [0, \pi] \longrightarrow \mathbb{R}^3,$$

$$\Sigma(\rho, \theta, \phi) = (\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi).$$

A triple (ρ, θ, ϕ) is a set of **spherical coordinates** for P if $P = \Sigma(\rho, \theta, \phi)$. Here ρ is the distance from the origin, θ is the azimuthal angle in the xy -plane, and ϕ is the angle measured down from the positive z -axis. Both angles are measured in radians.

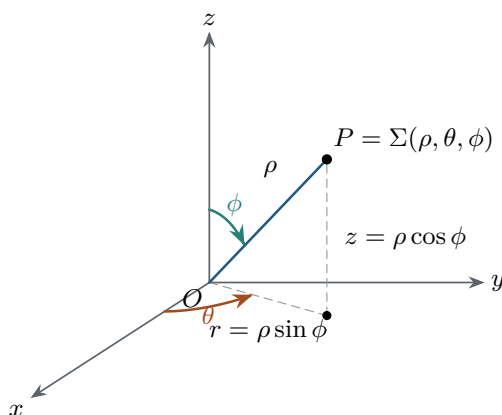


Figure 22.1: The distance from O to P is ρ . The right triangle gives $r = \rho \sin \phi$ and $z = \rho \cos \phi$. The angle θ is read in the xy -plane.

Transformations and uniqueness. The cylindrical radius is $r = \rho \sin \phi$, so

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi.$$

For $P \neq O$,

$$\rho = \sqrt{x^2 + y^2 + z^2}, \quad \phi = \arccos\left(\frac{z}{\rho}\right) \in [0, \pi].$$

Off the z -axis, the quadrant of (x, y) determines a unique $\theta \in [0, 2\pi)$. On the positive z -axis $\phi = 0$, and on the negative z -axis $\phi = \pi$, while θ is arbitrary on both. At the origin both angles are undefined. Thus the parameter face $\rho = 0$ collapses to O , and the faces $\phi = 0, \pi$ collapse to the two halves of the z -axis.

Restricting to $\rho > 0$, $0 < \theta < 2\pi$, and $0 < \phi < \pi$ gives a smooth one-to-one change of variables onto space with the half-plane $y = 0$, $x \geq 0$ removed. The seam, axis, and origin have volume zero. As before, an unbounded integrand needs a separate convergence check.

Reading coordinate surfaces. For $a > 0$, $\rho = a$ is a sphere. For fixed θ , we obtain a vertical half-plane. For $0 < \phi < \pi/2$, ϕ constant gives the upper nappe of a circular cone. The value $\phi = \pi/2$ gives the xy -plane, and $\pi/2 < \phi < \pi$ gives a lower nappe. For example, $z = \sqrt{x^2 + y^2}$ becomes $\phi = \pi/4$ away from O .

Example 22.1. Convert $(2, 2, \sqrt{8})$ to spherical coordinates.

Solution. The point lies in the first octant, so

$$\rho = \sqrt{4 + 4 + 8} = 4, \quad \theta = \frac{\pi}{4}, \quad \phi = \arccos\left(\frac{\sqrt{8}}{4}\right) = \frac{\pi}{4}.$$

Thus $(\rho, \theta, \phi) = (4, \pi/4, \pi/4)$. ■

22.3 Deriving the volume element

At a point with $\rho > 0$ and $0 < \phi < \pi$, consider a small parameter box. Its three coordinate directions are perpendicular:

- Changing ρ moves along a ray, giving length $d\rho$.
- Changing ϕ follows a meridian of radius ρ , giving length $\rho d\phi$.
- Changing θ follows a latitude circle of radius $\rho \sin \phi$, giving length $\rho \sin \phi d\theta$.

The image is a curved wedge, whose volume to first order is the product of these lengths. Hence

$$dV = \rho^2 \sin \phi d\rho d\phi d\theta.$$

For an analytical check, differentiate the coordinate map:

$$\begin{aligned} \Sigma_\rho &= (\sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi), \\ \Sigma_\theta &= (-\rho \sin \phi \sin \theta, \rho \sin \phi \cos \theta, 0), \\ \Sigma_\phi &= (\rho \cos \phi \cos \theta, \rho \cos \phi \sin \theta, -\rho \sin \phi). \end{aligned}$$

Their pairwise dot products are zero and their lengths are 1, $\rho \sin \phi$, and ρ , respectively. Thus

$$|\det D\Sigma| = |\Sigma_\rho \cdot (\Sigma_\theta \times \Sigma_\phi)| = \rho^2 \sin \phi.$$

The determinant is negative for the parameter order (ρ, θ, ϕ) , but volume uses its absolute value.

22.4 Setting up triple integrals

Let E be a bounded solid swept out once in its interior by

$$\alpha \leq \theta \leq \beta, \quad 0 \leq \phi_1(\theta) \leq \phi \leq \phi_2(\theta) \leq \pi,$$

$$0 \leq \rho_1(\phi, \theta) \leq \rho \leq \rho_2(\phi, \theta),$$

with piecewise continuous boundary functions. If f is continuous on the closure of E ,

$$\begin{aligned} \iiint_E f(x, y, z) dV &= \int_{\alpha}^{\beta} \int_{\phi_1(\theta)}^{\phi_2(\theta)} \int_{\rho_1(\phi, \theta)}^{\rho_2(\phi, \theta)} \\ &\quad f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi d\rho d\phi d\theta. \end{aligned}$$

Convert the boundary surfaces before choosing limits. A solid may need to be split if its radial or angular bounds change.

Example 22.2. Find the volume inside the sphere $x^2 + y^2 + z^2 \leq 9$ that lies *above* the cone $z = \sqrt{x^2 + y^2}$.

Solution. In spherical coordinates (ρ, θ, ϕ) with

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi,$$

the sphere is $\rho \leq 3$ and the cone $z = \sqrt{x^2 + y^2}$ is $\phi = \pi/4$. “Above the cone” means $0 \leq \phi \leq \pi/4$. The Jacobian is $\rho^2 \sin \phi$. Thus,

$$V = \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\pi/4} \int_{\rho=0}^3 \rho^2 \sin \phi d\rho d\phi d\theta.$$

Integrate in the order ρ, ϕ , then θ :

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^{\pi/4} 9 \sin \phi d\phi d\theta \\ &= \int_0^{2\pi} 9 \left(1 - \frac{\sqrt{2}}{2}\right) d\theta \\ &= 18\pi \left(1 - \frac{\sqrt{2}}{2}\right). \end{aligned}$$

Therefore

$$V = 9 \cdot \left(1 - \frac{\sqrt{2}}{2}\right) \cdot 2\pi = 18\pi \left(1 - \frac{\sqrt{2}}{2}\right) = 18\pi - 9\sqrt{2}\pi.$$

$$V = 18\pi \left(1 - \frac{\sqrt{2}}{2}\right)$$

Check (geometry argument): The cone with half-angle $\alpha = \pi/4$ subtends solid angle $\Omega = 2\pi(1 - \cos \alpha)$. The fraction of the sphere it captures is $\Omega/(4\pi) = (1 - \cos \alpha)/2$. Hence

$$V = \frac{1 - \cos(\pi/4)}{2} \cdot \frac{4}{3}\pi(3)^3 = \frac{1 - \frac{\sqrt{2}}{2}}{2} \cdot 36\pi = 18\pi \left(1 - \frac{\sqrt{2}}{2}\right),$$

matching the integral. ■

Example 22.3. Find the volume of the solid that lies *outside* the sphere $\rho = 1$ and *inside* the surface $\rho = 2 \cos \phi$.

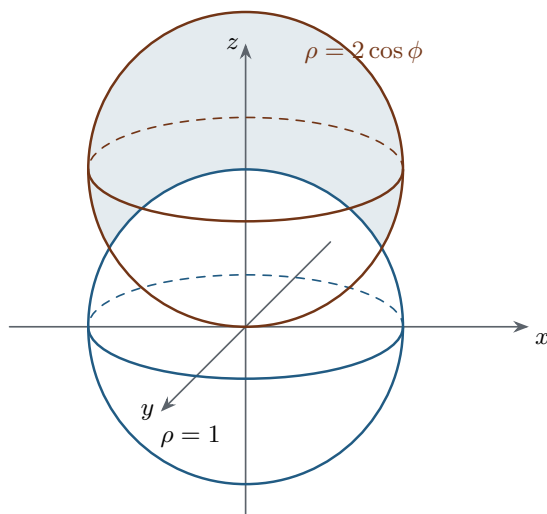


Figure 22.2: Two spheres meeting along $z = 1/2$. The desired solid is inside the sphere centered at $(0, 0, 1)$ and outside the sphere centered at O . The shading is schematic. The next figure gives the exact central cross-section.

Solution. As shown in Figure 22.2, $\rho = 2 \cos \phi$ is the sphere $x^2 + y^2 + z^2 = 2z$ (radius 1, center $(0, 0, 1)$). The intersection with $\rho = 1$ occurs when $1 = 2 \cos \phi \Rightarrow \phi = \pi/3$, i.e., along $z = \frac{1}{2}$ (see also Figure 22.3). Thus, in spherical coordinates

$$0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \frac{\pi}{3}, \quad 1 \leq \rho \leq 2 \cos \phi.$$

The volume is

$$V = \int_0^{2\pi} \int_0^{\pi/3} \int_1^{2 \cos \phi} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = 2\pi \int_0^{\pi/3} \frac{(2 \cos \phi)^3 - 1}{3} \sin \phi \, d\phi.$$

With $u = \cos \phi$,

$$V = \frac{2\pi}{3} \int_{1/2}^1 (8u^3 - 1) \, du = \frac{2\pi}{3} \left[2u^4 - u \right]_{1/2}^1 = \boxed{\frac{11\pi}{12}}.$$

■

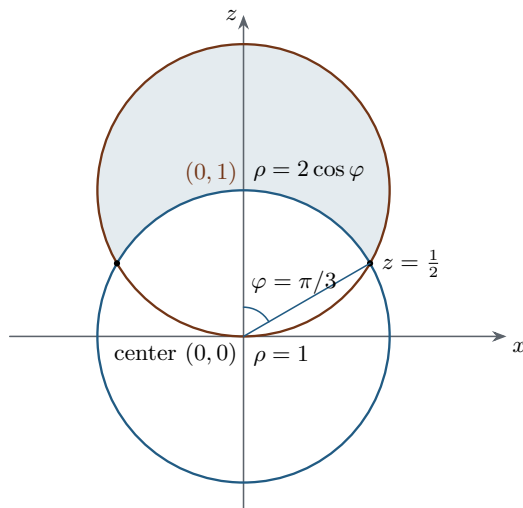


Figure 22.3: Cross-section in the xz -plane ($y = 0$): the part of the disk centered at $(0, 1)$ lying outside the disk centered at $(0, 0)$, both of radius 1. The boundary circles meet at $(x, z) = (\pm\sqrt{3}/2, 1/2)$. This is a cross-section, rather than the projection of the solid.

23 Integrals in Mass Calculations

23.1 Goals

- Compute mass from a (possibly variable) density.
- Find center of mass.
- Compute moments of inertia about coordinate axes.

Let $D \subset \mathbb{R}^2$ be a bounded lamina and $E \subset \mathbb{R}^3$ a bounded solid, each with piecewise smooth boundary. Assume the density ρ is continuous and nonnegative. For a center of mass, assume also that the total mass is positive. The same formulas apply to integrable densities when all the required integrals are finite.

23.2 Mass

$$M_{\text{lamina}} = \iint_D \rho(x, y) dA, \quad M_{\text{solid}} = \iiint_E \rho(x, y, z) dV.$$

We want to calculate the center of mass of objects in $\mathbb{R}^2$ and $\mathbb{R}^3$. For that, we first need to calculate the moments. Roughly speaking,

$$\text{first moment} = (\text{mass}) \cdot (\text{signed coordinate}).$$

For a point mass m at (x, y) (see [Figure 23.1](#)), the first moment about the y -axis is

$$M_y = mx.$$

Thus

$$M_y = mx.$$

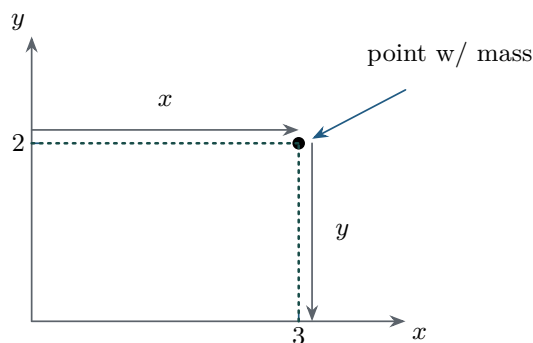


Figure 23.1: The x and y moments of a point (a, b) in $\mathbb{R}^2$.

Similarly, the moment with respect to the x -axis is given by

$$M_x = my.$$

Example 23.1. Consider the point $(2, 3)$ with mass $m = 5$. The moment with respect to the x -axis is

$$M_x = m \cdot y = 5 \cdot 3 = 15,$$

while the moment with respect to the y -axis is

$$M_y = m \cdot x = 5 \cdot 2 = 10.$$

The total mass is $M = 5$, so provided $0 < M < \infty$, the center of mass is

$$(\bar{x}, \bar{y}) = \left(\frac{M_y}{M}, \frac{M_x}{M} \right) = \left(\frac{10}{5}, \frac{15}{5} \right) = (2, 3).$$

We use the same point-mass idea to define moments of a region with variable density $\rho(x, y)$. Consider the figure:

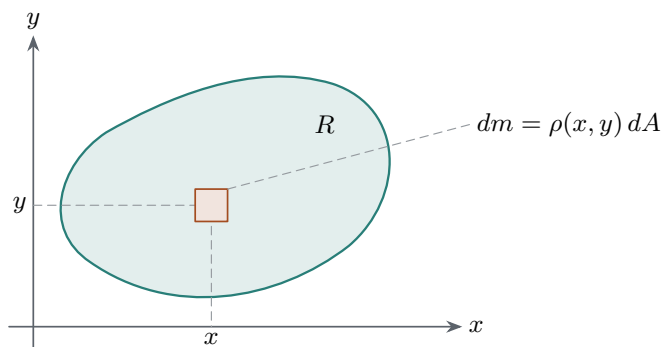


Figure 23.2: Breaking R into small pieces to build moments.

Break the region R into small rectangular pieces. A piece at (x, y) has mass $dm = \rho(x, y) dA$. Its signed coordinates are x relative to the y -axis and y relative to the x -axis. Its first-moment contributions are

$$dM_y = \rho(x, y) x dA, \quad dM_x = \rho(x, y) y dA.$$

Accumulating over R (i.e., integrating) gives the moments of the whole region:

$$M_y = \iint_R x \rho(x, y) dA, \quad M_x = \iint_R y \rho(x, y) dA.$$

With total mass

$$M = \iint_R \rho(x, y) dA,$$

provided $0 < M < \infty$, the center of mass is

$$\bar{x} = \frac{M_y}{M}, \quad \bar{y} = \frac{M_x}{M}.$$

Example 23.2. Find M , M_x , M_y , and the center of mass for the lamina occupying the rectangle $R = \{(x, y) \mid 0 \leq x \leq 3, 0 \leq y \leq 2\}$ with density $\rho(x, y) = x + y$.

Solution. Mass:

$$M = \iint_R (x + y) dA = \int_0^3 \int_0^2 (x + y) dy dx = \int_0^3 (2x + 2) dx = [x^2 + 2x]_0^3 = 15.$$

Moment about y :

$$\begin{aligned} M_y &= \iint_R x(x + y) dA = \int_0^3 \int_0^2 (x^2 + xy) dy dx = \int_0^3 (2x^2 + 2x) dx \\ &= \left[\frac{2}{3}x^3 + x^2 \right]_0^3 \\ &= 27. \end{aligned}$$

Moment about x :

$$\begin{aligned} M_x &= \iint_R y(x + y) dA = \int_0^3 \int_0^2 (xy + y^2) dy dx = \int_0^3 \left(2x + \frac{8}{3} \right) dx \\ &= \left[x^2 + \frac{8}{3}x \right]_0^3 \\ &= 17. \end{aligned}$$

Center of mass:

$$\bar{x} = \frac{M_y}{M} = \frac{27}{15} = \frac{9}{5}, \quad \bar{y} = \frac{M_x}{M} = \frac{17}{15}$$

■

23.3 3D version.

For a solid region $E \subset \mathbb{R}^3$ with density $\rho(x, y, z) \geq 0$, a small volume element has mass

$$dM = \rho(x, y, z) dV,$$

and its signed coordinates relative to the yz -, xz -, and xy -planes are x , y , and z , respectively. Its first-moment contributions are

$$dM_{yz} = x \rho dV, \quad dM_{xz} = y \rho dV, \quad dM_{xy} = z \rho dV.$$

Integrating over E :

$$M_{yz} = \iiint_E x \rho \, dV, \quad M_{xz} = \iiint_E y \rho \, dV, \quad M_{xy} = \iiint_E z \rho \, dV,$$

with total mass

$$M = \iiint_E \rho \, dV.$$

Provided $0 < M < \infty$, the center of mass $(\bar{x}, \bar{y}, \bar{z})$ is

$$\bar{x} = \frac{M_{yz}}{M}, \quad \bar{y} = \frac{M_{xz}}{M}, \quad \bar{z} = \frac{M_{xy}}{M}.$$

Quick sanity checks. If ρ is constant (uniform solid), $(\bar{x}, \bar{y}, \bar{z})$ is the geometric centroid. For a rectangular box $0 \leq x \leq a$, $0 \leq y \leq b$, $0 \leq z \leq c$ with constant ρ ,

$$(\bar{x}, \bar{y}, \bar{z}) = \left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2} \right).$$

Example 23.3. Find the center of mass of the tetrahedron in the first octant bounded by $2x + y + 3z = 6$ and the coordinate planes, with density $\rho(x, y, z) = 1 + z$.

Solution. At height z , the horizontal slice is the triangle $2x + y \leq 6 - 3z$ in the first quadrant. Its area is

$$A(z) = \frac{(6 - 3z)^2}{4} = \frac{9}{4}(2 - z)^2, \quad 0 \leq z \leq 2.$$

Since the density depends only on z , the mass is

$$M = \frac{9}{4} \int_0^2 (1 + z)(2 - z)^2 \, dz = 9.$$

The centroid of the slice has coordinates $x = (2 - z)/2$ and $y = 2 - z$. Thus

$$M_{yz} = \frac{9}{8} \int_0^2 (1 + z)(2 - z)^3 \, dz = \frac{63}{10},$$

$$M_{xz} = \frac{9}{4} \int_0^2 (1 + z)(2 - z)^3 \, dz = \frac{63}{5}, \quad M_{xy} = \frac{9}{4} \int_0^2 z(1 + z)(2 - z)^2 \, dz = \frac{27}{5}.$$

Therefore,

$$(\bar{x}, \bar{y}, \bar{z}) = \left(\frac{7}{10}, \frac{7}{5}, \frac{3}{5} \right).$$

The geometric centroid of the uniform tetrahedron has height $1/2$. The density $1 + z$ gives more weight to the upper slices, so the height $3/5$ is consistent with the model. ■

23.4 Moments of Inertia

A moment of inertia uses the *square* of the distance to an axis, rather than a signed coordinate. For a lamina,

$$I_x = \iint_D y^2 \rho \, dA, \quad I_y = \iint_D x^2 \rho \, dA, \quad I_z = \iint_D (x^2 + y^2) \rho \, dA = I_x + I_y.$$

For a solid,

$$I_x = \iiint_E (y^2 + z^2) \rho \, dV, \quad I_y = \iiint_E (x^2 + z^2) \rho \, dV, \quad I_z = \iiint_E (x^2 + y^2) \rho \, dV.$$

24 Vector Fields

24.1 Goals

- Define vector field functions.
- Sketch arrow plots for simple vector fields.
- Define conservative vector fields and potential functions.

Definition 24.1 (Vector Field)

A *vector field* on a subset $D \subseteq \mathbb{R}^n$ is a function

$$\mathbf{F} : D \rightarrow \mathbb{R}^n$$

that assigns to each point $p \in D$ a vector $\mathbf{F}(p) \in \mathbb{R}^n$, regarded as a vector based at p . The input is a location, and the output describes a direction and size there. An arrow plot places the tail of $\mathbf{F}(p)$ at p ; it is a visualization of the field, rather than its graph as a function.

- In $\mathbb{R}^2$, a vector field can be written as

$$\mathbf{F}(x, y) = \langle P(x, y), Q(x, y) \rangle,$$

where P and Q are scalar functions on $D \subseteq \mathbb{R}^2$.

- In $\mathbb{R}^3$, a vector field has the form

$$\mathbf{F}(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle.$$

$$\mathbf{F}(x, y) = \langle -y, x \rangle$$

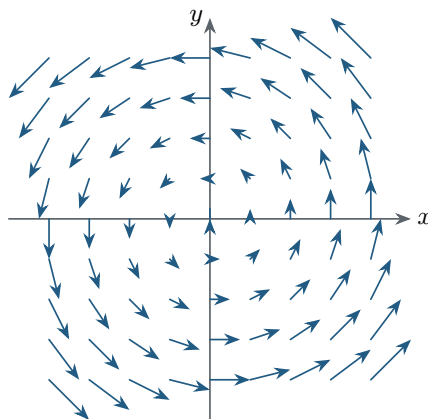
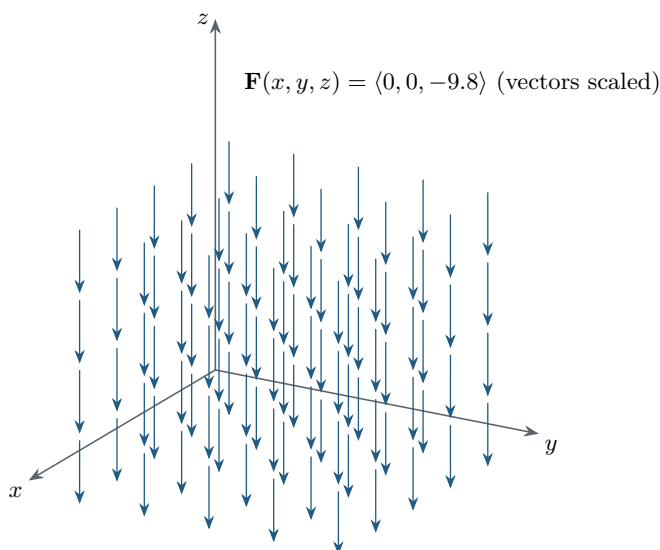


Figure 24.1: Counterclockwise rotating vector field about the origin.

Example 24.1.

1. $\mathbf{F}(x, y) = \langle -y, x \rangle$ is a planar vector field representing a counterclockwise rotation about the origin. See Figure 24.1.
2. $\mathbf{F}(x, y, z) = \langle 0, 0, -9.8 \rangle$ models the constant gravitational field near Earth's surface. See Figure 24.2.

Figure 24.2: Constant gravitational field *near* Earth's surface.**Definition 24.2 (Potential Function)**

Let $\mathbf{F}$ be a vector field on an open set $D \subseteq \mathbb{R}^n$. A differentiable scalar function $\varphi : D \rightarrow \mathbb{R}$ is called a *potential function* for $\mathbf{F}$ if

$$\nabla \varphi = \mathbf{F} \quad \text{on } D.$$

That is, the gradient of φ reproduces the vector field.

Definition 24.3 (Conservative Vector Field)

A vector field $\mathbf{F}$ on $D \subseteq \mathbb{R}^n$ is called *conservative* if there exists a potential function φ with

$$\mathbf{F} = \nabla \varphi.$$

Equivalently, $\mathbf{F}$ is conservative if it is the gradient of some scalar field.

Example 24.2.

1. $\mathbf{F}(x, y) = \langle 2x, 2y \rangle$ is conservative, since $\varphi(x, y) = x^2 + y^2$ satisfies $\nabla \varphi = \mathbf{F}$.
2. $\mathbf{F}(x, y) = \langle -y, x \rangle$ is *not* conservative, since no potential function exists (the field has nonzero curl).

$$\mathbf{F}(x, y) = \left\langle \frac{2x}{x^2+y^2}, \frac{2y}{x^2+y^2} \right\rangle$$

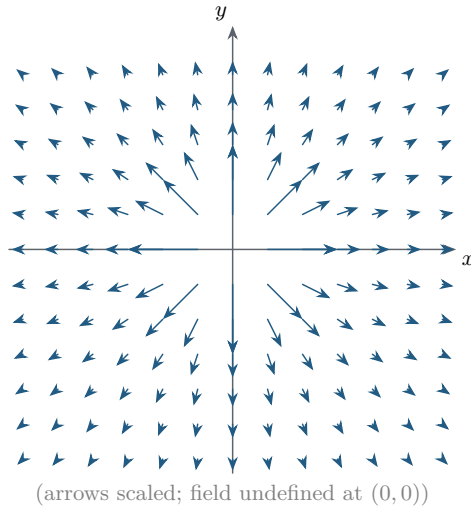


Figure 24.3: Example of a repulsive force vector field.

Example 24.3. Consider the vector field

$$\mathbf{F}(x, y) = \left\langle \frac{2x}{x^2 + y^2}, \frac{2y}{x^2 + y^2} \right\rangle,$$

see Figure 24.3. Notice that it looks like the gradient of

$$\varphi(x, y) = \ln(x^2 + y^2).$$

Indeed, compute

$$\nabla\varphi = \left\langle \frac{\partial}{\partial x} \ln(x^2 + y^2), \frac{\partial}{\partial y} \ln(x^2 + y^2) \right\rangle = \left\langle \frac{2x}{x^2 + y^2}, \frac{2y}{x^2 + y^2} \right\rangle.$$

Thus $\varphi(x, y) = \ln(x^2 + y^2)$ is a potential function for $\mathbf{F}$ on $\mathbb{R}^2 \setminus \{(0, 0)\}$. Both are undefined at the origin.

Remark 24.4 (Conservative Fields and Curl). If $\mathbf{F} = \nabla\varphi$ and φ has continuous second partial derivatives, then $\nabla \times \mathbf{F} = \mathbf{0}$. This follows from equality of mixed partials. For a planar field $\langle P, Q \rangle$, the corresponding condition is $Q_x - P_y = 0$.

The converse requires a condition on the domain. A curl-free field can have nonzero circulation around a hole. This does not occur for a field with a globally defined potential, even if its domain has holes.

25 Line Integrals of Functions (I)

25.1 Goals

- Define scalar line integrals over piecewise smooth curves.
- Interpret line integrals as “accumulated density along a curve.”

- Learn how to compute line integrals using parametrizations.

25.2 Motivation

In earlier lessons we integrated over regions (area) and solids (volume). However, what if we want to integrate a function *along a curve*? For example:

- Finding the mass of a thin wire lying along a curve C with variable linear density $\rho(x, y)$.
- Computing work done by a force field along a trajectory.

Definition 25.1

Let C be a piecewise smooth curve in $\mathbb{R}^2$ or $\mathbb{R}^3$, traced once by a piecewise C^1 parametrization $\mathbf{r}(t)$, $a \leq t \leq b$, with nonzero velocity on the interior of each smooth piece. If f is continuous on C , the *line integral of f along C with respect to arc length* is

$$\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| \, dt.$$

Note: The main difficulty in problems with line integrals is finding a “nice” parametrization for the curve C . Consider the following example.

Example 25.1. Let C be the directed line segment from $(0, 0)$ to $(5, -2)$. Evaluate

$$\int_C x e^{y^2} \, ds.$$

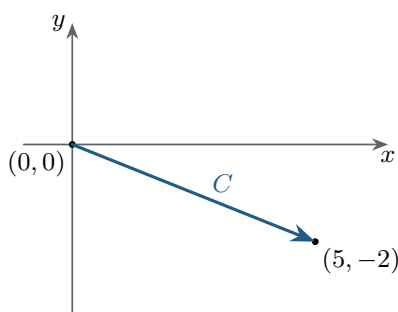


Figure 25.1: The straight line segment from $(0, 0)$ to $(5, -2)$.

Solution. Parametrize the segment (see Figure 25.1) by $\mathbf{r}(t) = \langle 5t, -2t \rangle$, $0 \leq t \leq 1$. Then

$$x = 5t, \quad y = -2t, \quad \|\mathbf{r}'(t)\| = \sqrt{5^2 + (-2)^2} = \sqrt{29}.$$

Hence

$$\int_C x e^{y^2} \, ds = \int_0^1 (5t) e^{4t^2} \sqrt{29} \, dt = 5\sqrt{29} \int_0^1 t e^{4t^2} \, dt.$$

Let $u = 4t^2$ so $du = 8t \, dt$:

$$5\sqrt{29} \int_0^1 t e^{4t^2} \, dt = \frac{5\sqrt{29}}{8} \int_0^4 e^u \, du = \frac{5\sqrt{29}}{8} (e^4 - 1).$$

■

Note

This parametrization is not unique. For example, we could have used

$$\mathbf{r}(t) = \langle 10t, -4t \rangle, \quad 0 \leq t \leq \frac{1}{2}.$$

Then

$$\mathbf{r}'(t) = \langle 10, -4 \rangle \implies \|\mathbf{r}'(t)\| = \sqrt{100 + 16} = 2\sqrt{29}.$$

Hence

$$\int_C x e^{y^2} ds = \int_0^{\frac{1}{2}} (10t) e^{(16t^2)} (2\sqrt{29}) dt = 20\sqrt{29} \int_0^{\frac{1}{2}} t e^{16t^2} dt.$$

Let $u = 16t^2$ so $du = 32t dt$. When $t : 0 \rightarrow \frac{1}{2}$, $u : 0 \rightarrow 4$. Then

$$20\sqrt{29} \int_0^{\frac{1}{2}} t e^{16t^2} dt = 20\sqrt{29} \cdot \frac{1}{32} \int_0^4 e^u du = \frac{5\sqrt{29}}{8} (e^4 - 1),$$

the same value as before. A regular change of parametrization does not change a scalar line integral, even if it reverses orientation. Retracing an arc counts its contribution again.

Example 25.2. Let C be the closed curve composed of two paths:

$$C : \begin{cases} (0, 0) \rightarrow (2, 4) \text{ along } y = x^2, \\ (2, 4) \rightarrow (0, 0) \text{ along } y = 2x. \end{cases}$$

Evaluate

$$\oint_C x ds.$$

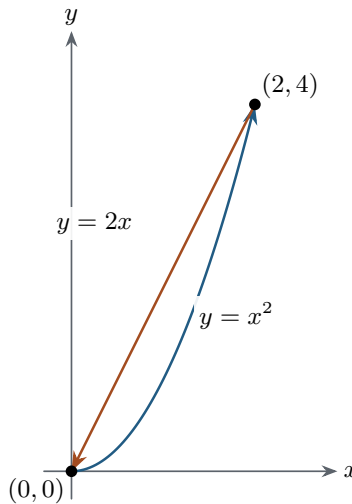


Figure 25.2: Closed path between two points composed by a parabola and a straight line segment.

Solution. Split over $C_1 : y = x^2, (0, 0) \rightarrow (2, 4)$ and $C_2 : y = 2x, (2, 4) \rightarrow (0, 0)$.

On C_1 set $\mathbf{r}(x) = \langle x, x^2 \rangle$, $0 \leq x \leq 2$. Then $ds = \sqrt{1 + (2x)^2} dx = \sqrt{1 + 4x^2} dx$, so

$$\int_{C_1} x ds = \int_0^2 x \sqrt{1 + 4x^2} dx.$$

Let $u = 1 + 4x^2$, $du = 8x dx$. Then

$$\int_0^2 x \sqrt{1 + 4x^2} dx = \frac{1}{8} \int_{u=1}^{17} u^{1/2} du = \frac{1}{8} \cdot \frac{2}{3} \left[u^{3/2} \right]_1^{17} = \frac{1}{12} (17^{3/2} - 1).$$

On C_2 set $\mathbf{r}(x) = \langle x, 2x \rangle$. A scalar line integral is independent of orientation, so for simplicity, we can parametrize the path from $x = 0$ to $x = 2$. Then the arc length element is $ds = \|\mathbf{r}'(x)\| dx = \sqrt{1^2 + 2^2} dx = \sqrt{5} dx$. The integral over C_2 is:

$$\int_{C_2} x ds = \int_0^2 x \sqrt{5} dx = \sqrt{5} \int_0^2 x dx = \sqrt{5} \left[\frac{x^2}{2} \right]_0^2 = \sqrt{5} \left(\frac{4}{2} - 0 \right) = 2\sqrt{5}.$$

Therefore, the total integral is the **sum** of the integrals over the two paths:

$$\oint_C x ds = \int_{C_1} x ds + \int_{C_2} x ds = \frac{1}{12} (17^{3/2} - 1) + 2\sqrt{5} = \frac{17\sqrt{17} - 1}{12} + 2\sqrt{5}.$$

■

26 Line Integrals of Vector Fields and the Fundamental Theorem

26.1 Goals

- Integrate vector fields over a line (calculate work).
- Understand the concept of Circulation.
- Recognize when a vector field is conservative and find potential functions.
- Apply the Fundamental Theorem of Line Integrals to compute integrals efficiently.

26.2 Line Integrals and Work

Definition 26.1

Let C be an oriented, piecewise smooth curve in $\mathbb{R}^2$ or $\mathbb{R}^3$, parametrized by $\mathbf{r}(t)$ for $a \leq t \leq b$, and let $\vec{F}$ be continuous on C . The line integral of $\vec{F}$ along C is

$$\int_C \vec{F} \cdot d\mathbf{r} = \int_a^b \vec{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt.$$

Note

This definition generalizes the idea of computing work. Recall that if a particle moves along a

curve C under the influence of a force field $\vec{F}$, then the work done by $\vec{F}$ is

$$W = \int_C \vec{F} \cdot d\mathbf{r}.$$

Where the speed is nonzero, $\vec{F} \cdot \mathbf{r}' = (\vec{F} \cdot \mathbf{T})\|\mathbf{r}'\|$. The tangential force component is $\vec{F} \cdot \mathbf{T}$, so $dW = (\vec{F} \cdot \mathbf{T}) ds$. Reversing the path changes the sign of the work integral.

Remark 26.1 (Alternative Notation). Suppose $\vec{F}(x, y) = \langle f(x, y), g(x, y) \rangle$. Then $\vec{F} \cdot d\vec{r} = f(x, y) dx + g(x, y) dy$, so line integrals may also be written as:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C f(x, y) dx + g(x, y) dy.$$

This compact form is often the preferred notation when applying Green's Theorem or working with differential forms.

Example 26.2. Find the line integral of $\vec{F} = \langle x^2, y - e^x \rangle$, as we move along the straight line segment from $A(0, 0)$ to $B(4, -5)$.

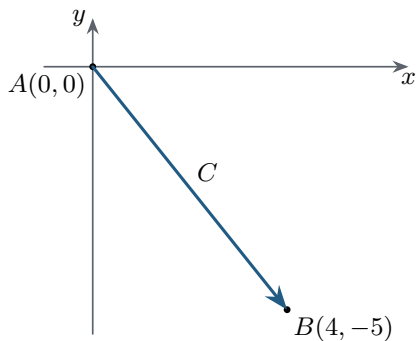


Figure 26.1: Straight line path from A to B .

Solution. Parametrize the line AB by $\mathbf{r}(t) = \langle 4t, -5t \rangle$, $0 \leq t \leq 1$, so $\mathbf{r}'(t) = \langle 4, -5 \rangle$. Along this path $x = 4t$, $y = -5t$, hence

$$\vec{F}(\mathbf{r}(t)) = \langle 16t^2, -5t - e^{4t} \rangle.$$

Thus

$$\vec{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 64t^2 + 25t + 5e^{4t}.$$

Therefore

$$\int_C \vec{F} \cdot d\mathbf{r} = \int_0^1 (64t^2 + 25t + 5e^{4t}) dt = \frac{64}{3} + \frac{25}{2} + \frac{5}{4}(e^4 - 1).$$

■

26.3 Circulation

Definition 26.2 (Circulation)

If C is a closed loop (so the starting and ending points coincide), the line integral

$$\oint_C \vec{F} \cdot d\mathbf{r} = \int_a^b \vec{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt, \quad \mathbf{r}(a) = \mathbf{r}(b)$$

is called the **circulation** of $\vec{F}$ around C . Intuitively, circulation measures the tendency of the vector field to “flow” or swirl around the curve.

Example 26.3. Consider $\vec{F} = \langle -y, x \rangle$ and let C be the boundary of the triangle with vertices $(0, 0)$, $(2, 0)$, and $(0, 3)$, oriented counterclockwise. Evaluate $\oint_C \vec{F} \cdot d\mathbf{r}$.

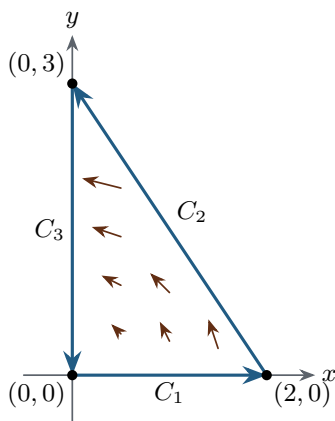


Figure 26.2: The triangular path $C = C_1 \cup C_2 \cup C_3$ oriented counterclockwise.

Solution. The curve C consists of three line segments: $C = C_1 \cup C_2 \cup C_3$.

1. **On C_1 :** $(0, 0) \rightarrow (2, 0)$. Parametrize $\mathbf{r}_1(t) = \langle t, 0 \rangle$, $0 \leq t \leq 2$.

$$\vec{F}(\mathbf{r}_1(t)) = \langle 0, t \rangle, \quad \mathbf{r}'_1(t) = \langle 1, 0 \rangle \implies \int_{C_1} \vec{F} \cdot d\mathbf{r} = \int_0^2 0 dt = 0.$$

2. **On C_2 :** $(2, 0) \rightarrow (0, 3)$. Parametrize $\mathbf{r}_2(t) = \langle 2 - 2t, 3t \rangle$, $0 \leq t \leq 1$.

$$\vec{F}(\mathbf{r}_2(t)) = \langle -3t, 2 - 2t \rangle, \quad \mathbf{r}'_2(t) = \langle -2, 3 \rangle.$$

$$\vec{F} \cdot \mathbf{r}'_2 = (-3t)(-2) + (2 - 2t)(3) = 6t + 6 - 6t = 6 \implies \int_{C_2} \vec{F} \cdot d\mathbf{r} = \int_0^1 6 dt = 6.$$

3. **On C_3 :** $(0, 3) \rightarrow (0, 0)$. Parametrize $\mathbf{r}_3(t) = \langle 0, 3 - 3t \rangle$, $0 \leq t \leq 1$.

$$\vec{F}(\mathbf{r}_3(t)) = \langle -3 + 3t, 0 \rangle, \quad \mathbf{r}'_3(t) = \langle 0, -3 \rangle \implies \int_{C_3} \vec{F} \cdot d\mathbf{r} = 0.$$

Therefore, the total circulation is:

$$\oint_C \vec{F} \cdot d\mathbf{r} = 0 + 6 + 0 = 6.$$

■

26.4 The Fundamental Theorem of Line Integrals

Given $\vec{F}$, a natural question is whether it is conservative (i.e., implies path independence), and if so, how to find the potential function ϕ .

Recall that $\vec{F}$ is *conservative* when $\vec{F} = \nabla\phi$ for a scalar potential ϕ .

Testing for Conservativity: Suppose $\vec{F} = \langle f(x, y), g(x, y) \rangle$ has continuous first partial derivatives on an open set D . If $\vec{F}$ is conservative, then by Clairaut's Theorem (equality of mixed partials), a necessary condition is:

$$\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}.$$

If D is simply connected, this condition is also sufficient. Here simply connected means that D is path connected and every closed loop can be continuously contracted to a point within D .

Theorem 26.1 (Fundamental Theorem of Line Integrals)

If ϕ is continuously differentiable on an open set D and C is a piecewise smooth curve in D from A to B , then

$$\int_C \vec{F} \cdot d\vec{r} = \phi(B) - \phi(A).$$

Thus, the integral depends only on the endpoints, not on the path.

Example 26.4. Determine whether $\vec{F} = \langle 2xy, x^2 \rangle$ is conservative, and if so, find a potential function ϕ and evaluate $\int_C \vec{F} \cdot d\vec{r}$ for any path from A to B .

Solution. Here $f = 2xy$ and $g = x^2$. We check the condition $f_y = g_x$:

$$f_y = \frac{\partial}{\partial y}(2xy) = 2x, \quad g_x = \frac{\partial}{\partial x}(x^2) = 2x.$$

Since $f_y = g_x$ on the simply connected domain $\mathbb{R}^2$, the field is conservative. To find ϕ , we solve $\phi_x = 2xy$ and $\phi_y = x^2$.

Integrate ϕ_x with respect to x :

$$\phi(x, y) = \int 2xy \, dx = x^2y + C(y).$$

Differentiate with respect to y and match with g :

$$\frac{\partial \phi}{\partial y} = x^2 + C'(y) = x^2 \implies C'(y) = 0.$$

Thus, $\phi(x, y) = x^2y$. By the FTLI:

$$\int_C \vec{F} \cdot d\vec{r} = \phi(B) - \phi(A) = x_B^2 y_B - x_A^2 y_A.$$

■

Example 26.5 (Non-Conservative Field). Test whether $\vec{F}(x, y) = \langle -yx, x^2 \rangle$ is conservative.

Solution. Here $f = -yx$ and $g = x^2$.

$$f_y = -x, \quad g_x = 2x.$$

Since $f_y \neq g_x$, $\vec{F}$ is not conservative. No potential ϕ exists, and line integrals will depend on the specific path taken. ■

Example 26.6 (Path Independence). A particle moves in the plane under the force field

$$\vec{F}(x, y) = \langle 2xe^{2y} - 2, 2x^2e^{2y} + 2y \rangle.$$

Compute the work done in moving from $A = (1, 0)$ to $B = (1, 1)$.

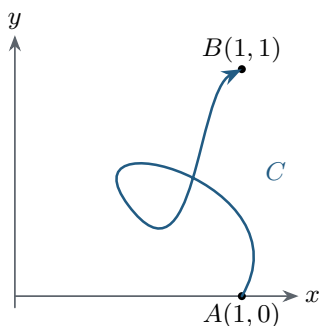


Figure 26.3: A smooth path C from A to B .

Solution. Check conservativity:

$$\frac{\partial}{\partial y}(2xe^{2y} - 2) = 4xe^{2y}, \quad \frac{\partial}{\partial x}(2x^2e^{2y} + 2y) = 4xe^{2y}.$$

The field is conservative. We could find ϕ , but we can also just pick a convenient path since the integral is path-independent. Let's use the vertical line $x = 1$: $\vec{r}(t) = \langle 1, t \rangle$, $0 \leq t \leq 1$.

$$\vec{r}'(t) = \langle 0, 1 \rangle, \quad \vec{F}(\vec{r}(t)) = \langle 2e^{2t} - 2, 2e^{2t} + 2t \rangle.$$

$$W = \int_0^1 \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt = \int_0^1 (2e^{2t} + 2t) dt = [e^{2t} + t^2]_0^1 = e^2.$$

■

Example 26.7 (Circulation of Conservative Fields). Consider $\vec{F}(x, y) = \langle ye^{xy} + 2xy, xe^{xy} + x^2 + 3y^2 \rangle$. Let C be a smooth, simple closed curve oriented counterclockwise. Compute $\oint_C \vec{F} \cdot d\vec{r}$.

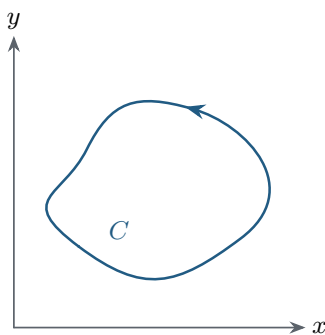


Figure 26.4: A smooth, simple closed curve C .

Solution. First check conservativity:

$$f_y = e^{xy} + xye^{xy} + 2x, \quad g_x = e^{xy} + xye^{xy} + 2x.$$

Since $f_y = g_x$ on the simply connected domain $\mathbb{R}^2$, $\vec{F}$ is conservative ($\vec{F} = \nabla\phi$). By the Fundamental Theorem of Line Integrals, since the start and end points coincide ($A = B$), the integral vanishes:

$$\oint_C \vec{F} \cdot d\vec{r} = \phi(A) - \phi(A) = 0.$$

■

27 Curl and Divergence of Vector Fields

27.1 Goals

- Define the nabla (del) operator ∇ .
- Define and give formulas for the curl and divergence of vector fields.
- Understand why the curl of a conservative vector field is zero.

Definition 27.1 (Gradient)

For a scalar function $f(x, y, z)$, the gradient is

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle.$$

- Input: a scalar function f (e.g., temperature, potential, height).
- Output: a vector field pointing in the direction of steepest increase of f .

The gradient operator ∇ is often called the *del operator*:

$$\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle.$$

This notation collects the coordinate differentiation operators. It is not a vector of numbers: the symbols $\partial_x, \partial_y, \partial_z$ act on functions.

27.2 Curl

For a vector field with continuous first partial derivatives, the notation $\nabla \times \vec{F}$ records a specific combination of derivatives. If

$$\vec{F} = \langle f, g, h \rangle,$$

then its *curl* is denoted by

$$\text{curl } \vec{F} = \nabla \times \vec{F}.$$

Definition 27.2 (Curl in Coordinates)

For a C^1 vector field $\vec{F} = \langle f, g, h \rangle$ on an open subset of $\mathbb{R}^3$,

$$\text{curl } \vec{F} = \langle h_y - g_z, f_z - h_x, g_x - f_y \rangle.$$

The determinant below is a mnemonic: expand along the first row, with derivative symbols acting on the components below them.

$$\text{curl } \vec{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & g & h \end{vmatrix}.$$

In 2D, if $\vec{F} = \langle f, g, 0 \rangle$, then

$$\text{curl } \vec{F} = \left\langle 0, 0, \frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right\rangle,$$

which will be used for the formula of Green's theorem. The scalar $\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}$ measures *rotation* in the plane.

Example 27.1 (Curl in 3D). Let $\vec{F} = \langle x^2y^3, x^3y^2, x^3y^3 \rangle$. Compute $\text{curl } \vec{F}$.

Solution.

$$\text{curl } \vec{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y^3 & x^3y^2 & x^3y^3 \end{vmatrix}.$$

Expanding,

$$\text{curl } \vec{F} = \left(\frac{\partial}{\partial y}(x^3y^3) - \frac{\partial}{\partial z}(x^3y^2), \frac{\partial}{\partial z}(x^2y^3) - \frac{\partial}{\partial x}(x^3y^3), \frac{\partial}{\partial x}(x^3y^2) - \frac{\partial}{\partial y}(x^2y^3) \right) = \langle 3x^3y^2, -3x^2y^3, 0 \rangle.$$

■

27.3 Curl of a Conservative Vector Field

If $\vec{F}$ is a gradient field, i.e. $\vec{F} = \nabla\varphi$, then

$$\text{curl } \vec{F} = \nabla \times (\nabla\varphi).$$

Theorem 27.1

For any scalar function φ with continuous second partial derivatives,

$$\nabla \times \nabla\varphi = \mathbf{0}.$$

Proof.

$$\nabla \times \nabla\varphi = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial\varphi}{\partial x} & \frac{\partial\varphi}{\partial y} & \frac{\partial\varphi}{\partial z} \end{vmatrix}.$$

Each component involves mixed second partial derivatives, which cancel because of Clairaut's theorem ($f_{xy} = f_{yx}$). Thus the determinant is zero. ■

Conclusion: The curl of a continuously differentiable conservative vector field is zero. This provides a necessary condition for a field to be conservative.

27.4 Divergence

Instead of a cross product, consider the dot product:

$$\operatorname{div} \vec{F} = \nabla \cdot \vec{F}.$$

Definition 27.3 (Divergence in Coordinates)

If $\vec{F} = \langle f, g, h \rangle$ is C^1 on an open subset of $\mathbb{R}^3$, then

$$\operatorname{div} \vec{F} = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z}.$$

In 2D, for $\vec{F} = \langle f, g \rangle$, we get

$$\operatorname{div} \vec{F} = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y}.$$

Example 27.2 (Divergence in 3D). For

$$\vec{F} = \langle x^2y^3, x^3y^2, x^3y^3 \rangle,$$

compute $\operatorname{div} \vec{F}$.

Solution.

$$\operatorname{div} \vec{F} = \frac{\partial}{\partial x}(x^2y^3) + \frac{\partial}{\partial y}(x^3y^2) + \frac{\partial}{\partial z}(x^3y^3).$$

That is,

$$\operatorname{div} \vec{F} = 2xy^3 + 2x^3y + 0 = 2xy^3 + 2x^3y.$$

■

27.5 Geometric Intuition

- $\operatorname{curl} \vec{F}$: measures local *rotation* of the field. The component of curl along a unit vector measures circulation per unit area in the perpendicular plane. A tiny paddle wheel illustrates this local rotation.
- $\operatorname{div} \vec{F}$: measures local *outflow* (expansion or compression). Divergence is the limiting net outward flux per unit volume through a small box around the point. Positive values indicate local expansion of the flow.

Note

Curl and divergence are the fundamental differential operators used in vector calculus.

- Curl detects local *rotation*.
- Divergence detects local *sources and sinks*.

28 Green's Theorem

28.1 Goals

- Understand the statement of Green's Theorem.
- Interpret circulation and flux in terms of double integrals.
- Prove Green's theorem for a region that is both Type I and Type II.

Theorem 28.1 (Green's Theorem)

Let C be a positively oriented, simple, closed, piecewise smooth curve in the plane, and let R be the region enclosed by C . If $\vec{F} = \langle P(x, y), Q(x, y) \rangle$ is a vector field whose components have continuous partial derivatives on an open neighborhood of the closed region R and its boundary, then

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

Why this is true (intuition):

$$\underbrace{\oint_C \vec{F} \cdot \mathbf{T} ds}_{\text{accumulated tangential flow along the boundary} \Rightarrow \text{circulation}} = \underbrace{\iint_R (Q_x - P_y) dA}_{\text{total "microscopic rotation" inside region } R}, \quad (\nabla \times \langle P, Q, 0 \rangle) \cdot \mathbf{k} = Q_x - P_y.$$

Think of R as tiled by many tiny squares. On a small tile, the circulation is approximately $(Q_x - P_y)$ times the tile's area. When you add tiles, all the interior edges are traversed twice with opposite orientations, so they cancel. What survives is only the outer boundary C . Hence the boundary integral equals the sum of all tiny curls in the interior.

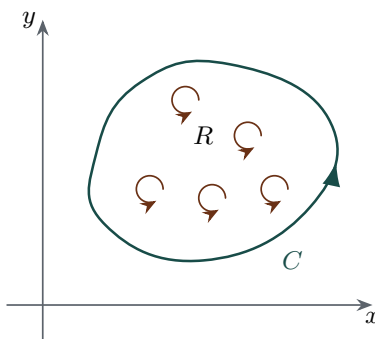


Figure 28.1: A region R with positively oriented boundary C ; interior swirls suggest $\text{curl } \vec{F}$.

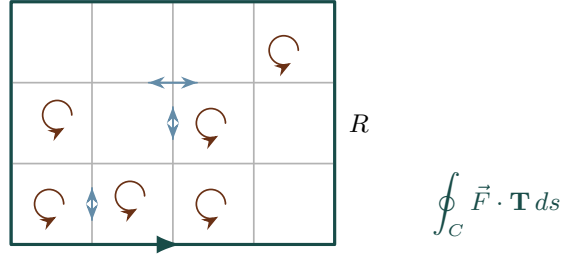


Figure 28.2: Tiling R into small cells: contributions on shared edges cancel; only the outer boundary remains.

28.2 Proof for a simple region

We prove the theorem for a region that is both Type I and Type II. Assume R is compact, its boundary C is a simple, closed, piecewise C^1 curve, and

$$\begin{aligned} R &= \{(x, y) : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\} \\ &= \{(x, y) : c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}. \end{aligned}$$

The boundary functions are continuous and piecewise C^1 on the interiors of their intervals. Assume their graph arcs admit piecewise C^1 parametrizations, including at the endpoints. This allows, for example, a disk whose graph derivatives become unbounded at the extreme points. Let P, Q be C^1 on an open neighborhood of R , and orient C counterclockwise.

Proof. Step 1: Compute the $P dx$ part using vertical slices. On the lower graph the positive orientation runs from a to b . On the upper graph it runs from b to a , since R must lie to the left. Any vertical boundary segments contribute zero because $dx = 0$. Therefore

$$\begin{aligned} \oint_C P dx &= \int_a^b P(x, g_1(x)) dx + \int_b^a P(x, g_2(x)) dx \\ &= \int_a^b (P(x, g_1(x)) - P(x, g_2(x))) dx. \end{aligned}$$

For each fixed x , the fundamental theorem of calculus gives

$$P(x, g_2(x)) - P(x, g_1(x)) = \int_{g_1(x)}^{g_2(x)} P_y(x, y) dy.$$

Substitution, followed by Fubini's theorem, yields

$$\oint_C P dx = - \int_a^b \int_{g_1(x)}^{g_2(x)} P_y(x, y) dy dx = - \iint_R P_y dA.$$

Step 2: Compute the $Q dy$ part using horizontal slices. The right graph is traversed from c to d , and the left graph from d to c . Horizontal boundary segments contribute zero because $dy = 0$. Consequently

$$\begin{aligned} \oint_C Q dy &= \int_c^d Q(h_2(y), y) dy + \int_d^c Q(h_1(y), y) dy \\ &= \int_c^d (Q(h_2(y), y) - Q(h_1(y), y)) dy \\ &= \int_c^d \int_{h_1(y)}^{h_2(y)} Q_x(x, y) dx dy = \iint_R Q_x dA. \end{aligned}$$

The third equality is again the fundamental theorem of calculus.

Step 3: Add the two identities. By linearity,

$$\oint_C (P dx + Q dy) = \iint_R (Q_x - P_y) dA.$$

All applications of Fubini are valid because the partial derivatives are continuous on the compact region. If a graph has an endpoint with a vertical or horizontal tangent, first truncate the graph away from that endpoint. The displayed identities pass to the limit by continuity and the finite length of each boundary arc. ■

Extension by subdivision

If a region is a finite union of regions satisfying these hypotheses, with disjoint interiors and compatible piecewise smooth boundaries, apply the proved identity on each piece. Each shared boundary arc has opposite orientations in the two adjacent pieces, so its line integrals cancel. The double integrals add, leaving Green's formula for the remaining boundary arcs with their induced orientation: counterclockwise around the outside and clockwise around any holes. This proves the theorem for such unions. The tiling picture explains this cancellation, but an approximation argument is needed for a general boundary that is not given by such a finite decomposition.

Two useful corollaries.

$$\begin{aligned} \text{(Area formula)} \quad \text{Area}(R) &= \iint_R 1 dA \\ &= \oint_C x dy \\ &= - \oint_C y dx \\ &= \frac{1}{2} \oint_C (x dy - y dx). \end{aligned}$$

$$\text{(Flux form)} \quad \oint_C \vec{F} \cdot \mathbf{n} ds = \iint_R \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA.$$

where $\mathbf{n}$ is the outward unit normal and $\mathbf{T}$ is the positively oriented unit tangent.

Regions with Holes

Green's theorem also applies to a region with finitely many holes, with the entire boundary oriented so that the region lies on your left. The outer boundary is counterclockwise and each hole boundary is clockwise. The field must be continuously differentiable near the region and all its boundary components.

Example 28.1. Evaluate $\oint_C (x^2 - y) dx + (x + y^2) dy$, where C is the circle $x^2 + y^2 = 4$ oriented counterclockwise.

Solution. We identify $f(x, y) = x^2 - y$, $g(x, y) = x + y^2$. Then

$$g_x = 1, \quad f_y = -1.$$

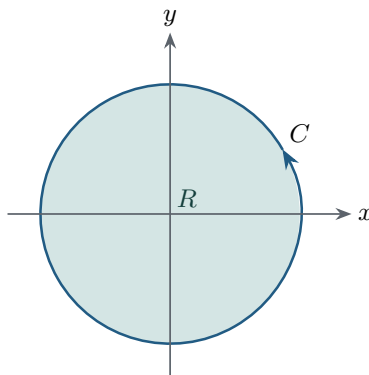


Figure 28.3: Region R : the disk enclosed by C .

So

$$g_x - f_y = 1 - (-1) = 2.$$

By Green's Theorem,

$$\oint_C f dx + g dy = \iint_R 2 dA = 2 \cdot \text{Area}(R).$$

Since R is the disk of radius 2, $\text{Area}(R) = \pi(2^2) = 4\pi$. Therefore,

$$\oint_C (x^2 - y) dx + (x + y^2) dy = 2 \cdot 4\pi = 8\pi.$$

■

Example 28.2. Green's Theorem

(a) Let C be the circle $x^2 + y^2 = 4$, oriented counterclockwise. Evaluate

$$\int_C (2y + x^4) dx + (3x + e^y) dy$$

(using Green's Theorem).

(b) Evaluate $\int_C y^2 dx - x^2 dy$ where C is the positively oriented circle of radius 3 centered at the origin.

(c) Use Green's Theorem to evaluate $\int_C xy dx + x^2 dy$ where C is the triangle with vertices $(0,0)$, $(0,3)$, $(2,0)$ with positive orientation.

Solution. (a) By Green's Theorem:

$$\oint_C P dx + Q dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

Here $P(x, y) = 2y + x^4$, $Q(x, y) = 3x + e^y$. So

$$\frac{\partial Q}{\partial x} = 3, \quad \frac{\partial P}{\partial y} = 2.$$

Thus

$$\iint_R (3 - 2) dA = \iint_R 1 dA = \text{Area}(R).$$

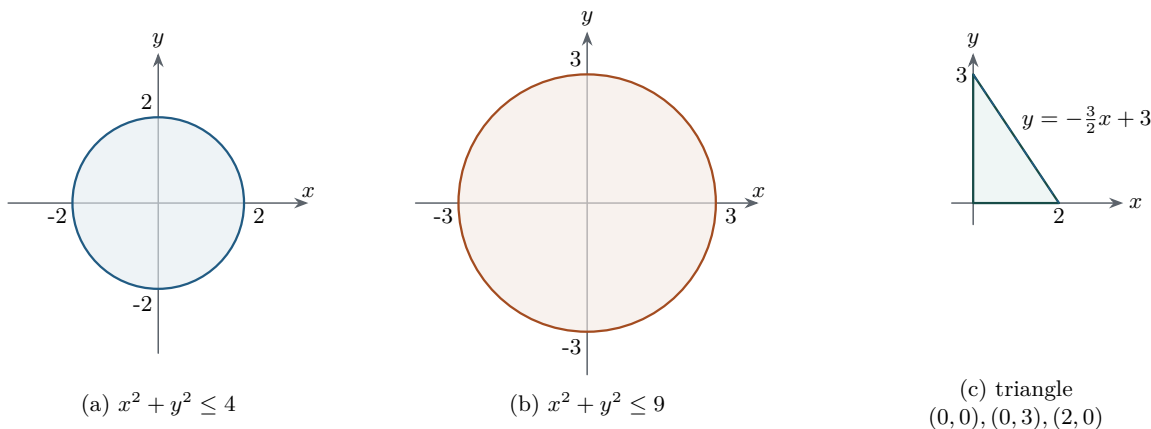


Figure 28.4: Regions for Green's Theorem examples.

Since R is the disk $x^2 + y^2 \leq 4$, its area is $\pi \cdot 2^2 = 4\pi$. So the value is

$$\int_C (2y + x^4) dx + (3x + e^y) dy = 4\pi.$$

(b) Here $P(x, y) = y^2$, $Q(x, y) = -x^2$. So

$$\frac{\partial Q}{\partial x} = -2x, \quad \frac{\partial P}{\partial y} = 2y.$$

Thus

$$\iint_R (-2x - 2y) dA.$$

Over the circle $x^2 + y^2 \leq 9$, the integrals of x and y vanish by symmetry. So the integral is 0.

(c) $P(x, y) = xy$, $Q(x, y) = x^2$. So

$$\frac{\partial Q}{\partial x} = 2x, \quad \frac{\partial P}{\partial y} = x.$$

Thus

$$\iint_R (2x - x) dA = \iint_R x dA.$$

R is the triangle with vertices $(0, 0)$, $(0, 3)$, $(2, 0)$. Equation of line from $(0, 3)$ to $(2, 0)$ is $y = -\frac{3}{2}x + 3$. So region is

$$0 \leq x \leq 2, \quad 0 \leq y \leq -\frac{3}{2}x + 3.$$

Thus

$$\begin{aligned} \int_0^2 \int_0^{-3x/2+3} x dy dx &= \int_0^2 x \left(-\frac{3}{2}x + 3 \right) dx \\ &= \int_0^2 \left(-\frac{3}{2}x^2 + 3x \right) dx = \left[-\frac{1}{2}x^3 + \frac{3}{2}x^2 \right]_0^2 = (-4 + 6) = 2. \end{aligned}$$

So the value is 2. ■

29 Surface Integrals (I) Scalar Functions

29.1 Goals

- Geometry and intuition of a surface integral; interpretation as accumulation over a surface.
- Formal definition and formula for surface integrals.
- Concrete examples and parametrizations of surfaces in $\mathbb{R}^3$.

We recall that the line integral of a scalar function $f(x, y, z)$ along a curve C

$$\int_C f \, ds$$

measures the *accumulation* of the values of f along the path C , weighted by arc length.

To compute this integral, we first parametrize the curve. If a curve C is described by a vector function

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle, \quad a \leq t \leq b,$$

then the differential of arc length is

$$ds = \|\mathbf{r}'(t)\| \, dt,$$

and hence

$$\int_C f(x, y, z) \, ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| \, dt.$$

Example 29.1. Consider the curve along the parabola $y = x^2$ from $(-1, 1)$ to $(4, 16)$. A convenient parametrization is

$$\mathbf{r}(t) = \langle t, t^2 \rangle, \quad -1 \leq t \leq 4.$$

In a completely parallel manner, the *surface integral* of a function $f(x, y, z)$ over a surface S , denoted by

$$\iint_S f(x, y, z) \, dS,$$

measures the accumulation of $f(x, y, z)$ over the surface S , weighted by surface area.

To compute this integral, we parametrize the surface. A surface S in $\mathbb{R}^3$ can be described by a vector function of two parameters:

$$\mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle, \quad (u, v) \in D,$$

where D is a region in the uv -plane. We require $\mathbf{r}$ to be C^1 on a neighborhood of D , with $\mathbf{r}_u \times \mathbf{r}_v \neq \mathbf{0}$ in its interior, and to cover the surface once apart from boundary seams. Such a parametrization is *regular*. A piecewise smooth surface is covered by finitely many regular patches whose interiors do not overlap. We integrate over the patches and add, so that no portion of positive area is counted twice.

Example 29.2. Parametrize the portion of the plane

$$4x + 3y + z = 8$$

that lies in the first octant.

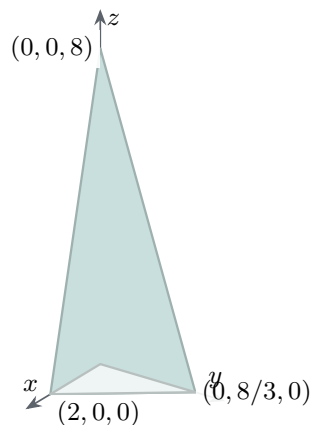


Figure 29.1: Part of the plane lying in the first octant.

Solution. To parametrize a surface, we need a way to locate every point on it using two parameters. From the equation of the plane, if we know x and y , then z is determined:

$$z = 8 - 4x - 3y.$$

Thus, we may take $u = x$ and $v = y$ as our parameters, and then

$$\mathbf{r}(u, v) = \langle u, v, 8 - 4u - 3v \rangle.$$

Next, we must determine the domain of (u, v) . Since we want the portion of the plane in the *first octant*, we require

$$u \geq 0, \quad v \geq 0, \quad z = 8 - 4u - 3v \geq 0.$$

The inequality $8 - 4u - 3v \geq 0$ rearranges to

$$v \leq \frac{8-4u}{3}.$$

Therefore, the domain is the triangular region

$$0 \leq u \leq 2, \quad 0 \leq v \leq \frac{8-4u}{3}.$$

So one parametrization of this surface is

$$\mathbf{r}(u, v) = \langle u, v, 8 - 4u - 3v \rangle, \quad 0 \leq u \leq 2, \quad 0 \leq v \leq \frac{8-4u}{3}.$$

■

Example 29.3. Parametrize the portion of the sphere

$$x^2 + y^2 + z^2 = 4$$

lying in the *first octant* with $1 \leq z \leq 2$.

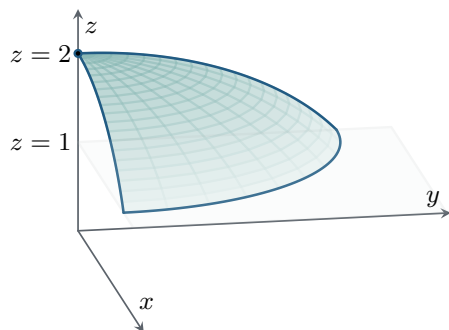


Figure 29.2: Spherical region of radius 2, centered at the origin, contained in the first octant between $z = 1$ and $z = 2$.

Solution. Geometry. On the sphere $x^2 + y^2 + z^2 = 4$, the top is at $z = 2$. The cut $z = 1$ gives $x^2 + y^2 = 4 - 1 = 3$, so the projection to the xy -plane is the quarter disk

$$D = \{(x, y) : x \geq 0, y \geq 0, x^2 + y^2 \leq 3\}.$$

Thus we are parametrizing a spherical cap above that quarter disk.

(A) Cartesian-style (use x, y as parameters). Solve for z on the upper hemisphere:

$$z = \sqrt{4 - x^2 - y^2}.$$

Take $u = x$, $v = y$. The first octant and $z \geq 1$ impose $u \geq 0$, $v \geq 0$, $u^2 + v^2 \leq 3$. An iterated description of this quarter-disk domain is

$$0 \leq u \leq \sqrt{3}, \quad 0 \leq v \leq \sqrt{3 - u^2}.$$

Hence

$$\mathbf{r}(u, v) = \langle u, v, \sqrt{4 - u^2 - v^2} \rangle, \quad 0 \leq u \leq \sqrt{3}, \quad 0 \leq v \leq \sqrt{3 - u^2}.$$

(B) Cylindrical-style (use r, θ as parameters). Write $x = r \cos \theta$, $y = r \sin \theta$, $z = \sqrt{4 - r^2}$. First octant $\Rightarrow 0 \leq \theta \leq \frac{\pi}{2}$. The condition $1 \leq z \leq 2$ becomes $1 \leq \sqrt{4 - r^2} \leq 2$, i.e. $0 \leq r \leq \sqrt{3}$. Thus

$$\mathbf{r}(u, v) = \langle u \cos v, u \sin v, \sqrt{4 - u^2} \rangle, \quad 0 \leq u \leq \sqrt{3}, \quad 0 \leq v \leq \frac{\pi}{2}.$$

(C) Spherical-angles (radius fixed). With spherical angles (ϕ, θ) (radius fixed at 2):

$$x = 2 \sin \phi \cos \theta, \quad y = 2 \sin \phi \sin \theta, \quad z = 2 \cos \phi.$$

First octant gives $0 \leq \theta \leq \frac{\pi}{2}$. The band $1 \leq z \leq 2$ means $1 \leq 2 \cos \phi \leq 2$, i.e. $\frac{1}{2} \leq \cos \phi \leq 1 \Rightarrow 0 \leq \phi \leq \frac{\pi}{3}$. Therefore

$$\mathbf{r}(\phi, \theta) = \langle 2 \sin \phi \cos \theta, 2 \sin \phi \sin \theta, 2 \cos \phi \rangle, \quad 0 \leq \phi \leq \frac{\pi}{3}, \quad 0 \leq \theta \leq \frac{\pi}{2}.$$

All three are equivalent; they describe the same surface! (B) and (C) have nicer bounds due to the spherical and cylindrical symmetry of the shape, while (A) mirrors the “graph above xy ” approach we used in the previous example. ■

For the surface-integral formula, assume the parametrization is continuously differentiable and $\mathbf{r}_u \times \mathbf{r}_v \neq \mathbf{0}$ in the interior of D . It should cover the surface once, apart from boundary seams of area zero. More general surfaces can be covered by patches whose integrals are added without counting overlaps twice.

A small parameter rectangle has tangent edges $\mathbf{r}_u du$ and $\mathbf{r}_v dv$. Their parallelogram area gives

$$dS = \|\mathbf{r}_u \times \mathbf{r}_v\| du dv,$$

where

$$\mathbf{r}_u = \frac{\partial \mathbf{r}}{\partial u}, \quad \mathbf{r}_v = \frac{\partial \mathbf{r}}{\partial v}.$$

Tangent parallelogram

$$\Delta S \approx \|\mathbf{r}_u \times \mathbf{r}_v\| \Delta u \Delta v$$

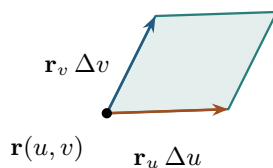


Figure 29.3: A small surface patch is approximated by a tangent parallelogram with sides $\mathbf{r}_u \Delta u$ and $\mathbf{r}_v \Delta v$. Its area gives the factor $\|\mathbf{r}_u \times \mathbf{r}_v\|$.

Thus the surface integral becomes

$$\iint_S f(x, y, z) dS = \iint_D f(\mathbf{r}(u, v)) \|\mathbf{r}_u \times \mathbf{r}_v\| du dv.$$

This formula is the exact surface-area analog of the line integral formula: just as $\|\mathbf{r}'(t)\| dt$ represents the infinitesimal length element of a curve, $\|\mathbf{r}_u \times \mathbf{r}_v\| du dv$ is the corresponding surface-area element. Consider the following example:

We can specialize this formula when the surface is given explicitly as $z = f(x, y)$. In this case we can use the “standard” parametrization

$$\mathbf{r}(x, y) = \langle x, y, f(x, y) \rangle, \quad (x, y) \in D \subseteq \mathbb{R}^2.$$

Then

$$\mathbf{r}_x = \langle 1, 0, f_x \rangle, \quad \mathbf{r}_y = \langle 0, 1, f_y \rangle,$$

so

$$\mathbf{r}_x \times \mathbf{r}_y = \det \begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & f_x \\ 0 & 1 & f_y \end{pmatrix} = \langle -f_x, -f_y, 1 \rangle.$$

Thus

$$\|\mathbf{r}_x \times \mathbf{r}_y\| = \sqrt{f_x^2 + f_y^2 + 1},$$

and therefore

$$\iint_S g dS = \iint_D g(x, y, f(x, y)) \sqrt{f_x^2 + f_y^2 + 1} dx dy.$$

Example 29.4. Let S be the part of the paraboloid $z = 4 - x^2 - y^2$ lying above the disk

$$D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 3\}.$$

Compute the surface integral $\iint_S 1 \, dS$ (i.e., find the surface area of S).

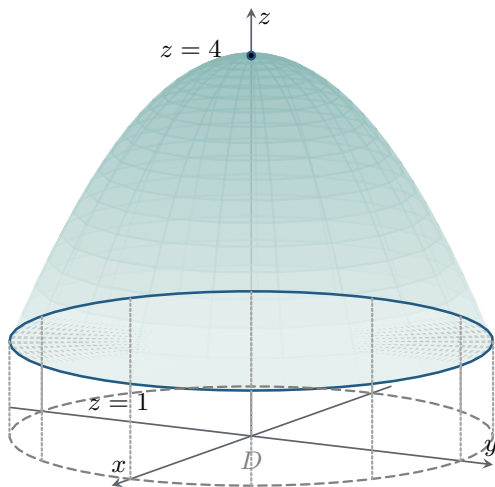


Figure 29.4: The paraboloid surface $z = 4 - x^2 - y^2$ between the planes $z = 1$ and $z = 4$.

Solution. For a graph $z = f(x, y)$, $dS = \sqrt{1 + (f_x)^2 + (f_y)^2} \, dA$. Here $f(x, y) = 4 - x^2 - y^2$, so

$$f_x = -2x, \quad f_y = -2y \quad \Rightarrow \quad \sqrt{1 + f_x^2 + f_y^2} = \sqrt{1 + 4x^2 + 4y^2}.$$

Thus

$$\iint_S 1 \, dS = \iint_D \sqrt{1 + 4x^2 + 4y^2} \, dA.$$

Switch to polar coordinates $(x, y) = (r \cos \theta, r \sin \theta)$ with $0 \leq r \leq \sqrt{3}$, $0 \leq \theta \leq 2\pi$:

$$\iint_D \sqrt{1 + 4r^2} \, r \, dr \, d\theta = \left(\int_0^{2\pi} d\theta \right) \left(\int_0^{\sqrt{3}} r \sqrt{1 + 4r^2} \, dr \right).$$

For the radial integral, let $u = 1 + 4r^2$, $du = 8r \, dr$, so $r \, dr = \frac{1}{8} du$:

$$\int_0^{\sqrt{3}} r \sqrt{1 + 4r^2} \, dr = \frac{1}{8} \int_{u=1}^{u=1+4 \cdot 3} u^{1/2} \, du = \frac{1}{8} \cdot \frac{2}{3} \left[u^{3/2} \right]_1^{13} = \frac{1}{12} (13^{3/2} - 1).$$

Multiplying by $\int_0^{2\pi} d\theta = 2\pi$, we get

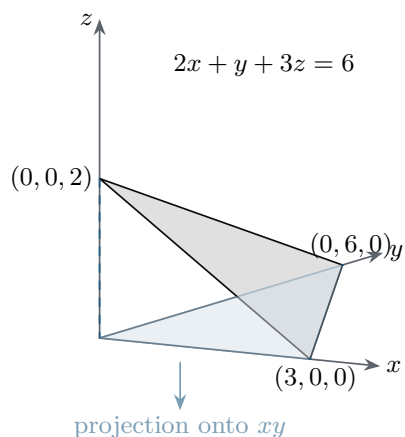
$$\boxed{\iint_S 1 \, dS = \frac{\pi}{6} (13^{3/2} - 1).}$$

■

Problem 29.5. Compute the surface integral

$$\iint_S (x^2 y + z) dS$$

where S is the part of the plane $2x + y + 3z = 6$ contained in the first octant.



$$D : 0 \leq x \leq 3, 0 \leq y \leq 6 - 2x$$

Figure 29.5: Triangular region bounded by $2x + y + 3z = 6$ in the first octant.

Solution. Solving for z gives

$$z = f(x, y) = \frac{6 - 2x - y}{3}.$$

So we parametrize:

$$\mathbf{r}(x, y) = \langle x, y, f(x, y) \rangle,$$

with (x, y) lying in the triangular domain bounded by the coordinate axes and the line $2x + y = 6$ in the first quadrant.

The surface element is

$$\|\mathbf{r}_x \times \mathbf{r}_y\| = \sqrt{\left(-\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + 1^2} = \sqrt{\frac{14}{9}} = \frac{\sqrt{14}}{3}.$$

Hence

$$\iint_S (x^2 y + z) dS = \iint_D \left(x^2 y + \frac{6-2x-y}{3}\right) \cdot \frac{\sqrt{14}}{3} dx dy.$$

The domain D is the triangle with vertices $(0, 0)$, $(3, 0)$, $(0, 6)$.

Thus the integral is

$$\frac{\sqrt{14}}{3} \iint_D \left(x^2 y + \frac{6-2x-y}{3}\right) dx dy.$$

At this point, one can proceed with explicit evaluation over D .

$$\begin{aligned}
 \iint_S (x^2 y + z) dS &= \frac{\sqrt{14}}{3} \iint_D \left(x^2 y + \frac{6 - 2x - y}{3} \right) dx dy, \\
 &= \frac{\sqrt{14}}{3} \int_0^3 \int_0^{6-2x} \left(x^2 y + \frac{6 - 2x - y}{3} \right) dy dx \\
 &= \frac{\sqrt{14}}{3} \int_0^3 \left[\frac{x^2}{2} (6 - 2x)^2 + \frac{1}{6} (6 - 2x)^2 \right] dx \\
 &= \frac{\sqrt{14}}{3} \int_0^3 (6 - 2x)^2 \left(\frac{x^2}{2} + \frac{1}{6} \right) dx \\
 &= \frac{\sqrt{14}}{3} \cdot \frac{111}{5} = \frac{37}{5} \sqrt{14}.
 \end{aligned}$$

■

Example 29.6. Find the surface area of the cone $z^2 = x^2 + y^2$ for $0 \leq z \leq 2$.

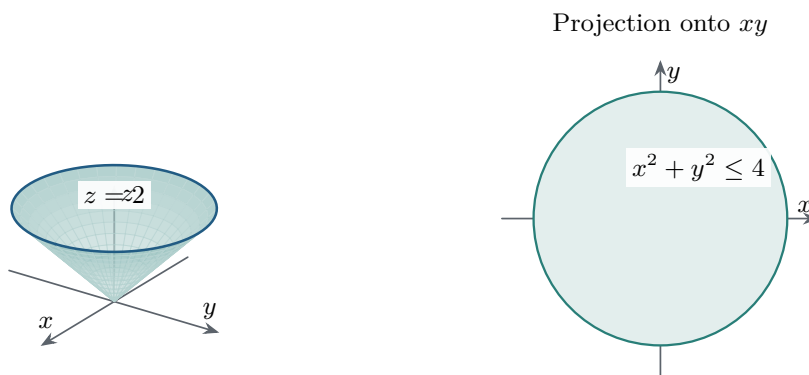


Figure 29.6: Left: the cone $z^2 = x^2 + y^2$ for $0 \leq z \leq 2$. Right: its projection onto the xy -plane.

Solution. The vertex is not a smooth point, but a single point has zero surface area. Apply the graph formula away from the vertex and take the limit. On the upper nappe, we may write

$$z = f(x, y) = \sqrt{x^2 + y^2}.$$

The projection onto the xy -plane is the disk

$$R = \{(x, y) : x^2 + y^2 \leq 4\}.$$

Since

$$f_x = \frac{x}{\sqrt{x^2 + y^2}}, \quad f_y = \frac{y}{\sqrt{x^2 + y^2}}, \quad 1 + f_x^2 + f_y^2 = 2,$$

we have

$$dS = \sqrt{2} dA.$$

Therefore,

$$\begin{aligned}\iint_S dS &= \iint_R \sqrt{2} dA \\ &= \sqrt{2} \iint_R dA.\end{aligned}$$

We can now compute this area in several ways.

Method 1: Geometric interpretation.

$$\begin{aligned}\sqrt{2} \iint_R dA &= \sqrt{2} \cdot (\text{area of } R) \\ &= \sqrt{2} \cdot \pi(2)^2 \\ &= 4\pi\sqrt{2}.\end{aligned}$$

Method 2: Polar coordinates. In polar, $R = \{0 \leq r \leq 2, 0 \leq \theta \leq 2\pi\}$ and $dA = r dr d\theta$. Thus

$$\begin{aligned}\sqrt{2} \iint_R dA &= \sqrt{2} \int_0^{2\pi} \int_0^2 r dr d\theta \\ &= \sqrt{2} \int_0^{2\pi} \left[\frac{r^2}{2} \right]_0^2 d\theta \\ &= \sqrt{2} \int_0^{2\pi} 2 d\theta \\ &= 2\sqrt{2} [\theta]_0^{2\pi} \\ &= 4\pi\sqrt{2}.\end{aligned}$$

Method 3: Direct parametrization. Let $u = r$, $v = \theta$, with $0 \leq u \leq 2$, $0 \leq v \leq 2\pi$. A parametrization is

$$\mathbf{r}(u, v) = \langle u \cos v, u \sin v, u \rangle.$$

Then

$$\mathbf{r}_u = \langle \cos v, \sin v, 1 \rangle, \quad \mathbf{r}_v = \langle -u \sin v, u \cos v, 0 \rangle.$$

Their cross product has norm

$$\|\mathbf{r}_u \times \mathbf{r}_v\| = \sqrt{2} u.$$

Hence

$$\begin{aligned}\text{Area}(S) &= \int_0^{2\pi} \int_0^2 \sqrt{2} u du dv \\ &= 4\pi\sqrt{2}.\end{aligned}$$

■

29.2 Average Value on a Surface

We can also compute the *average value* of a function $f(x, y, z)$ on a surface S of finite positive area. For a continuous function on a compact, piecewise smooth surface, define

$$f_{\text{avg}} = \frac{\iint_S f(x, y, z) dS}{\iint_S dS}.$$

This is natural, since

$$\iint_S f(x, y, z) dS = f_{\text{avg}} \iint_S dS,$$

meaning that the total “accumulation” of f over S is the same as if we replaced f by its constant average value.

Example 29.7. Find the average distance from the origin to points on the paraboloid

$$z = x^2 + y^2, \quad 1 \leq z \leq 4.$$

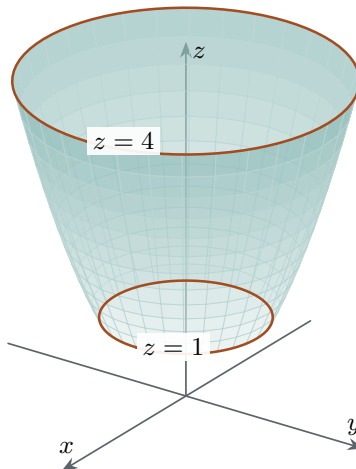


Figure 29.7: Part of the paraboloid $z = x^2 + y^2$ between the planes $z = 1$ and $z = 4$.

Solution. The distance from (x, y, z) to the origin is

$$d = \sqrt{x^2 + y^2 + z^2}.$$

Step 1: Parametrization. Using cylindrical coordinates, the projection of the paraboloid onto the xy -plane is the annulus $1 \leq r^2 \leq 4$, i.e. $1 \leq u \leq 2$ if we let $u = r$, $v = \theta$. Thus we parametrize:

$$\mathbf{r}(u, v) = \langle u \cos v, u \sin v, u^2 \rangle, \quad 1 \leq u \leq 2, \quad 0 \leq v \leq 2\pi.$$

Step 2: Surface element. We compute

$$\mathbf{r}_u = \langle \cos v, \sin v, 2u \rangle, \quad \mathbf{r}_v = \langle -u \sin v, u \cos v, 0 \rangle.$$

Their cross product has norm

$$\|\mathbf{r}_u \times \mathbf{r}_v\| = u\sqrt{1 + 4u^2}.$$

So

$$dS = u\sqrt{1 + 4u^2} du dv.$$

Step 3: Distance function. Since $z = u^2$, we get

$$d = \sqrt{u^2 + u^4}.$$

Step 4: Average value. Thus

$$f_{\text{avg}} = \frac{\iint_S f \, dS}{\iint_S dS}.$$

The numerator is

$$\iint_S dS = \int_0^{2\pi} \int_1^2 \sqrt{u^2 + u^4} (u\sqrt{1 + 4u^2}) \, du \, dv.$$

The denominator is

$$\iint_S dS = \int_0^{2\pi} \int_1^2 u\sqrt{1 + 4u^2} \, du \, dv.$$

Factoring out 2π , the ratio evaluates numerically to

$$f_{\text{avg}} \approx 3.09775 \approx 3.10.$$

■

30 Surface Integrals (II) Vector Fields

30.1 Goals

- The surface integral of a vector field.
- Surface orientations.
- Flux integrals.

In the previous lesson we discussed surface integrals of scalar fields (functions $f(x, y, z)$), which are computed as

$$\iint_S f \, dS = \iint_D f(\mathbf{r}(u, v)) \|\mathbf{r}_u \times \mathbf{r}_v\| \, dA.$$

Here, $\mathbf{r}_u \times \mathbf{r}_v$ gives a normal vector to the surface S . For scalar fields, only its magnitude matters. For *vector fields*, however, the direction of the normal is essential.

Orientation in Flux Integrals. In scalar surface integrals, the order of the cross product is irrelevant:

$$\|\mathbf{r}_u \times \mathbf{r}_v\| = \|\mathbf{r}_v \times \mathbf{r}_u\|.$$

But in vector field integrals, the direction of the normal matters:

$$\mathbf{r}_u \times \mathbf{r}_v = -(\mathbf{r}_v \times \mathbf{r}_u).$$

An *orientation* is a continuous choice of unit normal on a smooth surface, with compatible choices across smooth pieces. For a graph, upward and downward are the two choices. For the boundary of a solid, we usually choose outward. Follow the orientation specified in the problem. Some surfaces, such as a Möbius strip, have no global orientation.

Vector surface element. The oriented surface element is

$$d\vec{S} = \vec{n} \, dS,$$

where $\vec{n}$ is the chosen unit normal. Equivalently,

$$d\vec{S} = (\mathbf{r}_u \times \mathbf{r}_v) du dv,$$

taking $\mathbf{r}_u \times \mathbf{r}_v$ in the orientation consistent with the problem.

For a continuous vector field $\vec{F}$ on a compact oriented surface S ,

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \vec{n} dS = \iint_D \vec{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) dA.$$

$$\iint_S \vec{F} \cdot \vec{n} dS \text{ measures flow through } S$$

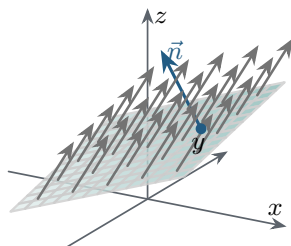


Figure 30.1: Surface integral measuring the “flow” of a vector field across a surface.

Special case: $z = f(x, y)$. If S is given by $z = f(x, y)$ with positive z -orientation (upward normal), then $\mathbf{r}(x, y) = \langle x, y, f(x, y) \rangle$ yields

$$\mathbf{r}_x \times \mathbf{r}_y = \langle -f_x, -f_y, 1 \rangle.$$

So

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_R (-F_1 f_x - F_2 f_y + F_3) dx dy,$$

Here each F_i is evaluated at $(x, y, f(x, y))$, and R is the projection of S onto the xy -plane.

Example 30.1. Compute the flux of $\vec{F} = \langle x, y, z \rangle$ across the part of the plane

$$2x + y + 3z = 6$$

lying in the first octant. Take the orientation to be upward (i.e., positive z -component).

Solution. Solve for z :

$$z = f(x, y) = \frac{6 - 2x - y}{3}, \quad (x, y) \in D = \{x \geq 0, y \geq 0, 2x + y \leq 6\}.$$

Use the standard parametrization $\mathbf{r}(x, y) = \langle x, y, f(x, y) \rangle$. Then

$$\mathbf{r}_x = \langle 1, 0, -\frac{2}{3} \rangle, \quad \mathbf{r}_y = \langle 0, 1, -\frac{1}{3} \rangle,$$

so (upward normal)

$$\mathbf{r}_x \times \mathbf{r}_y = \langle \frac{2}{3}, \frac{1}{3}, 1 \rangle.$$

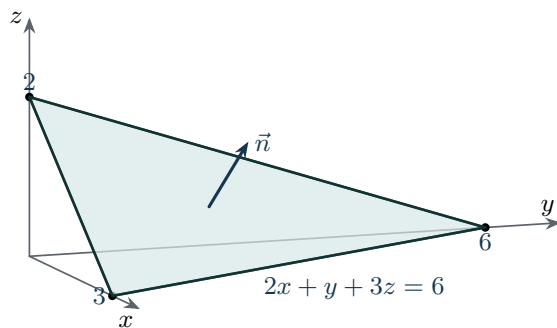


Figure 30.2: Triangular region contained in the first octant. The domain D in the xy -plane is bounded by $2x + y = 6$.

Now

$$\vec{F}(\mathbf{r}(x, y)) = \langle x, y, f(x, y) \rangle = \left\langle x, y, \frac{6-2x-y}{3} \right\rangle,$$

and the flux integrand is

$$\vec{F}(\mathbf{r}) \cdot (\mathbf{r}_x \times \mathbf{r}_y) = \frac{2}{3}x + \frac{1}{3}y + \frac{6-2x-y}{3} = 2.$$

Thus the flux is just

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_D 2 \, dx \, dy = 2 \cdot \text{Area}(D).$$

The domain D is the right triangle with vertices $(0, 0)$, $(3, 0)$, $(0, 6)$, so

$$\text{Area}(D) = \frac{1}{2} \cdot 3 \cdot 6 = 9,$$

and therefore

$$\boxed{\iint_S \vec{F} \cdot d\vec{S} = 18.}$$

Orientation check. If you chose the *downward* normal instead, the answer would be -18 . ■

Example 30.2. Compute the flux of

$$\vec{F}(x, y, z) = -\frac{\langle x, y, z \rangle}{(x^2 + y^2 + z^2)^{3/2}}$$

across the sphere $S : x^2 + y^2 + z^2 = a^2$, where $a > 0$, with outward orientation.

Solution. Parametrize the sphere with spherical angles:

$$\mathbf{r}(u, v) = \langle a \sin u \cos v, a \sin u \sin v, a \cos u \rangle, \quad 0 \leq u \leq \pi, \quad 0 \leq v \leq 2\pi.$$

Then

$$\mathbf{r}_u = \langle a \cos u \cos v, a \cos u \sin v, -a \sin u \rangle,$$

$$\mathbf{r}_v = \langle -a \sin u \sin v, a \sin u \cos v, 0 \rangle,$$

and

$$\mathbf{r}_u \times \mathbf{r}_v = a^2 \sin u \, \hat{\mathbf{r}}, \quad \text{where } \hat{\mathbf{r}} = \langle \sin u \cos v, \sin u \sin v, \cos u \rangle$$

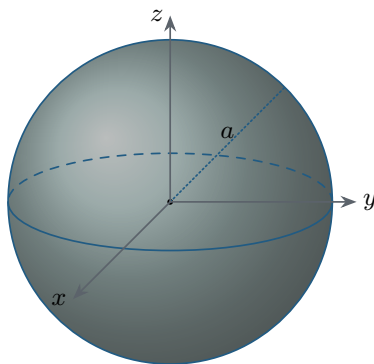


Figure 30.3: Sphere of radius a centered at $(0,0,0)$.

is the outward unit radial vector.

On the sphere $|\mathbf{r}| = a$, so

$$\vec{F}(\mathbf{r}) = -\frac{\mathbf{r}}{|\mathbf{r}|^3} = -\frac{1}{a^2} \hat{\mathbf{r}}.$$

Hence the flux integrand is

$$\vec{F}(\mathbf{r}) \cdot (\mathbf{r}_u \times \mathbf{r}_v) = \left(-\frac{1}{a^2} \hat{\mathbf{r}} \right) \cdot (a^2 \sin u \hat{\mathbf{r}}) = -\sin u.$$

Therefore

$$\begin{aligned} \iint_S \vec{F} \cdot d\vec{S} &= \int_0^{2\pi} \int_0^\pi (-\sin u) du dv \\ &= \left(-\int_0^\pi \sin u du \right) \left(\int_0^{2\pi} dv \right) \\ &= (-2)(2\pi) \\ &= \boxed{-4\pi}. \end{aligned}$$

Remarks.

- The value is independent of the radius a .
- If you reverse the orientation (inward), the answer becomes $+4\pi$.
- For an outward-oriented, piecewise smooth boundary enclosing the origin, the flux is -4π . If the origin lies outside the bounded solid, the flux is zero. The origin must not lie on the surface. Since the field is singular at the origin, the first assertion uses the divergence theorem after removing a small ball around the origin.

■

31 Stokes' Theorem

31.1 Goals

- State and interpret Stokes' Theorem, including surface and boundary orientation.

- Understand the physical meaning of curl and its relation to circulation (paddle wheel interpretation).
- Apply the theorem to convert line integrals into surface integrals (and vice versa).
- Develop geometric intuition regarding surface independence.

In previous lessons, we studied surface integrals of vector fields, which measure the *flux* of a field across a surface. Stokes' Theorem connects this idea with circulation along a boundary curve. It is the three-dimensional analogue of Green's Theorem from the plane, generalizing the relationship between rotation inside a region and circulation along its edge.

31.2 Statement of the Theorem

Let S be a compact, oriented, piecewise smooth surface with piecewise smooth boundary ∂S . Assume $\mathbf{F}$ has continuous first partial derivatives on an open neighborhood of S . With the induced boundary orientation, Stokes' theorem states

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}.$$

- S is an oriented, smooth surface.
- ∂S is the boundary curve of S .
- The orientations of S and ∂S are linked by the *right-hand rule*.

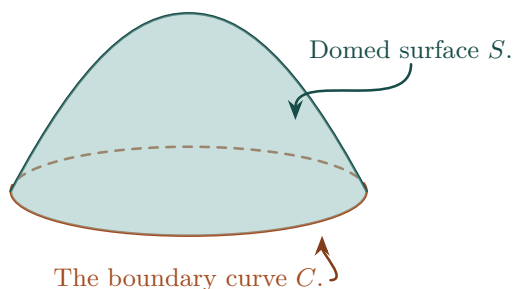


Figure 31.1: An open surface S with boundary curve C .

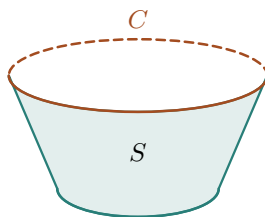


Figure 31.2: A surface S (cup) and its boundary curve C (the rim).

31.3 Orientation Rule

The key to applying Stokes' Theorem is consistency in orientation.

- If the unit normal $\hat{\mathbf{n}}$ is chosen for the surface, then traversing the boundary ∂S with your right-hand fingers curling in the direction of motion makes your thumb point along $\hat{\mathbf{n}}$.
- Equivalently: walk along ∂S with your head pointing in the direction of $\hat{\mathbf{n}}$. Then the surface S lies to your **left**.

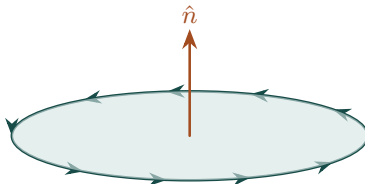


Figure 31.3: Boundary orientation consistent with an upward unit normal $\hat{\mathbf{n}}$.

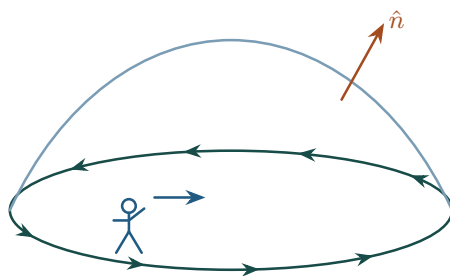


Figure 31.4: Walking along the boundary with head pointing along $\hat{\mathbf{n}}$: the surface lies on the left.

31.4 Physical Meaning: Curl and Circulation

To understand *why* we integrate $(\nabla \times \mathbf{F})$, recall that the curl measures the tendency of the field $\mathbf{F}$ to *rotate* locally.

Imagine placing a tiny paddle wheel in the vector field $\mathbf{F}$, with its axis oriented in some direction $\hat{\mathbf{n}}$ (see Figure 31.5).

- The *direction* of $\nabla \times \mathbf{F}$ is normal to the plane of greatest induced circulation (the axis about which the spin is maximized).
- The component of curl along a unit normal is the limiting circulation per unit area of a small loop normal to that direction. For a velocity field, half the curl is the local angular-velocity vector of the rotational part of the motion.
- The dot product $(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}}$ measures the spin specifically around the axis $\hat{\mathbf{n}}$.

Examples of curl:

- $\mathbf{F} = \langle x, y, 0 \rangle \implies \nabla \times \mathbf{F} = \mathbf{0}$. (Radial field; no spin).
- $\mathbf{F} = \langle -y, x, 0 \rangle \implies \nabla \times \mathbf{F} = \langle 0, 0, 2 \rangle$. (Swirling field; maximum spin about the z -axis).

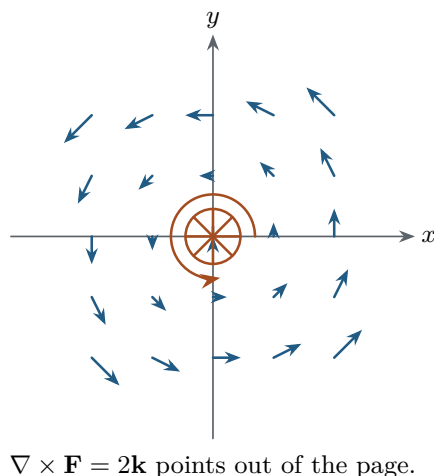


Figure 31.5: Rotation in the vector field $\mathbf{F} = \langle -y, x, 0 \rangle$. The orange paddle wheel is oriented to maximize the induced circulation; its spin axis aligns with $\nabla \times \mathbf{F}$, which here points along the z -axis.

31.5 Intuition: Why the Theorem Works

Stokes' Theorem relates the microscopic rotation (curl) accumulated over the surface to the macroscopic circulation around the boundary:

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}.$$

- The left-hand side sums the local *spin* of infinitesimal paddle wheels across the surface S .
- The right-hand side records the total circulation of $\mathbf{F}$ along the boundary curve ∂S .
- Subdivide the surface into small patches. Boundary integrals cancel along shared edges because adjacent patches induce opposite orientations there (see Figure 31.6). Only the integral along ∂S remains.

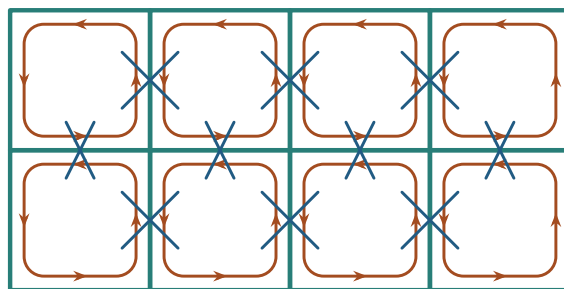


Figure 31.6: Interior contributions cancel; only the circulation along the boundary remains.

31.6 Computational Tools

To compute the surface integral, we use the oriented surface element:

$$d\mathbf{S} = \hat{\mathbf{n}} dS.$$

For common parametrizations:

$$d\mathbf{S} = \begin{cases} \langle 0, 0, 1 \rangle dx dy & (\text{upward normal in the } xy\text{-plane}), \\ \langle -f_x, -f_y, 1 \rangle dx dy & (\text{upward normal for } z = f(x, y)). \end{cases}$$

Example 31.1 (Verification). Let S be the upper hemisphere of the unit sphere $x^2 + y^2 + z^2 = 1$ with outward (upward) orientation, and let $C = \partial S$ be the unit circle in the xy -plane, oriented counterclockwise as viewed from $+z$ (See Figure 31.7). For the vector field

$$\mathbf{F}(x, y, z) = \langle -y, x, z \rangle,$$

verify Stokes' Theorem by computing both sides of the equality.

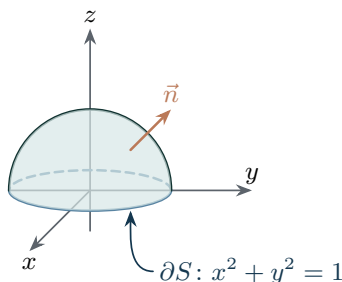


Figure 31.7: Upper hemisphere of unit sphere centered at $(0, 0, 0)$ with upwards orientation.

Solution. **1) Surface integral** $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS$.

Compute the curl:

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ -y & x & z \end{vmatrix} = \langle 0, 0, 2 \rangle.$$

On the upper hemisphere, the upward unit normal has $n_z = \cos \varphi$ in spherical coordinates, and $dS = \sin \varphi d\varphi d\theta$ with $0 \leq \theta \leq 2\pi$, $0 \leq \varphi \leq \frac{\pi}{2}$. Thus $(\nabla \times \mathbf{F}) \cdot \mathbf{n} = 2 \cos \varphi$.

$$\begin{aligned} \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS &= \int_0^{2\pi} \int_0^{\pi/2} 2 \cos \varphi \sin \varphi d\varphi d\theta \\ &= 2 \left[\frac{1}{2} \sin^2 \varphi \right]_0^{\pi/2} \int_0^{2\pi} d\theta = 2\pi. \end{aligned}$$

2) Line integral $\oint_C \mathbf{F} \cdot d\mathbf{r}$.

Parametrize the boundary circle C by $\mathbf{r}(\theta) = \langle \cos \theta, \sin \theta, 0 \rangle$ for $0 \leq \theta \leq 2\pi$. Then $d\mathbf{r} = \langle -\sin \theta, \cos \theta, 0 \rangle d\theta$.

$$\mathbf{F} \cdot d\mathbf{r} = \langle -\sin \theta, \cos \theta, 0 \rangle \cdot \langle -\sin \theta, \cos \theta, 0 \rangle d\theta = (\sin^2 \theta + \cos^2 \theta) d\theta = d\theta.$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} d\theta = 2\pi.$$

Conclusion: Both integrals equal 2π , verifying the theorem. ■

Remark 31.2. We may replace the hemisphere S with any oriented, piecewise smooth surface spanning the same oriented boundary, provided $\mathbf{F}$ is continuously differentiable near both surfaces. For instance, using the flat unit disk in the xy -plane ($z = 0$) simplifies the surface integral immensely, as $\hat{\mathbf{n}} = \mathbf{k}$ and $dS = dA$:

$$\iint_D (\nabla \times \mathbf{F}) \cdot \mathbf{k} dA = \iint_D 2 dA = 2(\text{Area}) = 2\pi.$$

Example 31.3 (Surface Independence). Let $a > 0$ and let S be the upper half of the ellipsoid

$$x^2 + y^2 + \frac{z^2}{a^2} = 1,$$

oriented outward. For

$$\mathbf{F} = \langle x - y, y + z, z - x \rangle,$$

evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$. Does replacing S by another surface with the same oriented boundary change the value?

Solution. The curl is

$$\nabla \times \mathbf{F} = \langle -1, 1, 1 \rangle.$$

The boundary is the unit circle in $z = 0$, oriented counterclockwise from above. By Stokes' theorem, replace S by the upward-oriented unit disk D . Then

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \iint_D \langle -1, 1, 1 \rangle \cdot \mathbf{k} dA = \boxed{\pi}.$$

Every oriented surface spanning this boundary gives the same value, since $\mathbf{F}$ is smooth on all of $\mathbb{R}^3$. Changing the spanning surface cannot maximize or minimize this flux. ■

32 Stokes' Theorem (II)

32.1 Goals

- Choose a convenient spanning surface and apply Stokes' theorem with consistent orientation.

Example 32.1. Let $\mathbf{F}(x, y, z) = \cos(y^2 z^3) \mathbf{i} - xz^8 \mathbf{j} + z^{16} \mathbf{k}$, and let S be the semi-ellipsoid $\frac{z}{5} = \sqrt{1 - \frac{x^2}{4} - \frac{y^2}{9}}$ with upward unit normal $\hat{\mathbf{n}}$. Evaluate

$$\iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} dS.$$

Solution. By Stokes' Theorem:

$$\iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} dS = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

The boundary curve C is the ellipse in the xy -plane where $z = 0$:

$$\frac{x^2}{4} + \frac{y^2}{9} = 1, \quad z = 0.$$

On C , we evaluate the vector field $\mathbf{F}$ at $z = 0$:

$$\mathbf{F}(x, y, 0) = \langle \cos(0), -x(0)^8, 0^{16} \rangle = \langle 1, 0, 0 \rangle.$$

The line integral becomes:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C \langle 1, 0, 0 \rangle \cdot \langle dx, dy, dz \rangle = \oint_C dx.$$

Since C is a closed loop, the integral of the exact differential dx is zero:

$$\oint_C dx = 0.$$

Thus, the value of the integral is 0. ■

Example 32.2. Compute $\iint_S (\nabla \times \vec{F}) \cdot \vec{n} dS$ for $\vec{F} = \langle 2x + z, 2y + x, 2z + y \rangle$, where S is the portion of the surface $z = \sqrt{3 + x^2 + y^2}$ within the cylinder $x^2 + y^2 = 1$, oriented with an upward unit normal.

Solution. By Stokes' Theorem, the surface integral is equal to the line integral over the boundary curve C :

$$\iint_S (\nabla \times \vec{F}) \cdot \vec{n} dS = \oint_C \vec{F} \cdot d\vec{r}$$

The boundary C is the intersection of $z = \sqrt{3 + x^2 + y^2}$ and $x^2 + y^2 = 1$. Substituting the cylinder equation into the surface equation yields:

$$z = \sqrt{3 + 1} = 2$$

The boundary is the unit circle $x^2 + y^2 = 1$ in the plane $z = 2$. We parametrize C as:

$$\vec{r}(t) = \langle \cos t, \sin t, 2 \rangle, \quad d\vec{r} = \langle -\sin t, \cos t, 0 \rangle dt, \quad t \in [0, 2\pi]$$

Evaluating $\vec{F}$ on the boundary C :

$$\vec{F}(\vec{r}(t)) = \langle 2 \cos t + 2, 2 \sin t + \cos t, 4 + \sin t \rangle$$

The line integral is:

$$\begin{aligned} \oint_C \vec{F} \cdot d\vec{r} &= \int_0^{2\pi} \langle 2 \cos t + 2, 2 \sin t + \cos t, 4 + \sin t \rangle \cdot \langle -\sin t, \cos t, 0 \rangle dt \\ &= \int_0^{2\pi} (-2 \sin t \cos t - 2 \sin t + 2 \sin t \cos t + \cos^2 t) dt \\ &= \int_0^{2\pi} (-2 \sin t + \cos^2 t) dt \end{aligned}$$

Evaluating the integral:

$$\int_0^{2\pi} -2 \sin t dt + \int_0^{2\pi} \frac{1 + \cos(2t)}{2} dt = 0 + \pi = \boxed{\pi}$$
■

Example 32.3. Let $F = \langle -y, x, z^{2026} \rangle$, and S be the part of the sphere $x^2 + y^2 + z^2 = 9$ above the plane $z = 1$, with upwards orientation. Find

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS.$$

Solution. By **Stokes' Theorem**, we evaluate the flux of the curl through the flat disk $D = \{(x, y, z) : x^2 + y^2 \leq 8, z = 1\}$ which shares the boundary C :

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS = \iint_D (\nabla \times \vec{F}) \cdot \mathbf{k} dA$$

Calculating the curl of $\vec{F} = \langle -y, x, z^{2026} \rangle$:

$$\nabla \times \vec{F} = \det \begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ -y & x & z^{2026} \end{pmatrix} = \langle 0, 0, 1 - (-1) \rangle = \langle 0, 0, 2 \rangle$$

On the disk D , the upward unit normal is $\hat{n} = \mathbf{k}$, so $(\nabla \times \vec{F}) \cdot \mathbf{k} = 2$.

$$\iint_D 2 dA = 2 \cdot \text{Area}(D) = 2 \cdot \pi(\sqrt{8})^2 = 16\pi$$

■

33 The Divergence Theorem

33.1 Goals

- Interpret the divergence of a vector field as the density of “sources” or “sinks” at a point.
- State Gauss' Divergence Theorem and understand its physical intuition.
- Apply the Divergence Theorem to calculate flux across closed surfaces efficiently.

33.2 Intuition

Consider a fluid flowing through a solid region V . If we want to measure the total net flow of fluid **out** of the region (the flux through the boundary S), we have two options:

1. Measure the flow crossing the boundary surface directly.
2. Sum up all the sources (where fluid is created) and sinks (where fluid is destroyed) inside the region.

The Divergence Theorem states that these two quantities are equal. Since $\nabla \cdot \vec{F}$ represents the expansion rate of the fluid at a point, integrating $\nabla \cdot \vec{F}$ over the entire volume gives the total net expansion, which must equal the total flux out of the boundary.

Theorem 33.1 (Gauss' Divergence Theorem)

Let $V \subset \mathbb{R}^3$ be a bounded solid region whose boundary $S = \partial V$ is piecewise smooth and oriented outward. Let $\vec{F}$ be a continuously differentiable vector field defined on an open neighborhood of the closed solid V and its boundary. Then

$$\iint_S \vec{F} \cdot \vec{n} dS = \iiint_V \nabla \cdot \vec{F} dV,$$

where $\vec{n}$ is the outward-pointing unit normal vector on S .

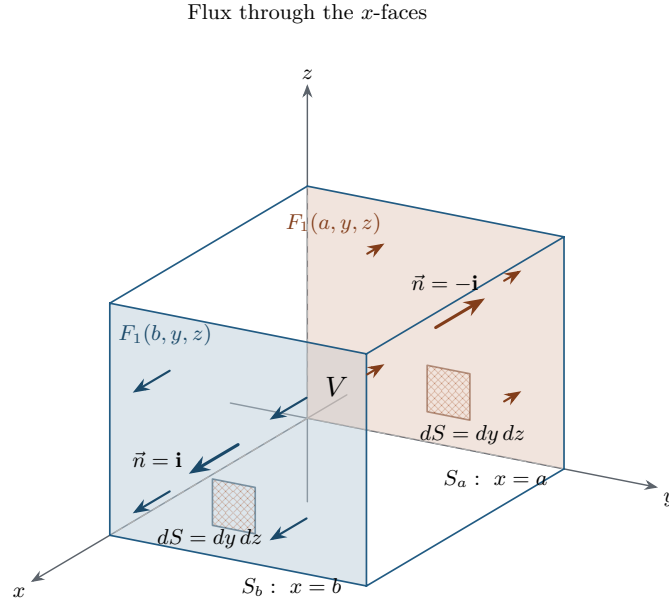


Figure 33.1: Visualizing the flux through a box. The net flux in the x -direction is the difference between the flow out of the right face and the flow into the left face: $\iint (F_1(b) - F_1(a)) dy dz$.

Proof for a Rectangular Box. We provide the argument for a rectangular box $V = [a, b] \times [c, d] \times [e, f]$ to illustrate the mechanism of the theorem. Let $\vec{F} = \langle F_1, F_2, F_3 \rangle$. Then

$$\nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}.$$

We compute the volume integral of the first term, $\frac{\partial F_1}{\partial x}$:

$$\iiint_V \frac{\partial F_1}{\partial x} dV = \int_e^f \int_c^d \left[\int_a^b \frac{\partial F_1}{\partial x} dx \right] dy dz.$$

By the Fundamental Theorem of Calculus, the inner integral is $F_1(b, y, z) - F_1(a, y, z)$. Thus,

$$\iiint_V \frac{\partial F_1}{\partial x} dV = \int_e^f \int_c^d F_1(b, y, z) dy dz - \int_e^f \int_c^d F_1(a, y, z) dy dz.$$

This result corresponds exactly to the surface integral of $\vec{F} \cdot \vec{n}$ over the two faces perpendicular to the x -axis (where $\vec{n} = \vec{i}$ at $x = b$ and $\vec{n} = -\vec{i}$ at $x = a$).

Analogous computations for the y - and z -derivatives yield the flux through the other pairs of faces. Summing these results gives:

$$\iiint_V \nabla \cdot \vec{F} \, dV = \iiint_V \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) dV = \iint_S \vec{F} \cdot \vec{n} \, dS.$$

■

Why internal flux cancels

For a finite union of boxes with disjoint interiors, the fluxes on shared faces cancel because the outward normals point in opposite directions. Summing the box identities therefore proves the theorem for such a union. Approximating a general solid by boxes suggests the same result, but passing to the limit requires control of the boundary and its normals. The box calculation alone is not a proof for an arbitrary solid.

Remark 33.1. The Divergence Theorem is a special case of the generalized Stokes' Theorem, $\int_{\partial M} \omega = \int_M d\omega$, applied to a 3-dimensional region with the differential 2-form associated to $\vec{F}$.

Example 33.2. Compute the flux integral $\iint_{\Sigma} \vec{F} \cdot \vec{n} \, dS$ where

$$\vec{F}(x, y, z) = \langle e^{2x} \sin(2y), e^{2x} \cos(2y), y^2 z^2 \rangle,$$

and Σ is the boundary of the rectangular box

$$0 \leq x \leq 1, \quad 0 \leq y \leq 1, \quad 0 \leq z \leq 2,$$

oriented outward.

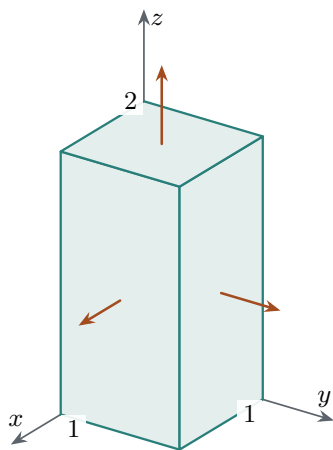


Figure 33.2: The region of integration is the box $[0, 1] \times [0, 1] \times [0, 2]$.

Solution. Calculating the surface integral directly would require parametrizing 6 different faces. Instead, we use the Divergence Theorem:

$$\iint_{\Sigma} \vec{F} \cdot \vec{n} \, dS = \iiint_V \nabla \cdot \vec{F} \, dV.$$

First, we compute the divergence of $\vec{F}$:

$$\begin{aligned}\nabla \cdot \vec{F} &= \frac{\partial}{\partial x}(e^{2x} \sin(2y)) + \frac{\partial}{\partial y}(e^{2x} \cos(2y)) + \frac{\partial}{\partial z}(y^2 z^2) \\ &= 2e^{2x} \sin(2y) + (-2e^{2x} \sin(2y)) + 2y^2 z \\ &= \underbrace{2e^{2x} \sin(2y) - 2e^{2x} \sin(2y)}_0 + 2y^2 z \\ &= 2y^2 z.\end{aligned}$$

The complicated exponential terms cancel, simplifying the integral significantly. We now integrate $2y^2 z$ over the box:

$$\begin{aligned}\text{Flux} &= \int_0^1 \int_0^1 \int_0^2 2y^2 z \, dz \, dy \, dx \\ &= \left(\int_0^1 dx \right) \left(\int_0^1 2y^2 \, dy \right) \left(\int_0^2 z \, dz \right) \\ &= [x]_0^1 \cdot \left[\frac{2}{3} y^3 \right]_0^1 \cdot \left[\frac{1}{2} z^2 \right]_0^2 \\ &= (1) \cdot \left(\frac{2}{3} \right) \cdot (2) \\ &= \frac{4}{3}.\end{aligned}$$

Therefore, the total outward flux is $\boxed{\frac{4}{3}}$. ■

34 The Divergence Theorem (II)

34.1 Goals

- Use the divergence theorem to compute flux, including surfaces that must be closed by adding a cap.

Example 34.1. Find the flux of the vector field $\vec{F}(x, y, z) = \langle y^5, -x^3, z^2 \rangle$ through the part of the paraboloid $z = x^2 + y^2$ that lies below $z = 2$ oriented downward.

Solution. To use the Divergence Theorem, we must have a closed surface. Let S_1 be the paraboloid $z = x^2 + y^2$ (oriented downward) and let S_2 be the disk $x^2 + y^2 \leq 2$ at the plane $z = 2$ (oriented upward). Together, $S_1 \cup S_2$ forms the boundary ∂V of the solid region V . By the Divergence Theorem:

$$\iint_{S_1} \vec{F} \cdot d\vec{S} + \iint_{S_2} \vec{F} \cdot d\vec{S} = \iiint_V \nabla \cdot \vec{F} \, dV.$$

First, we compute the divergence of $\vec{F} = \langle y^5, -x^3, z^2 \rangle$:

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(y^5) + \frac{\partial}{\partial y}(-x^3) + \frac{\partial}{\partial z}(z^2) = 0 + 0 + 2z = 2z.$$

Next, we evaluate the triple integral over the volume V using cylindrical coordinates ($x^2 + y^2 = r^2$):

$$\begin{aligned}
 \iiint_V 2z \, dV &= \int_0^{2\pi} \int_0^{\sqrt{2}} \int_{r^2}^2 2z \cdot r \, dz \, dr \, d\theta \\
 &= 2\pi \int_0^{\sqrt{2}} r \left[z^2 \right]_{r^2}^2 \, dr \\
 &= 2\pi \int_0^{\sqrt{2}} (4r - r^5) \, dr \\
 &= 2\pi \left[2r^2 - \frac{1}{6}r^6 \right]_0^{\sqrt{2}} = 2\pi \left(4 - \frac{8}{6} \right) = \frac{16\pi}{3}.
 \end{aligned}$$

Now, we compute the flux through the lid S_2 . On S_2 , the upward normal is $\vec{n} = \langle 0, 0, 1 \rangle$ and $z = 2$:

$$\begin{aligned}
 \iint_{S_2} \vec{F} \cdot \vec{n} \, dS &= \iint_{S_2} \langle y^5, -x^3, 2^2 \rangle \cdot \langle 0, 0, 1 \rangle \, dA \\
 &= \iint_{S_2} 4 \, dA = 4 \cdot \text{Area}(S_2) = 4(2\pi) = 8\pi.
 \end{aligned}$$

Finally, solving for the flux through the paraboloid S_1 :

$$\begin{aligned}
 \text{Flux}_{S_1} &= \iiint_V \nabla \cdot \vec{F} \, dV - \iint_{S_2} \vec{F} \cdot d\vec{S} \\
 &= \frac{16\pi}{3} - 8\pi = \frac{16\pi - 24\pi}{3} = -\frac{8\pi}{3}.
 \end{aligned}$$

Therefore, the downward flux is $\boxed{-\frac{8\pi}{3}}$. ■

Example 34.2. Let S be the surface $z = (9 - x^2 - y^2)^{11/4}$ with upwards pointing normal vector orientation. Evaluate the flux integral

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, dS,$$

where $\vec{F}(x, y, z) = \langle e^{x^{11}} - y, x + \sin(y^3), z^{13} \rangle$.

Solution. The surface S is defined by $z = (9 - x^2 - y^2)^{11/4}$. Evaluating the flux of a curl directly is computationally intensive, so we apply **Stokes' Theorem**:

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, dS = \oint_C \vec{F} \cdot d\vec{r},$$

where C is the boundary curve of S . The boundary occurs where the surface meets the xy -plane ($z = 0$):

$$0 = (9 - x^2 - y^2)^{11/4} \implies x^2 + y^2 = 9.$$

The curve C is a circle of radius 3 in the $z = 0$ plane. With the upward orientation of S , C is oriented counter-clockwise. On C , $z = 0$ and $dz = 0$, so the vector field simplifies to $\vec{F} = \langle e^{x^{11}} - y, x + \sin(y^3), 0 \rangle$. The line integral is:

$$\oint_C (e^{x^{11}} - y) \, dx + (x + \sin(y^3)) \, dy.$$

Applying **Green's Theorem** over the disk D enclosed by C ($x^2 + y^2 \leq 9$):

$$\begin{aligned}
 \text{Flux} &= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA \\
 &= \iint_D \left(\frac{\partial}{\partial x}(x + \sin(y^3)) - \frac{\partial}{\partial y}(e^{x^{11}} - y) \right) dA \\
 &= \iint_D (1 - (-1)) dA \\
 &= \iint_D 2 dA.
 \end{aligned}$$

The complicated transcendental terms disappear, leaving a constant integrand. The integral is simply twice the area of the disk D with radius $R = 3$:

$$\begin{aligned}
 \text{Flux} &= 2 \cdot \text{Area}(D) \\
 &= 2 \cdot (\pi \cdot 3^2) \\
 &= 18\pi.
 \end{aligned}$$

Therefore, the value of the flux integral is $\boxed{18\pi}$. ■

Remark 34.3. For the preceding example, we can also close the surface and use the divergence theorem. If $\vec{F}$ has continuous second partial derivatives near the enclosed solid, then the mixed partials cancel to give

$$\nabla \cdot (\nabla \times \vec{F}) = 0.$$

If we *close* the surface S by adding the disk D at $z = 0$ (oriented downward), the Divergence Theorem states:

$$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} + \iint_D (\nabla \times \vec{F}) \cdot d\vec{S}_{\text{down}} = \iiint_V \nabla \cdot (\nabla \times \vec{F}) dV = 0.$$

This implies:

$$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \iint_D (\nabla \times \vec{F}) \cdot d\vec{S}_{\text{up}}.$$

This equality requires matching induced boundary orientations and smoothness of the field on the required surfaces and region. Calculating the curl's flux through the flat disk D is often much easier than the original surface:

$$\nabla \times \vec{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ e^{x^{11}} - y & x + \sin(y^3) & z^{13} \end{vmatrix} = \langle 0, 0, 1 - (-1) \rangle = \langle 0, 0, 2 \rangle.$$

The flux through the disk D (where $\vec{n} = \langle 0, 0, 1 \rangle$) is simply $\iint_D 2 dA = 2 \cdot \text{Area}(D) = 18\pi$.